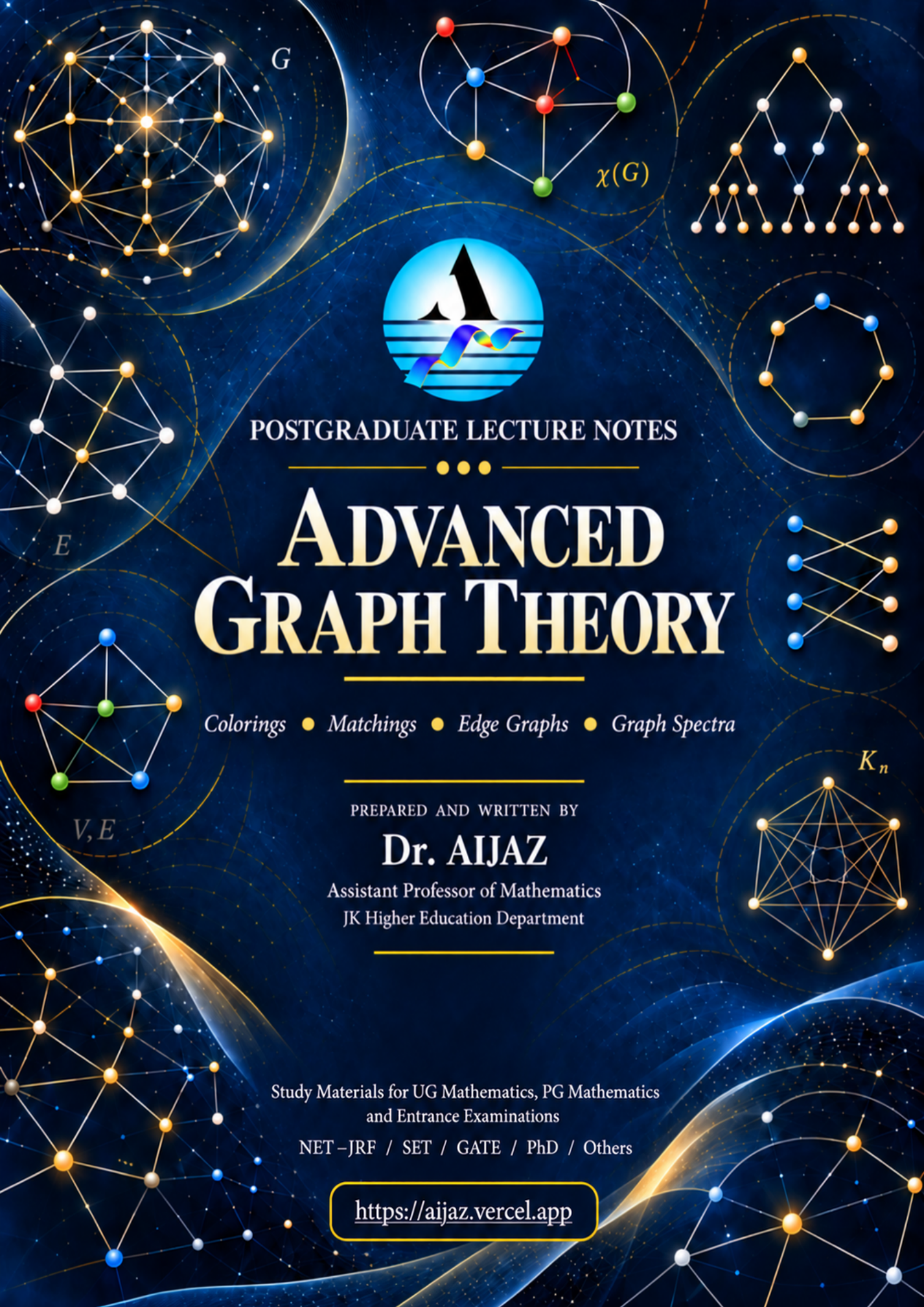


POSTGRADUATE LECTURE NOTES

ADVANCED GRAPH THEORY

Colorings • Matchings • Edge Graphs • Graph Spectra

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Chapter 1

Coloring of Graphs

Introduction

Graph coloring originated with the celebrated **Four Color Problem**. In 1852, Francis Guthrie observed that four colors appeared sufficient to color the regions of a map so that adjacent regions received different colors. The problem was later popularized by Augustus De Morgan and Arthur Cayley and became one of the central questions in graph theory.
Alfred Kempe proposed an influential proof in 1879. Although the proof was later shown to be incorrect, several of its ideas survived. In 1890, Percy Heawood used related techniques to prove the **Five Color Theorem** and to extend map-coloring ideas to surfaces of higher genus. Later developments shifted attention from planar maps to vertex and edge colorings of general graphs. Brooks' theorem (1941) related the chromatic number $\chi(G)$ to the maximum degree $\Delta(G)$, while Vizing's theorem (1964) showed that the edge chromatic number of a simple graph is either $\Delta(G)$ or $\Delta(G)+1$.
The Four Color Theorem was finally proved by Appel and Haken in 1976 using computer-assisted methods. The ideas developed around the problem—proper colorings, critical graphs, degree bounds, and structural characterizations—remain fundamental in modern graph theory.

1.1 Vertex Coloring

1.1.1 Basic Definitions

Definition 1.1.1 (Vertex Coloring)**.** Let $G = (V, E)$ be an undirected graph. A **proper vertex coloring** with k colors is a function $c : V \to \{1, 2, \ldots, k\}$ such that $uv \in E$ implies $c(u) \neq c(v)$. A proper coloring using at most k colors is called a **k-coloring**.

The color labels themselves are irrelevant; positive integers are used only for convenience.

Definition 1.1.2 (Chromatic Number)**.** The **chromatic number** of G, denoted by $\chi(G)$, is the minimum number of colors required for a proper coloring; equivalently,

$$\chi(G) = \min\{k : G \text{ admits a proper } k\text{-coloring}\}.$$

Elementary Examples

- If G has no edges and at least one vertex, then $\chi(G) = 1$.
- If G contains at least one edge, then $\chi(G) \geq 2$.

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- For the complete graph K_n, $\chi(K_n) = n$.
- For the cycle C_n $(n \geq 3)$,

$$\chi(C_n) = \begin{cases} 2, & n \text{ even}, \\ 3, & n \text{ odd}. \end{cases}$$

Proof for cycles. Label the vertices $v_1, \ldots, v_n$ consecutively. If n is even, alternating two colors around the cycle gives a proper coloring, and since C_n contains an edge, $\chi(C_n) = 2$. If n is odd, alternating two colors forces v_n and v_1 to receive the same color although they are adjacent, so two colors are impossible; three colors suffice. Hence $\chi(C_n) = 3$. ■

1.1.2 Trees and Bipartite Graphs

Theorem 1.1.3. *Every tree with at least two vertices is 2-chromatic.*

Proof. Let T be a tree with at least two vertices. Since T contains an edge, $\chi(T) \geq 2$. Choose a root $r \in V(T)$ and color each vertex u according to the parity of its distance from r: use color 1 if $d(r,u)$ is even and color 2 if it is odd.

If $xy \in E(T)$, then the unique paths from r to x and r to y differ in length by exactly one. Thus $d(r,x)$ and $d(r,y)$ have opposite parity, so adjacent vertices receive different colors. Hence T is 2-colorable and therefore $\chi(T) = 2$. ■

Theorem 1.1.4. *A graph G is 2-colorable if and only if it contains no odd cycle.*

Proof. If G is 2-colorable, the colors must alternate around every cycle; therefore every cycle has even length.

Conversely, suppose G has no odd cycle. It suffices to consider each connected component separately. Choose a root r in a connected component and define

$$V_1 = \{u : d(r,u) \text{ is even}\}, \qquad V_2 = \{u : d(r,u) \text{ is odd}\}.$$

No two vertices in the same set can be adjacent, for otherwise the corresponding shortest paths from r, together with that edge, would create an odd cycle. Hence every edge joins V_1 to V_2, so coloring V_1 and V_2 with two different colors gives a proper 2-coloring. ■

Theorem 1.1.5. *A nonempty graph G is 2-chromatic if and only if G is bipartite.*

Proof. If G is bipartite with bipartition $V(G) = V_1 \cup V_2$, color V_1 with color 1 and V_2 with color 2. Since G is nonempty, it contains an edge, so $\chi(G) = 2$.

Conversely, if G has a proper 2-coloring, let V_1 and V_2 be the sets of vertices receiving the two colors. No two vertices within the same set are adjacent, so every edge joins V_1 to V_2. Hence G is bipartite. ■

Remark 1.1.6. Thus, for a nonempty graph,

$$G \text{ is bipartite} \iff G \text{ is 2-colorable} \iff G \text{ has no odd cycle.}$$

Theorem 1.1.7 (A General Upper Bound). *For every graph G, $\chi(G) \leq \Delta(G) + 1$.*

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Proof. We induct on $n = |V(G)|$. The case $n = 1$ is immediate. Assume the result holds for graphs with $n - 1$ vertices, and choose $v \in V(G)$. By induction,

$$\chi(G - v) \leq \Delta(G - v) + 1 \leq \Delta(G) + 1.$$

Thus $G - v$ can be colored with at most $\Delta(G) + 1$ colors. Since v has at most $\Delta(G)$ neighbors, at most $\Delta(G)$ colors are forbidden at v, leaving at least one available color. Hence the coloring extends to G, proving $\chi(G) \leq \Delta(G) + 1$. ■

1.1.3 Color-Critical Graphs

Definition 1.1.8 (Color-Critical Graph)**.** A graph G is **k-critical** if $\chi(G) = k$ and $\chi(G - v) = k - 1$ for every $v \in V(G)$.

Theorem 1.1.9. *If G is k-critical, then:*

(a) G is connected;

(b) $\delta(G) \geq k - 1$;

(c) there do not exist proper subgraphs G_1, G_2 such that $G = G_1 \cup G_2$ and $G_1 \cap G_2$ is complete;

(d) if $k > 1$, then $G - v$ is connected for every $v \in V(G)$.

Proof. a. Assume G is not connected. Since $\chi(G) = k$, there is a component G_1 of G such that $\chi(G_1) = k$. If v is any vertex of G which is not in G_1, then G_1 is a component of the subgraph $G - v$. Therefore, $\chi(G - v) = \chi(G_1) = k$. This contradicts the fact that G is k-critical.

b. Assume to the contrary that v be a vertex of G so that $d(v) < k - 1$. Since G is k-critical, the subgraph $G - v$ has a $(k - 1)$-colouring. As v has at most $k - 2$ neighbours, these neighbours use at most $k - 2$ colours in this $(k - 1)$-colouring of $G - v$. Now, colour v with the unused colour and this gives a $(k - 1)$-colouring of G. This contradicts the given assumption that $\chi(G) = k$. Hence every vertex v has degree at least $k - 1$.

c. Let $G = G_1 \cup G_2$, where G_1 and G_2 are subgraphs with $G_1 \cap G_2 = K_r$. Since G is k-critical, therefore G_1 and G_2 both have chromatic number at most $k - 1$. Consider a $(k - 1)$-colouring of G_1 and a $(k - 1)$-colouring of G_2. As $G_1 \cap G_2$ is complete, in the overlap, every vertex in $G_1 \cap G_2$ has a different colour (in each of the $(k - 1)$-colourings). This implies that colours in the $(k - 1)$-colouring of G_2 can be rearranged such that it assigns the same colour to each vertex in $G_1 \cap G_2$, as is given by the colouring of G_1. Combining the two colourings then produces a $(k - 1)$-colouring of all of G. This is impossible, since $\chi(G) = k$. Thus no subgraphs of the type G_1 and G_2 exist.

d. Assume $G - v$ is disconnected, for some vertex v of G. Then $G - v$ has a subgraph H_1 and H_2 with $H_1 \cup H_2 = G - v$ and $H_1 \cap H_2 = \emptyset$. Let G_1 and G_2 be the subgraphs of G, where G_1 is induced by H_1 and v, while G_2 is induced by H_2 and v. Then $G = G_1 \cup G_2$ and $G_1 \cap G_2 = K_1$ (with K_1 as a single vertex). This contradicts (c) and thus $G - v$ is connected.

■

Corollary 1.1.10. *Every k-chromatic graph has at least k vertices of degree at least $k - 1$.*

Proof. Starting from a k-chromatic graph G, repeatedly delete vertices whenever the chromatic number remains k. This process ends with a k-critical subgraph H. By Theorem 1.1.9, $\delta(H) \geq k - 1$, and since

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every k-chromatic graph has at least k vertices, H contains at least k vertices of degree at least $k-1$. Their degrees in G are at least as large. ■

Theorem 1.1.11. *Let*

$$k = \max\{\delta(G') : G' \text{ is a nonempty subgraph of } G\}.$$

Then $\chi(G) \leq k+1$.

Proof. Let $\chi(G) = r$ and let H be an r-critical subgraph of G. By Theorem 1.1.9, $\delta(H) \geq r-1$. Since H is a subgraph of G, the definition of k gives $\delta(H) \leq k$. Hence $r-1 \leq k$, so $\chi(G) = r \leq k+1$. ■

Theorem 1.1.12. *Let G have degree sequence $d_1 \geq d_2 \geq \cdots \geq d_n$. Then*

$$\chi(G) \leq \max_{1 \leq i \leq n} \min\{i, d_i + 1\}.$$

Proof. Let $\chi(G) = k$. By Corollary 1.1.10, G has at least k vertices of degree at least $k-1$, so $d_k \geq k-1$. Hence $\min\{k, d_k+1\} = k$, and therefore

$$\max_{1 \leq i \leq n} \min\{i, d_i + 1\} \geq k = \chi(G).$$

■

Algorithm 1.1.13 (The Greedy Coloring Algorithm)**.** The **greedy coloring algorithm** is a simple method for graph vertex coloring that processes vertices in a specific order and assigns each vertex the smallest possible color not used by its already-colored neighbors.

1. Order the vertices $v_1, v_2, \ldots, v_n$
2. For $i = 1$ to n:

   Assign v_i the smallest color $c \in \mathbb{N}$ such that no adjacent vertex already has color c.

The algorithm produces a proper coloring but may not use the minimum number of colors (chromatic number $\chi(G)$). The number of colors used depends heavily on the vertex ordering. (See Figures 1.1 and 1.2).

Algorithm 1.1.14. A **Kempe chain** is a maximal connected component of a graph induced by vertices colored with two given colors.

The **Kempe chain argument** is a recoloring technique used in graph coloring where colors are swapped along an alternating path between two colors. By swapping these colors along the entire component, one obtains another proper coloring.

This method was famously used in:

- The proof of the **Five Color Theorem** for planar graphs
- Alfred Kempe's flawed but influential attempt at the **Four Color Theorem**
- Various graph coloring algorithms and proofs

How to use Kempe Chain Argument: Let G be a graph that uses at least two colors for its proper coloring. Choose two colors, say i & j and construct a subgraph H of G induced by the colors i & j. Let

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K be a connected component in H. Interchange the colors i and j in K. This produces a new coloring of G. In this process, the subgraph K of G is known as Kempe chain and this technique to obtain a new coloring of G is called the Kempe chain argument. (See Figures 1.1.3, 1.1.3).

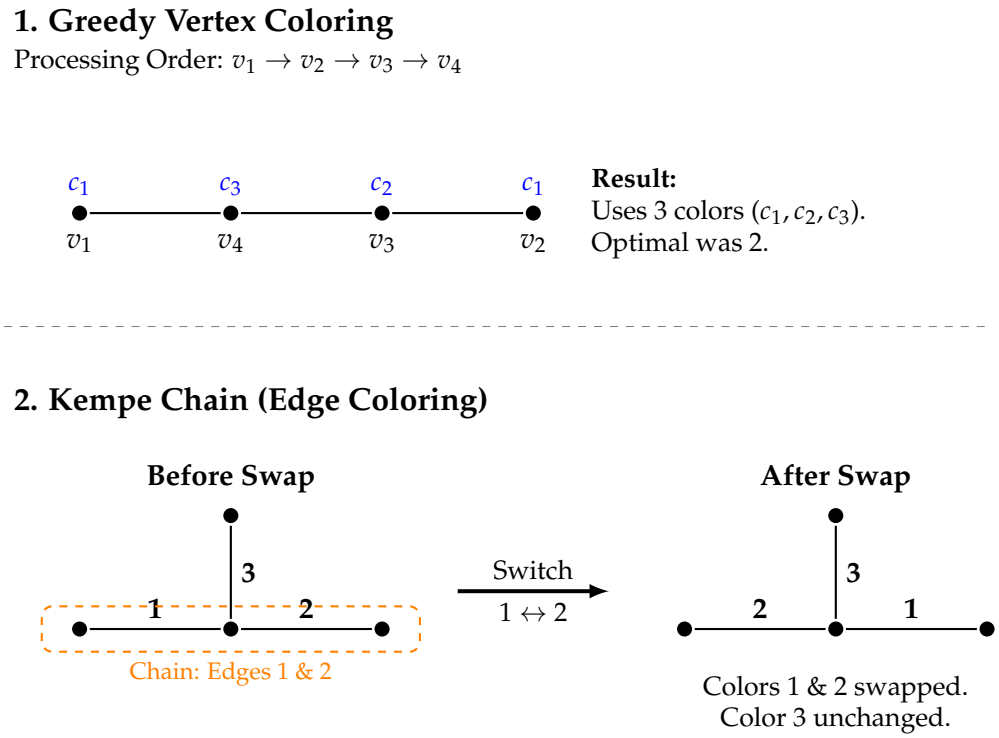


Figure 1.1: Top: Greedy Vertex Coloring. Bottom: Kempe Chain Argument on Edge Coloring.

Theorem 1.1.15 (Brooks' Theorem). *Let G be a connected graph with maximum degree Δ. If G is neither a complete graph nor an odd cycle, then*

$$\chi(G) \leq \Delta.$$

Proof. If $\Delta \leq 2$, then G is a path or a cycle. Paths and even cycles are 2-colorable, while odd cycles are excluded. Hence assume

$$\Delta \geq 3.$$

Case 1: G **is not** Δ**-regular.**

Choose a vertex v with $\deg(v) < \Delta$. Order the vertices by decreasing distance from v, so that v is colored last.

Color greedily using the colors $1, \ldots, \Delta$. Every vertex $x \neq v$ has a neighbor closer to v, and this neighbor occurs later in the ordering. Thus, when x is colored, at most $\Delta - 1$ of its neighbors have already been colored. Hence a color is available for x. Finally, since $\deg(v) < \Delta$, a color is also available for v. Therefore

$$\chi(G) \leq \Delta.$$

We may now assume that G is Δ-regular.

Case 2: G **has a cut-vertex.**

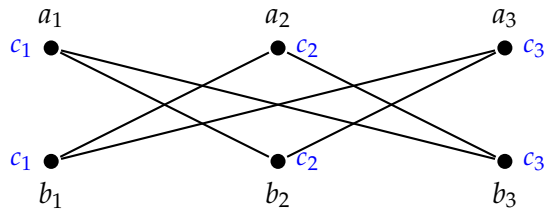
Let v be a cut-vertex, and let

$$C_1, \ldots, C_k$$

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1. Greedy Coloring (Crown Graph)

Order: $a_1, b_1, a_2, b_2, a_3, b_3$



Result:
Used 3 Colors (c_1, c_2, c_3)
(Optimal is 2)

2. Kempe Chain (Edge Coloring)

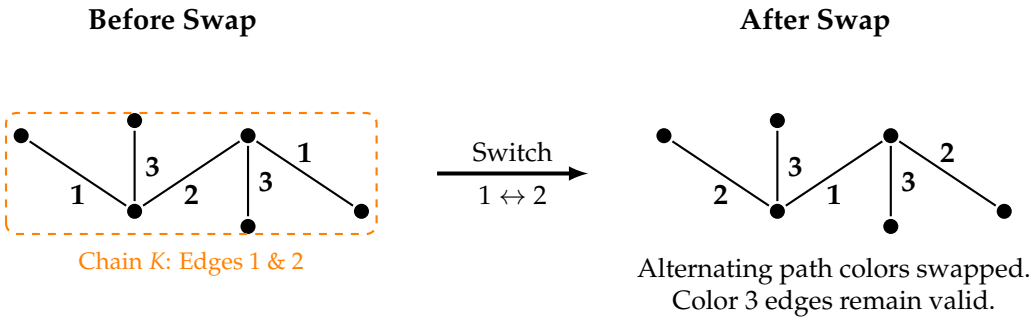


Figure 1.2: Top: Greedy coloring on a Crown Graph (6 vertices) forcing a sub-optimal 3rd color. Bottom: Kempe Chain argument on a larger path, swapping edge colors 1 and 2.

be the components of $G - v$. For each i, consider

$$G_i = G[C_i \cup \{v\}].$$

Since v has neighbors in at least two components,

$$\deg_{G_i}(v) < \Delta.$$

Applying the greedy argument of Case 1 to each G_i, we obtain a Δ-coloring of every G_i. By permuting the colors, we may assume that v receives the same color in every G_i. These colorings then combine to give a proper Δ-coloring of G.

Thus it remains to consider the case in which G is Δ-regular and 2-connected.

We first identify three useful vertices. We claim that there exist vertices u, v, w such that

$$uv, vw \in E(G), \qquad uw \notin E(G),$$

and

$$G - \{u, w\}$$

is connected.

If G is 3-connected, choose any induced path $u - v - w$; such a path exists because G is connected and not complete. Since G is 3-connected, deleting u and w leaves a connected graph.

Suppose instead that G is 2-connected but not 3-connected. Then there are vertices v and x such that $G - \{v, x\}$ is disconnected. Hence $G - v$ is connected but has a cut-vertex.

Choose two distinct endblocks B_1 and B_2 of $G - v$, and let b_i be the cut-vertex through which B_i is attached to the rest of $G - v$. Since G itself has no cut-vertex, v must have a neighbor in each set

$$B_i - \{b_i\}.$$

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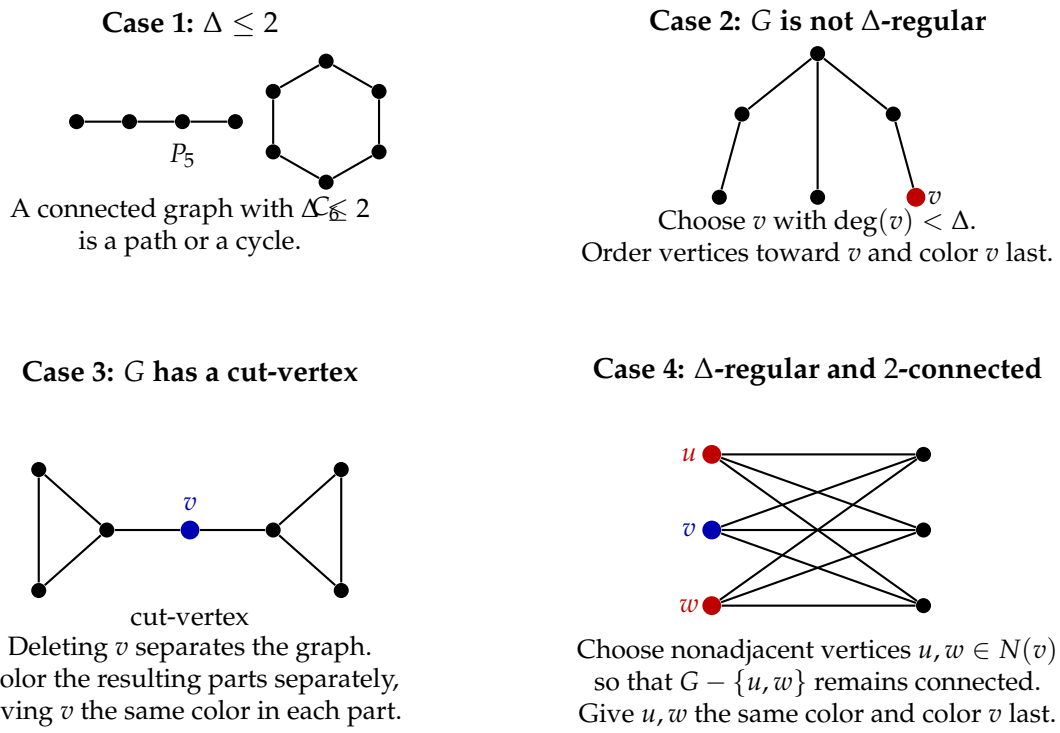


Figure 1.3: Graphical illustration of the main cases in the proof of Brooks' theorem.

Choose

$$u \in B_1 - \{b_1\}, \qquad w \in B_2 - \{b_2\}$$

such that

$$uv, vw \in E(G).$$

Because u and w belong to distinct endblocks, they are not adjacent.

Moreover, u and w are non-cut vertices lying in distinct endblocks. Deleting them does not disconnect $G - v$. Thus

$$G - \{u, v, w\}$$

is connected. Since $\deg(v) = \Delta \geq 3$, the vertex v has a neighbor other than u and w, so adding v back shows that

$$G - \{u, w\}$$

is connected. This proves the claim.

Now give the nonadjacent vertices u and w the same color, say color 1.

Since

$$H = G - \{u, w\}$$

is connected, order the vertices of H by decreasing distance from v, with v last. Color them greedily with the colors $1, \ldots, \Delta$.

Every vertex $x \neq v$ has a neighbor closer to v, hence a neighbor that is still uncolored when x is considered. Therefore at most $\Delta - 1$ of its neighbors have already been colored, and a color is available.

Finally, consider v. It has exactly Δ neighbors, but two of them, namely u and w, have the same color. Hence the neighbors of v use at most

$$\Delta - 1$$

distinct colors. Thus one of the Δ colors is available for v.

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Consequently, G has a proper Δ-coloring, and therefore

$$\chi(G) \leq \Delta.$$

■

Theorem 1.1.16. *For any graph G of order n,*

$$\frac{n}{\chi(G)} \leq \alpha_0(G) \leq n - \chi(G) + 1.$$

Proof. Let $\chi(G) = k$. Then the vertices of G can be partitioned into k colour classes

$$V_1, V_2, \ldots, V_k,$$

where each V_i is an independent set.

Let $|V_i| = n_i$. Since $\alpha_0(G)$ is the maximum size of an independent set in G, we have

$$n_i \leq \alpha_0(G) \qquad \text{for each } i.$$

Therefore,

$$n = \sum_{i=1}^{k} n_i \leq \sum_{i=1}^{k} \alpha_0(G) = k\alpha_0(G).$$

Since $k = \chi(G)$, it follows that

$$\boxed{\frac{n}{\chi(G)} \leq \alpha_0(G)}.$$

For the upper bound, let S be a maximum independent set of G, so that

$$|S| = \alpha_0(G).$$

Removing S can reduce the chromatic number by at most one, because all vertices of S can be assigned a single colour. Hence

$$\chi(G - S) \geq \chi(G) - 1.$$

On the other hand, $G - S$ has $n - \alpha_0(G)$ vertices, and therefore

$$\chi(G - S) \leq n - \alpha_0(G).$$

Combining these inequalities gives

$$\chi(G) - 1 \leq \chi(G - S) \leq n - \alpha_0(G).$$

Thus,

$$\chi(G) - 1 \leq n - \alpha_0(G),$$

and hence

$$\boxed{\alpha_0(G) \leq n - \chi(G) + 1}.$$

■

Theorem 1.1.17. *For any graph of order n,*

$$2\sqrt{n} \leq \chi + \overline{\chi} \leq n + 1, \text{ and } n \leq \chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2, \tag{1.1}$$

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where $\chi = \chi(G)$ and $\overline{\chi} = \chi(\overline{G})$.

Proof. Let G be k-chromatic and let $v_1, v_2, \ldots, v_k$ be the colour classes of G, where $|V_i| = n_i$. Then $\sum n_i = n$ and $\max n_i \geq n/k$. Since each V_i induces a complete subgraph of $\overline{G}$, $\overline{\chi} \geq \max n_i/n_i$ so that $\overline{\chi} \geq n/n_i$. Thus for each i, $n_i \geq n/\overline{\chi}$. Since the geometric mean of two positive numbers is always less or equal to their arithmetic mean, it follows that $\chi + \overline{\chi} \geq 2\sqrt{n}$.

To prove $\chi + \overline{\chi} \leq n + 1$, we induct on n. Clearly, the equality holds for $n = 1$. We assume that $\chi(G) + \overline{\chi}(G) \leq n$ for all graphs G having fewer than n vertices. Let H and $\overline{H}$ be complementary graphs with n vertices and let v be a vertex of H. Then $G = H - v$ and $\overline{G} = \overline{H} - v$ are complementary graphs with $n - 1$ vertices. Let the degree of v in H be d, so that degree of v in $\overline{H}$ is $n - d - 1$. Clearly, $\chi(H) \leq \chi(G) + 1$ and $\overline{\chi}(H) \leq \overline{\chi}(G) + 1$. If either $\chi(H) < \chi(G) + 1$ or $\overline{\chi}(H) < \overline{\chi}(G) + 1$, then $\chi(H) + \overline{\chi}(H) \leq n + 1$. If $\chi(H) = \chi(G) + 1$ and $\overline{\chi}(H) = \overline{\chi}(G) + 1$, then the removal of v from H, producing G, decreases the chromatic number, so that $d \geq \chi(G)$. Similarly, $n - d - 1 \geq \overline{\chi}(G)$. Thus, $\chi(G) + \overline{\chi}(G) \leq n - 1$. Therefore we always have $\chi(H) + \overline{\chi}(H) \leq n + 1$. Applying now the inequality $4\chi\overline{\chi} \leq (\chi + \overline{\chi})^2$, we get $\chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2$. ■

2nd Proof. Let G be a graph of order n. Let $\chi = \chi(G)$ and $\overline{\chi} = \chi(\overline{G})$.

(i) $n \leq \chi\overline{\chi}$**:** Consider a χ-coloring of G with color classes $V_1, \ldots, V_\chi$. Each V_i is an independent set in G. Consider a $\overline{\chi}$-coloring of $\overline{G}$ with color classes $\overline{V}_1, \ldots, \overline{V}_{\overline{\chi}}$. Each $\overline{V}_j$ is an independent set in $\overline{G}$, which means it is a clique in G. Every vertex $v \in V(G)$ belongs to some V_i and some $\overline{V}_j$. If u, v are two distinct vertices such that they are in the same color class V_i in G AND in the same color class $\overline{V}_j$ in $\overline{G}$, then u and v are not adjacent in G (because V_i is an independent set) and u and v are not adjacent in $\overline{G}$ (because $\overline{V}_j$ is an independent set). This is impossible, as any two distinct vertices must be adjacent in either G or $\overline{G}$. Therefore, each vertex must be assigned a unique ordered pair of colors (i, j) where $1 \leq i \leq \chi$ and $1 \leq j \leq \overline{\chi}$. The total number of such unique pairs is $\chi\overline{\chi}$. Since there are n vertices, $n \leq \chi\overline{\chi}$.

(ii) $2\sqrt{n} \leq \chi + \overline{\chi}$**:** Using the AM-GM inequality: $\frac{\chi+\overline{\chi}}{2} \geq \sqrt{\chi\overline{\chi}}$. From the previous part, $n \leq \chi\overline{\chi}$, so $\sqrt{n} \leq \sqrt{\chi\overline{\chi}}$. Combining these, $\frac{\chi+\overline{\chi}}{2} \geq \sqrt{n}$, which implies $2\sqrt{n} \leq \chi + \overline{\chi}$.

(iii) $\chi + \overline{\chi} \leq n + 1$**:** This is a standard result, often attributed to Nordhaus and Stewart (1963). The maximum value of $\chi(G) + \chi(\overline{G})$ is $n + 1$, achieved by complete graphs (K_n) and empty graphs ($\overline{K_n}$). For K_n, $\chi(K_n) = n$ and $\overline{\chi}(K_n) = 1$, so $\chi + \overline{\chi} = n + 1$. For $\overline{K_n}$, $\chi(\overline{K_n}) = 1$ and $\overline{\chi}(\overline{K_n}) = n$, so $\chi + \overline{\chi} = n + 1$.

(iv) $\chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2$**:** From $\chi + \overline{\chi} \leq n + 1$, and using the property that for any real numbers a, b, $4ab \leq (a + b)^2$: $4\chi\overline{\chi} \leq (\chi + \overline{\chi})^2 \leq (n + 1)^2$. Dividing by 4, we get $\chi\overline{\chi} \leq \frac{(n+1)^2}{4} = \left(\frac{n+1}{2}\right)^2$. ■

1.2 Edge Colouring

An edge colouring of a non-empty graph G is a labelling $f : E(G) \to \{1, 2, \ldots\}$ such that adjacent edges are assigned different colours. The labels are called *colours*. A k-edge colouring of G is an edge colouring of G that uses exactly k different colours, and in this case, G is said to be k-edge colourable.

This definition implies that a k-edge colouring of a graph $G(V, E)$ partitions the edge set E into k independent sets $E_1, E_2, \ldots, E_k$ such that $E = E_1 \cup E_2 \cup \ldots \cup E_k$. These independent sets E_i, for $1 \leq i \leq k$, are called the *colour classes*. The function $f : E(G) \to \{1, 2, \ldots, k\}$ such that $f(e) = i$ for each $e \in E_i$ (where $1 \leq i \leq k$) is called the *colour function*. The minimum number k for which a k-colouring of G exists is called the *edge chromatic number* (or *edge chromatic index*) and is denoted by $\chi'(G)$.

We have the following observations:

1. If H is a subgraph of a graph G, then $\chi'(H) \leq \chi'(G)$.

2. For any graph G, $\chi'(G) \geq \Delta(G)$, where $\Delta(G)$ is the maximum degree of G.

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If v is any vertex of G with degree $d(v) = \Delta(G)$, then the $\Delta(G)$ edges incident with v must all have different colours in any edge colouring of G.

3. For a cycle graph C_n of order n:

$$\chi'(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

Definition: Let i be a colour used in an edge colouring of a graph G. If there is an edge coloured i incident to a vertex v of G, we say i is *present* at v. If there is no edge coloured i incident to v, we say i is *absent* from v.

Theorem 1.2.1 (König, 1916). *For a non-empty bipartite graph G, $\chi'(G) = \Delta(G)$.*

Proof. The proof is by induction on the number of edges of G. If G has only one edge, the result is trivial since $\chi'(G) = 1$ and $\Delta(G) = 1$.

Assume that G has more than one edge, and that the theorem holds for all non-empty bipartite graphs having fewer edges than G. Since we know $\Delta(G) \leq \chi'(G)$ (Observation 2), it suffices to prove that G has a $\Delta(G)$-edge colouring. Let $\Delta(G) = k$.

Consider an arbitrary edge $e = uv$ in G. The graph $G - e$ is bipartite and has fewer edges than G. By the induction hypothesis, $G - e$ has a $\Delta(G - e)$-edge colouring. Since $\Delta(G - e) \leq \Delta(G) = k$, $G - e$ has a k-edge colouring. We will show how to extend this k-edge colouring of $G - e$ to a k-edge colouring of G by assigning a colour to e.

Since $d_G(u) \leq k$ and $d_G(v) \leq k$, in the k-edge colouring of $G - e$, there is at least one colour absent at vertex u and at least one colour absent at vertex v.

1. **Case 1: There is a colour c that is absent at both u and v.** In this case, we can assign colour c to edge $e = uv$. This results in a k-edge colouring of G.

2. **Case 2: There is no colour absent at both u and v.** This implies that for every colour c, if c is absent at u, it must be present at v, and vice versa. Let i be a colour absent at v (and thus present at u) and j be a colour absent at u (and thus present at v).

   Consider the subgraph $H(i, j)$ formed by all edges in $G - e$ coloured either i or j. Let K be the connected component of $H(i, j)$ containing u. We claim that v does not belong to K.

   Proof of claim: Assume, for contradiction, that v belongs to K. Then there exists a path P in K between u and v. Since u and v are adjacent in G (via edge e), they belong to different partitions in the bipartite graph G. Therefore, any path between u and v in G must have odd length. The path P is composed of alternately coloured edges i and j. Since colour i is present at u, the first edge of P incident to u must be coloured i. As P has an odd number of edges and alternates colours, its last edge incident to v must also be coloured i. This implies that colour i is present at v, which contradicts our initial assumption that i is absent from v. Thus, v does not belong to the Kempe chain K.

   Now, we apply the Kempe chain argument to the component K. We interchange the colours i and j on all edges in K.

   - At vertex u: Since $u \in K$ and the colour i was present at u, after the recolouring, colour j becomes present at u and colour i becomes absent at u.
   - At vertex v: Since $v \notin K$, the colours of edges incident to v are unaffected by this interchange. Therefore, i remains absent from v.

   After this recolouring, colour i is now absent at both u and v. We are now back to Case 1, where we can assign colour i to edge $e = uv$. This yields a k-edge colouring of G.

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Therefore, in all cases, G has a $\Delta(G)$-edge colouring. ■

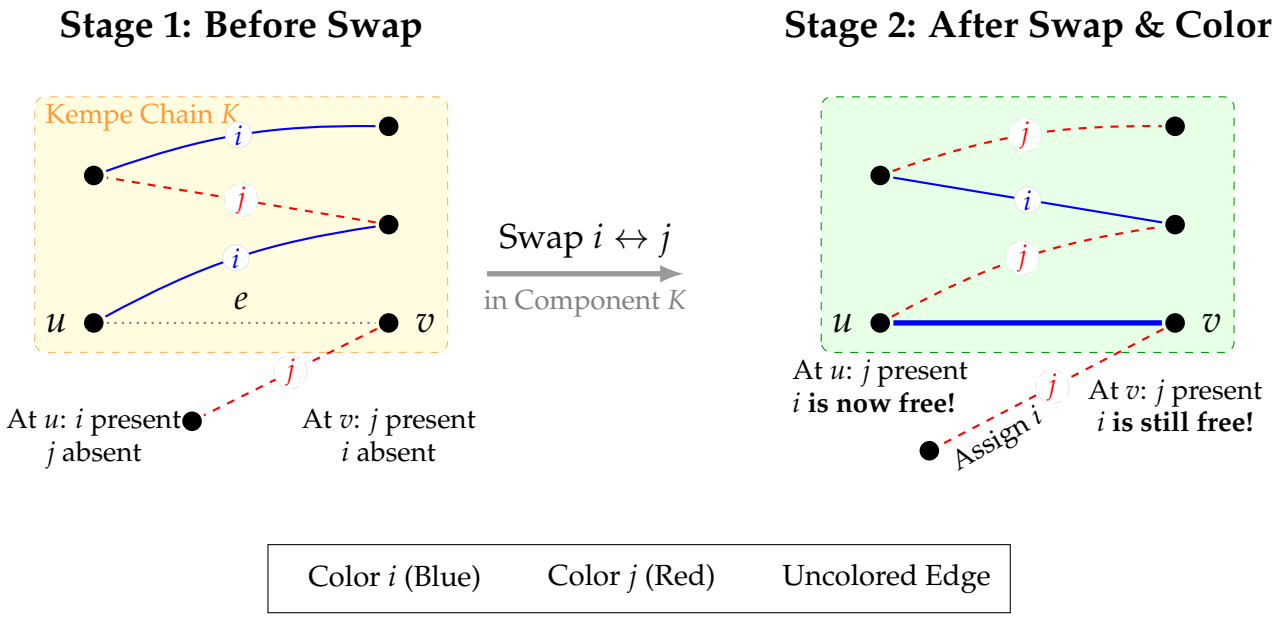


Figure 1.4: Visualization of the Kempe Chain swap. By drawing the highlight background first and using standard arrow syntax, this code is robust and compatible with standard LaTeX installations.

Theorem 1.2.2. *If $G = K_n$ is a complete graph with n vertices, $n \geq 2$, then*

$$\chi'(G) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Proof. (See Figure 1.5).

Since every vertex of K_n has degree $n-1$, we have $\Delta(K_n) = n-1$. Hence, $\chi'(K_n) \geq n-1$.

Case 1: n is odd. Suppose that K_n can be edge-coloured with $n-1$ colours. Each colour class is a matching, and since n is odd, a matching contains at most $(n-1)/2$ edges. Therefore, $n-1$ colours can colour at most

$$(n-1)\frac{n-1}{2} = \frac{(n-1)^2}{2}$$

edges.

However,

$$|E(K_n)| = \frac{n(n-1)}{2} > \frac{(n-1)^2}{2}.$$

This is a contradiction. Hence $\chi'(K_n) \geq n$.

To show that n colours are sufficient, label the vertices by $0, 1, \ldots, n-1$, with arithmetic modulo n. Colour the edge joining i and j with the colour

$$i + j \pmod{n}.$$

If two edges ij and ik incident with the same vertex i received the same colour, then $i + j \equiv i + k \pmod{n}$, and hence $j = k$. Thus adjacent edges receive different colours. Therefore K_n has a proper n-edge colouring.

Hence

$$\chi'(K_n) = n$$

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when n is odd.

Case 2: n is even. We already know that $\chi'(K_n) \geq n-1$. It remains to show that $n-1$ colours are sufficient.

Place $n-1$ vertices at the vertices of a regular $(n-1)$-gon and place the remaining vertex at the centre. Join the central vertex to one outer vertex, and pair the remaining $n-2$ outer vertices symmetrically. These edges form a perfect matching, so they may all be assigned the same colour.

Now rotate this arrangement successively through the $n-1$ positions of the outer polygon. Each rotation gives a new perfect matching, and the resulting $n-1$ perfect matchings together contain every edge of K_n exactly once. Assigning a different colour to each matching gives a proper edge colouring of K_n with $n-1$ colours.

Therefore

$$\chi'(K_n) = n-1$$

when n is even.

Thus,

$$\chi'(K_n) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

■

Theorem 1.2.3 (Vizing). *For any graph G,*

$$\Delta(G) \leq \chi'(G) \leq \Delta(G)+1.$$

Proof. Let G be a simple graph. We always have $\Delta(G) \leq \chi'(G)$. It remains to prove that $\chi'(G) \leq \Delta(G)+1$.

We use induction on the number of edges of G. Let $\Delta(G) = k$. If G has only one edge, then $k = 1 = \chi'(G)$. Therefore, assume that G has more than one edge and that the result is true for all graphs having fewer edges than G.

Let $e = v_1v_2$ be an edge of G. By the induction hypothesis, the subgraph $G-e$ has a $(k+1)$-edge colouring. Let the colours used be $1, 2, \ldots, k+1$.

Since $d(v_1) \leq k$ and $d(v_2) \leq k$, out of these $k+1$ colours, at least one colour is absent from v_1 and at least one colour is absent from v_2. If there is a common colour absent from both v_1 and v_2, then we use this colour to colour e and obtain a $(k+1)$-edge colouring of G. Therefore, in this case, $\chi'(G) \leq k+1$.

We now assume that there is a colour, say 1, absent from v_1 but present at v_2, and a colour, say 2, absent from v_2 but present at v_1. We start from v_1 and v_2 and construct a sequence of distinct vertices $v_1, v_2, \ldots, v_j$, where each v_i, for $i \geq 2$, is adjacent to v_1.

Let v_1v_3 be coloured 2. This vertex v_3 exists because colour 2 is present at v_1. We observe that not all the $k+1$ colours are present at v_3, and assume that colour 3 is absent from v_3. If colour 3 is present at v_1, choose the vertex v_4 so that v_1v_4 is coloured 3. Continuing in this way, we choose a new colour i absent from v_i but present at v_1, so that v_1v_{i+1} is the edge coloured i.

In this way, we obtain a sequence of vertices $v_1, v_2, \ldots, v_{j-1}, v_j$ such that

(a) v_i is adjacent to v_1 for each $i > 1$;

(b) colour i is absent from v_i for $i = 1, 2, \ldots, j-1$;

(c) the edge v_1v_{i+1} is coloured i for $i = 1, 2, \ldots, j-1$.

Since $d(v_1) \leq k$, condition (a) implies that such a sequence has at most $k+1$ terms; that is, $j \leq k+1$. Assume that $v_1, v_2, \ldots, v_j$ is a longest such sequence; that is, it is not possible to find a new colour j, absent from v_j, together with a new neighbour v_{j+1} of v_1 such that v_1v_{j+1} is coloured j.

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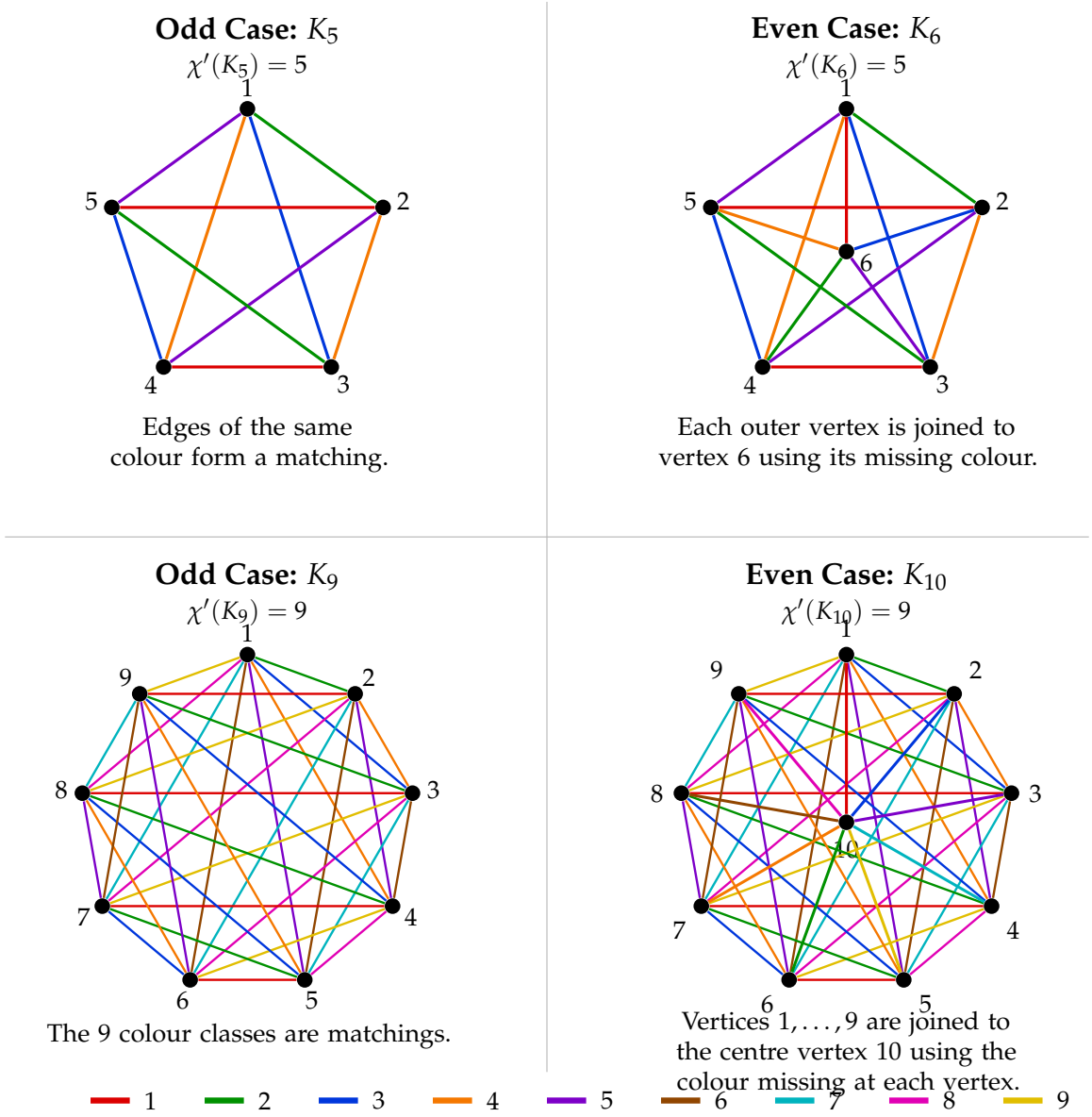


Figure 1.5: Proper edge colourings of K_5, K_6, K_9, and K_{10}.

We first assume that, for some colour j absent from v_j, there is no edge of colour j incident with v_1. We colour the edge $e = v_1v_2$ by 2, and then recolour the edges v_1v_i by i, for $i = 3, \ldots, j-1$. Since colour i is absent from v_i for each $i = 2, \ldots, j-1$, this gives a $(k+1)$-edge colouring of the subgraph $G - v_1v_j$. Since colour j is absent from both v_1 and v_j, we colour v_1v_j by j. This gives a $(k+1)$-edge colouring of G.

Now assume that whenever colour j is absent from v_j, it is present at v_1. If v_1v_{j+1} is an edge joining v_1 to a new neighbour v_{j+1} and is coloured j, then we can extend our sequence to $v_1, v_2, \ldots, v_j, v_{j+1}$, contradicting the assumption that $v_1, v_2, \ldots, v_j$ is a longest sequence. Thus one of the edges $v_1v_3, \ldots, v_1v_{j-1}$ must be coloured j; say v_1v_ℓ, where $3 \leq \ell \leq j-1$.

Now colour $e = v_1v_2$ by 2, and for $i = 3, \ldots, \ell-1$, recolour each edge v_1v_i by i, while leaving the colours of the edges v_1v_i, for $i = \ell+1, \ldots, j$, unchanged. Removing colour j from v_1v_ℓ, we obtain a $(k+1)$-edge colouring of the edge-deleted subgraph $G - v_1v_\ell$.

Let $H(1, j)$ represent the subgraph of G induced by the edges coloured 1 or j in this partial colouring of G. Since the degree of every vertex in $H(1, j)$ is at most 2, each component of $H(1, j)$ is either a path or a cycle. Since colour 1 is absent from v_1 and colour j is absent from both v_j and v_ℓ, it follows that

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these three vertices do not all belong to the same connected component of $H(1, j)$. Therefore, if K and L represent the corresponding Kempe chains containing v_j and v_ℓ, respectively, then either $v_1 \notin K$ or $v_1 \notin L$.

Suppose that $v_1 \notin L$. Interchanging colours 1 and j on L, the Kempe chain argument gives a $(k+1)$-edge colouring of $G - v_1v_\ell$ in which colour 1 is absent from both v_1 and v_ℓ. Colouring v_1v_ℓ by 1 gives a $(k+1)$-edge colouring of G.

Now suppose that $v_1 \notin K$. Colour the edge v_1v_ℓ by ℓ, recolour the edges v_1v_i by i, for $i = \ell, \ldots, j-1$, and remove colour $j-1$ from v_1v_j. Then, from the definition of the sequence $v_1, v_2, \ldots, v_j$, we obtain a $(k+1)$-edge colouring of $G - v_1v_j$ without affecting the two-coloured subgraph $H(1, j)$. Using the Kempe chain argument to interchange the colours on K, we obtain a $(k+1)$-edge colouring of $G - v_1v_j$ in which colour 1 is absent from both v_1 and v_j. Therefore, colouring v_1v_j by 1 gives a $(k+1)$-edge colouring of G.

Hence $\chi'(G) \leq k+1 = \Delta(G)+1$. Together with $\Delta(G) \leq \chi'(G)$, we obtain

$$\Delta(G) \leq \chi'(G) \leq \Delta(G)+1.$$

■

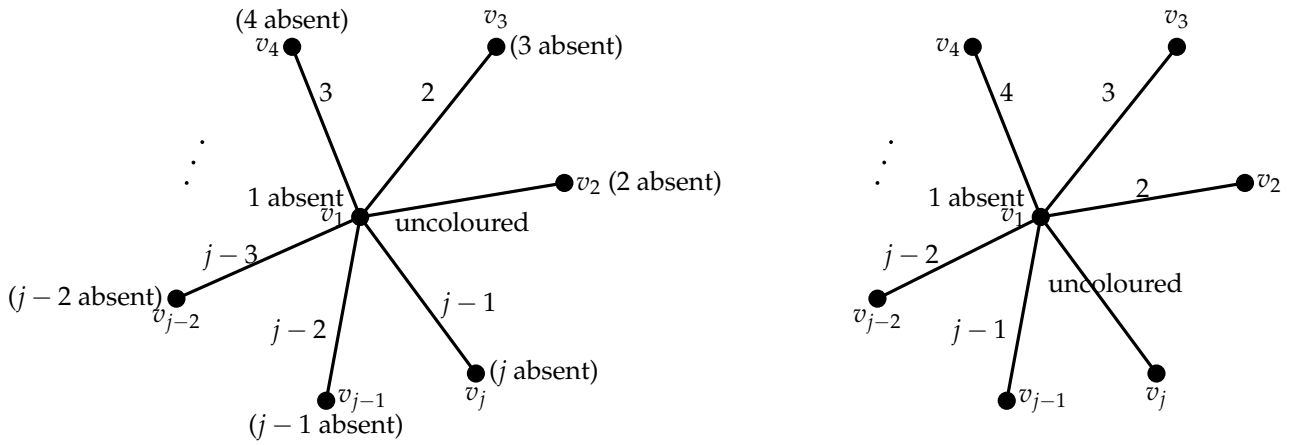


Figure 1.6: Illustration of Vizing's theorem.

Second Proof. The lower bound $\Delta(G) \leq \chi'(G)$ is straightforward. At any vertex v with degree $\Delta(G)$, there are $\Delta(G)$ edges incident with it. Since all these edges must receive distinct colours in a proper edge colouring, at least $\Delta(G)$ colours are necessary. Thus $\chi'(G) \geq \Delta(G)$.

The more challenging part is to prove the upper bound $\chi'(G) \leq \Delta(G)+1$. This part of the proof typically involves a constructive algorithm or an inductive argument. Here, we provide a high-level constructive proof sketch.

Let $\Delta = \Delta(G)$. We want to show that G can be $(\Delta+1)$-edge coloured. We proceed by induction on the number m of edges of G. The base case is trivial for $m = 0$ or $m = 1$.

Assume that any graph with fewer than m edges can be $(\Delta+1)$-edge coloured. Consider a graph G with m edges. Let $e = uv$ be an uncoloured edge of G. We try to colour e with a colour $c \in \{1, \ldots, \Delta+1\}$. If we can colour all the edges one by one, we obtain a $(\Delta+1)$-edge colouring.

Consider a partial edge colouring c' of G, where some edges may be uncoloured. For any vertex $x \in V(G)$, let $C(x)$ be the set of colours used on edges incident with x, and let

$$M(x) = \{1, \ldots, \Delta+1\} \setminus C(x)$$

be the set of colours missing at x. Since the degree of every vertex is at most Δ and we have $\Delta+1$ available colours, every vertex x has at least one missing colour; that is, $M(x) \neq \emptyset$.

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Now let $e_0 = uv$ be an uncoloured edge. We want to assign a colour to e_0. If there is a colour $c \in M(u) \cap M(v)$, we can assign colour c to e_0. If no such colour exists, then $M(u)$ and $M(v)$ are disjoint. Since $|M(u)| \geq 1$ and $|M(v)| \geq 1$, we have $|C(u)| \leq \Delta$ and $|C(v)| \leq \Delta$, while $|M(u)| \geq (\Delta+1) - \Delta = 1$ and $|M(v)| \geq (\Delta+1) - \Delta = 1$.

If $M(u) \cap M(v) = \emptyset$, we try to rearrange the colours to make a colour available for uv. Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there is an edge vw coloured a. Since $b \notin M(u)$, there is an edge ux coloured b. We consider an (a, b)-alternating path or cycle starting from v. Such a path has edges whose colours alternate between a and b.

The proof involves a technique called colour shifting or fan colouring. Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there exists a vertex w such that the edge vw is coloured a. Consider the path P starting at w whose edges alternate between colours a and b. If this path does not reach u, we can interchange colours a and b along P. This frees colour a at v and allows us to colour uv with a.

The essential point is that interchanging the two colours along an alternating path does not create a conflict. The argument considers the possible (a, b)-alternating paths and cycles arising from the vertices involved. Since $\Delta + 1$ colours are available while the maximum degree is only Δ, there is sufficient flexibility to rearrange the colours and make a colour available for the uncoloured edge.

By systematically applying this procedure, colouring one uncoloured edge at a time and using colour shifting whenever necessary, we obtain a proper $(\Delta + 1)$-edge colouring of G. Hence

$$\chi'(G) \leq \Delta(G) + 1.$$

Together with $\Delta(G) \leq \chi'(G)$, this proves the result. ■

Theorem 1.2.4. *Let G be a graph of order n and let $\overline{G}$ be the complement of G. Then*

(a) if n is even, then

$$n - 1 \leq \chi'(G) + \chi'(\overline{G}) \leq 2(n-1),$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n-1)^2;$$

(b) if n is odd, then

$$n \leq \chi'(G) + \chi'(\overline{G}) \leq 2n - 3,$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n-1)(n-2).$$

Further, the bounds are best possible for every positive integer $n \neq 2$.

Proof. Let G be a graph of order n and let $\overline{G}$ be its complement. Clearly,

$$\Delta(G) \geq n - 1 - \Delta(\overline{G}),$$

and hence

$$\Delta(G) + \Delta(\overline{G}) \geq n - 1.$$

Since $\chi'(G) \geq \Delta(G)$ and $\chi'(\overline{G}) \geq \Delta(\overline{G})$, it follows that

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1.$$

Part 1: Even n. Let n be even. Then

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1$$

and

$$\chi'(G)\chi'(\overline{G}) \geq 0.$$

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The lower bounds are attained when $G = K_n$ or $G = \overline{K_n}$.

For the upper bounds, both G and $\overline{G}$ are subgraphs of K_n. Since $\chi'(K_n) = n - 1$ for even n, we have

$$\chi'(G) \leq n - 1 \qquad \text{and} \qquad \chi'(\overline{G}) \leq n - 1.$$

Therefore,

$$\chi'(G) + \chi'(\overline{G}) \leq 2(n - 1)$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)^2.$$

These upper bounds are attained for suitable graphs, including the complete bipartite graph $K_{1,n-1}$ when $n \neq 2$.

Part 2: Odd n. Let n be odd. We claim that

$$\chi'(G) + \chi'(\overline{G}) \geq n.$$

For example, if $G = K_n$, then $\chi'(K_n) = n$, while $\overline{G}$ is the empty graph and has edge chromatic number 0. Thus

$$\chi'(G) + \chi'(\overline{G}) = n.$$

Similarly, the same value is obtained when G is the empty graph.

Clearly,

$$\chi'(G)\chi'(\overline{G}) \geq 0,$$

and this lower bound is also attained for complete or empty graphs.

We now prove the upper bounds

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

For $n = 1$ and $n = 3$, the inequalities are immediate. Thus assume that $n \geq 5$. If

$$\Delta(G) + \Delta(\overline{G}) \leq 2n - 5,$$

then by Vizing's theorem,

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2 \leq 2n - 3.$$

Otherwise, we consider the following cases.

(i) Suppose $\Delta(G) = n - 1$ and $\Delta(\overline{G}) = n - 2$. Then G has a pendant vertex v. Also, $\Delta(G - v) \geq n - 2$ and hence $\chi'(G - v) \geq n - 2$. Since $\overline{G - v}$ is a subgraph of K_{n-1},

$$\chi'(\overline{G - v}) \leq \chi'(K_{n-1}) = n - 2.$$

As $\overline{G}$ is the disjoint union of an isolated vertex and a subgraph of K_{n-1}, we obtain the required bound.

(ii) Suppose $\Delta(G) = n - 2$ and $\Delta(\overline{G}) = n - 2$. Again, G has a pendant vertex v, and as before,

$$\chi'(G - v) \leq n - 2 \qquad \text{and} \qquad \chi'(\overline{G}) \leq n - 2.$$

Similarly, $\chi'(\overline{G}) = n - 2$. Thus

$$\chi'(G) + \chi'(\overline{G}) = 2n - 4 < 2n - 3.$$

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(iii) Suppose $\Delta(G) = n-1$ and $\Delta(\overline{G}) = n-3$. In this case, G has a vertex v of degree 2, and so $\chi'(G-v) = n-2$.

Let vu and vw be the edges incident with v in G, where $d(u) = n-1$. In an $(n-2)$-edge colouring of $G-v$, change the colour i of an edge uu' to a new colour $n-1$. Now colour vu by i and vw by $n-1$. This gives an $(n-1)$-edge colouring of G. By Vizing's theorem,

$$\chi'(\overline{G}) \leq n-2.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq 2n-3.$$

Since

$$\chi'(G) + \chi'(\overline{G}) \leq 2n-3,$$

the possible largest product under the stated bounds is

$$\chi'(G)\chi'(\overline{G}) \leq (n-1)(n-2).$$

The upper bounds are attained, for example, by suitable complete bipartite graphs such as $K_{1,n-1}$.

Thus all the stated inequalities follow. ■

A more complete argument for the upper bound when n is odd involves considering the possible maximum degrees of G and $\overline{G}$. By Vizing's theorem,

$$\chi'(G) \leq \Delta(G)+1 \qquad \text{and} \qquad \chi'(\overline{G}) \leq \Delta(\overline{G})+1.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2.$$

For every graph G of order n,

$$\Delta(G) + \Delta(\overline{G}) \geq n-1.$$

Also,

$$\Delta(G) + \Delta(\overline{G}) \leq 2n-2.$$

The sharper bound $2n-3$ for odd n requires a more detailed examination of the possible degree configurations in G and $\overline{G}$.

Similarly, the maximum value

$$\chi'(G)\chi'(\overline{G}) = (n-1)(n-2)$$

is obtained when the two edge chromatic numbers take the values $n-1$ and $n-2$, respectively. The determination of such sharp bounds belongs to the general study of graph invariants of complementary graphs.

1.3 Region Colouring (Map Colouring)

A region colouring of a planar graph is a labelling of its regions $f : R(G) \to \{1, 2, \ldots\}$; the labels called *colours*, such that no two adjacent regions get the same colour. A k-region colouring of a planar graph G consists of k different colours and G is then called k-region colourable. From the definition, it follows that the k-region colouring of a planar graph G partitions the region set R into k independent sets $R_1, R_2, \ldots, R_k$, so that $R = R_1 \cup R_2 \cup \ldots \cup R_k$. The independent sets are called the *colour classes*, and the function $f : R(G) \to \{1, 2, \ldots, k\}$ such that $f(r) = i$, for each $r \in R_i, 1 \leq i \leq k$, is called the *colour function*. The minimum number k for which there is a k-region colouring of the planar graph G is called the *region-chromatic number* of G, and is denoted by $\chi''(G)$. The colouring of regions is also called *map colouring*, because of the fact that in an atlas different countries are coloured such that countries with common boundaries are shown in different colours.

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Theorem 1.3.1. *1. A planar graph G is k-region colourable if and only if its dual G^* is k-vertex colourable.*

2. If G is a plane connected graph without loops, then G has a k-vertex colouring if and only if its dual G^* has a k-region colouring.

Proof. a. Let the regions and edges of G be respectively denoted by $r_1, \ldots, r_t$ and $e_1, \ldots, e_m$. Let the vertices of G^* be $r_1^*, \ldots, r_t^*$ and edges be $e_1^*, \ldots, e_m^*$. Then the vertices and edges of G^* are in one-to-one correspondence with the regions and edges of G, and two vertices r^* and s^* in G^* are joined by an edge e^* if and only if the corresponding regions r and s in G have the corresponding edge e as a common edge on their boundary.

Let G be k-region colourable. We colour the vertices in G^* such that each vertex r^* in G^* gets the same colour as assigned to the region r in G. Since the vertices r^* and s^* are only adjacent in G^* if the corresponding regions r and s are adjacent in G, G^* is k-vertex colourable.

Conversely, let G^* be k-vertex colourable. Now colour the regions of G such that the region r in G gets the same colour as the vertex r^* in G^*. This gives a k-region colouring of G, since the regions r and s are adjacent in G only if the corresponding vertices r^* and s^* are adjacent in G^*.

b. Since G has no loops its dual G^* has no bridges, and therefore G^* is planar. Thus by (a), G^* is k-region colourable if and only if the double dual G^{**} is k-vertex colourable. Since G is connected, G is isomorphic to G^{**} and hence the result follows.

■

Theorem 1.3.2. *A planar graph is bicolourable if and only if it is an Euler graph.*

Proof. Suppose G is bicolourable. Around each vertex, the colours of the incident faces alternate. Hence every vertex has even degree, so G is an Euler graph.

Conversely, suppose G is an Euler graph. Then every vertex of G has even degree, so every face of the dual graph G^* has even length. Hence G^* is bipartite. Therefore its vertices can be coloured with two colours, which gives a two-colouring of the faces of G.

Thus G is bicolourable if and only if it is an Euler graph. ■

Second Proof. Suppose G is bicolourable, that is, its dual G^* is 2-vertex-colourable. So G^* is bipartite, so every face of G^* has even length. Thus, every vertex of the dual of G^* has even degree. Since the bidual G^{**} is isomorphic to G, every vertex of G has even degree. Hence G is an Euler graph.

Conversely, suppose G is an Euler graph. Since $G^{**} \cong G$, every vertex of G^{**} has even degree. Thus every face of G^* has even length, and hence G^* is bipartite. Therefore the faces of G can be coloured with two colours, so G is bicolourable. ■

Theorem 1.3.3. *Every planar graph is 6-colourable.*

Proof. We use induction on the number of vertices. Every planar graph has a vertex v of degree at most 5. By induction, $G - v$ is 6-colourable. Since v has at most 5 neighbours, at most 5 colours appear on them, so one of the 6 colours is available for v. Hence G is 6-colourable. ■

Theorem 1.3.4 (Heawood)**.** *Every planar graph is 5-colourable.*

Proof. (See Figure 1.7). Let G be a planar graph with n vertices. We use induction on n, the order of G. The result is obvious for $n \leq 5$. So, let $n \geq 6$. Assume the result is true for all planar graphs with fewer than n vertices.

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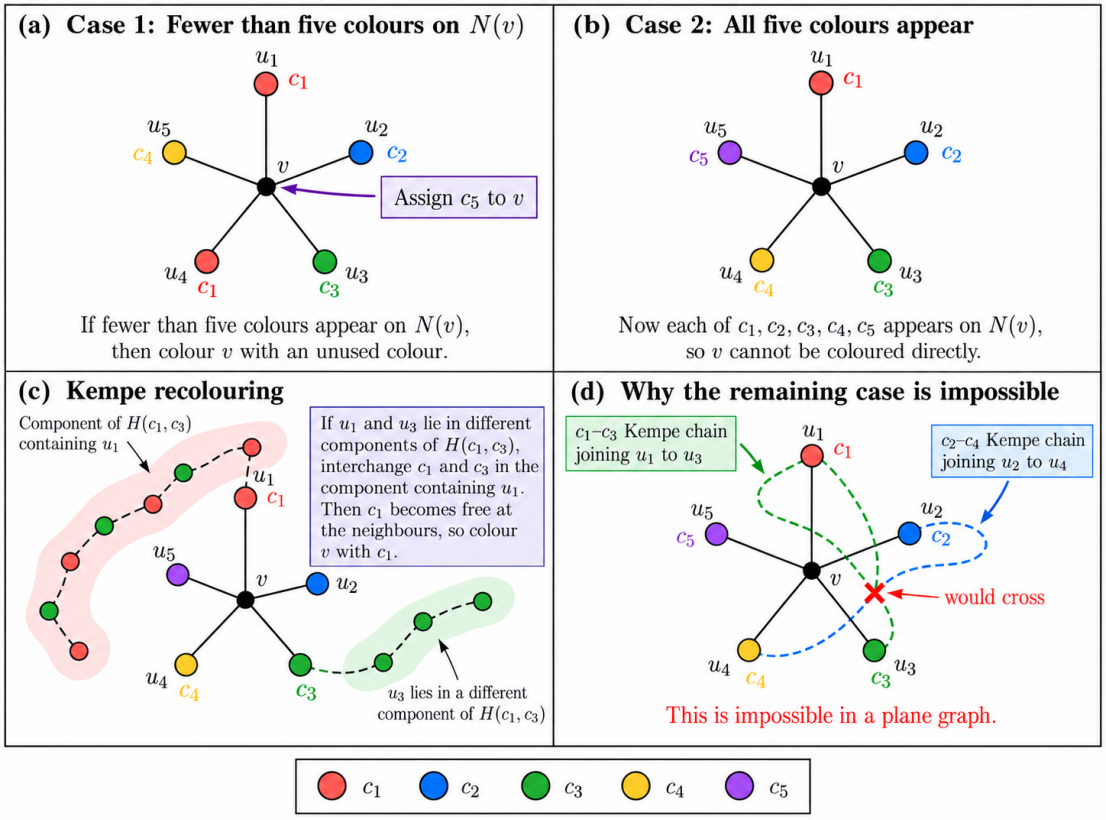


Figure 1.7: Heawood 5-color Theorem

Let G' be the graph obtained from G by deleting the vertex v and removing all the edges incident with v. The graph G' with order $n-1$ is clearly planar and by induction hypothesis is 5-colourable. Let the colours used to colour G' be c_1, c_2, c_3, c_4, c_5. We know for a planar graph with $n \geq 6$ vertices, there exists a vertex, say v, such that $d(v) \leq 5$. Thus v has utmost five neighbours in G and all of these neighbours are the already coloured vertices in G'. If in G' less than five colours are used to colour these neighbours, then the 5-colouring of G is obtained by using the colouring for G' on all vertices, and by colouring v with the colour not used to colour the neighbours of v.

Let all the five colours be used in G', to colour the neighbours of v. This implies that there are exactly five neighbours of v, say u_1, u_2, u_3, u_4 and u_5. Assume without loss of generality that u_i is coloured with c_i, for each i, $1 \leq i \leq 5$.

Consider all the vertices of G' that are coloured with colour c_1 and c_3. If u_1 and u_3 are in different components of the Kempe subgraph $H(c_1, c_3)$ induced by those vertices coloured c_1 and c_3, then using the Kempe chain argument to interchange the colours c_1 and c_3 in the component containing u_1 leaves c_1 unused on the set $\{u_1, u_2, u_3, u_4, u_5\}$. We colour v by c_1 to get a 5-colouring of G. Finally, if u_1 and u_3 are in the same component of $H(c_1, c_3)$, and u_2 and u_4 are in the same component of $H(c_2, c_4)$, then any component of $H(c_2, c_4)$ can not contain both u_2 and u_4 since otherwise the $c_1 - c_3$ Kempe chain from u_1 to u_3 crosses the $c_2 - c_4$ Kempe chain from u_2 to u_4. This is impossible as the graph is a plane graph. Thus, we always have a pair of colors, say c_i and c_j, such that the induced graph $H(c_i, c_j)$ has a component K which does not contain both u_i and u_j. Interchanging the colors on any such component releases a color for the last uncolored vertex. ■

Theorem 1.3.5. *Every planar graph is four colourable if and only if every cubic bridgeless planar graph is 4-colourable.*

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Proof. We observe that every planar graph is four colourable if and only if every bridgeless planar graph (without cut edges) is four colourable. This is because if G has a bridge e and G' is obtained from G by contracting e, then $\chi''(G') = \chi''(G)$ (as the elementary contraction of identifying the end vertices of a bridge affects neither the number of regions in the planar graph nor the adjacency of any of the regions). We now prove that every bridgeless planar graph is 4-colourable if and only if every cubic bridgeless planar graph is 4-colourable. If every bridgeless planar graph is 4-colourable, then evidently every cubic bridgeless planar graph is 4-colourable. Conversely, let all cubic bridgeless planar graphs be 4-colourable. Let G be a bridgeless planar graph. We obtain G' from G by performing the following operations. In case G has a vertex v of degree two, let xv and yv be the edges incident with v. Subdivide xv at u and yv at w. Take two new vertices a and b and add the edges au, aw, bu, bw and ab. Now remove v (and uv and uw). In doing so, we have replaced v by a $K_4 - e$ and we see that each new vertex has degree three.

If G has a vertex v so that $d(v) \geq 4$, let $vx_1, vx_2, \ldots, vx_d$ be the edges incident with v. Subdivide each vx_i producing a new vertex v_i, for $1 \leq i \leq d$. We then remove v and add the new edges $v_1v_2, v_2v_3, \ldots, v_{d-1}v_d, v_dv_1$. Here we have replaced v by C_d and again the degree of each new vertex is three.

In both cases the graph G' formed is a bridgeless cubic graph. If these $K_4 - e$ and C_d introduced are contracted, we get the original graph G. Hence G is 4-colourable, since we have assumed that G' is 4-colourable. ■

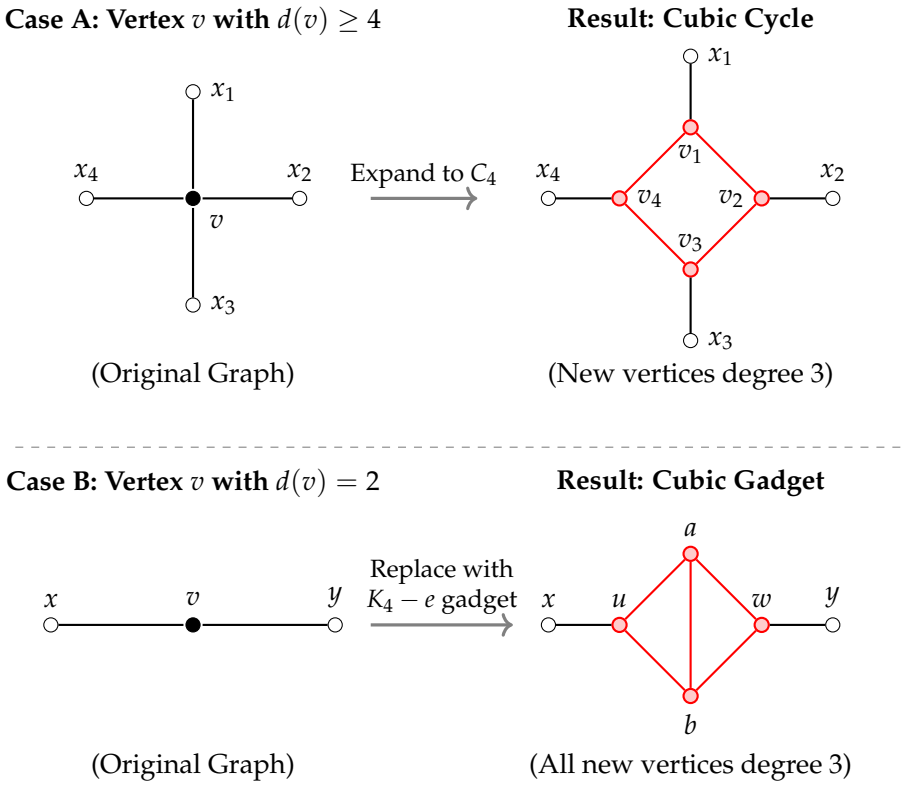


Figure 1.8: Visualisation of the operations described in Theorem 1.3.5. Top: Expanding a vertex of degree 4 into a cycle. Bottom: Replacing a vertex of degree 2 with a specialized gadget.

Theorem 1.3.6 (Tait's Theorem). *A cubic bridgeless, planar graph G is 4 region colorable if and only if, G is $3-$edge colorable.*

Proof. **Suppose G is 4-region colorable.** Let the 4-region coloring of G be given, where the colors are

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the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$. We now define an edge coloring of G. We define the color of an edge as the sum of colors of regions adjacent to it, noting that each edge lies on the boundary of at most two regions (because the given graph is cubic and bridgeless). Thus, we are done.

Conversely: [Refer to Figure above]. Suppose that G is 3-edge colorable. Let a 3-edge coloring of G be given using the colors $c_1 = (0,1), c_2 = (1,0), c_3 = (1,1)$. This produces a partition of $E(G)$ into three perfect matchings E_1, E_2, E_3 where E_i is the set of edges colored c_i. Let G_1 be a spanning subgraph of G with edge set $E(G_1) = E_1 \cup E_2$ and G_2 be the spanning subgraph of G with edge set $E(G_2) = E_2 \cup E_3$. Then, both G_1 and G_2 are 2-regular spanning subgraphs of G so each of G_1 and G_2 is the disjoint union of even cycles. For $i = 1, 2$ each region of G_i is the union of regions of G. For each cycle C in the graph G_i, every region of G either lies interior or exterior to C. Furthermore, each edge of G belongs to a cycle C' of G_i and is on the boundary of two distinct regions of G_i, one of which lies interior to C' and the other exterior to C'.

We now define a 4-region coloring of G. We assign to region R of G the color (a_1, a_2) with $a_i \in \{0, 1\}$ for $i = 1, 2$, where $a_i = 0$ if R lies interior to even number of cycles in G_i and $a_i = 1$ if R lies interior to odd number of cycles in G_i.

Next, we shall show that this 4-region coloring of G is proper. Let R_1 and R_2 be two adjacent regions with common edge e. Then clearly either e is in G_1 or in G_2 or in both. Thus, e lies on a cycle in G_1 or a cycle in G_2 or both. Let C be a cycle in G_1 containing the edge e. Exactly, one of R_1 and R_2 lies interior to C, while for every other cycle C', either both lie interior to C' or both exterior to C. Hence, the first coordinate of the colours of R_1 and R_2 are different. Therefore, the colors of every two adjacent regions of G differ in the first coordinate or the second coordinate or both. Hence, the 4-region coloring of G is proper. ■

Theorem 1.3.7 (Tait's Theorem). *A cubic bridgeless planar graph G is 4-region colorable if and only if G is 3-edge colorable.*

Proof. We prove the result in both directions.

Suppose first that G is 4-region colorable.

Let the four region colors be identified with the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$, namely

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1).$$

We define a color for each edge of G as follows. If an edge e separates two regions having colors a and b, assign to e the color

$$a + b,$$

where the addition is taken in $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Since the region coloring is proper, the two regions adjacent to an edge have different colors. Therefore, $a \neq b$, and hence

$$a + b \neq (0,0).$$

Thus every edge receives one of the three nonzero elements

$$(0,1), \qquad (1,0), \qquad (1,1).$$

We now show that the three edges incident with any vertex receive three different colors.

Let v be a vertex of G. Since G is cubic, exactly three edges are incident with v. Let the three regions occurring around v have colors a, b, c. Because the region coloring is proper, these three colors are pairwise distinct.

The three incident edges receive the colors

$$a + b, \qquad b + c, \qquad c + a.$$

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These three colors are nonzero and pairwise distinct. For example, if

$$a + b = b + c,$$

then, since every element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is its own additive inverse, cancellation gives $a = c$, which is impossible. The same argument applies to the other pairs.

Hence the three edges incident with every vertex receive three distinct colors. Therefore, G is 3-edge colorable.

Conversely, suppose that G is 3-edge colorable.

Let the three edge colors be

$$c_1 = (0,1), \qquad c_2 = (1,0), \qquad c_3 = (1,1).$$

Let

$$E_i = \{e \in E(G) : e \text{ has color } c_i\}, \qquad i = 1,2,3.$$

Since G is cubic and the edge coloring is proper, exactly one edge of each color is incident with every vertex. Hence each set E_i is a perfect matching.

Now define two spanning subgraphs G_1 and G_2 of G by

$$E(G_1) = E_1 \cup E_2$$

and

$$E(G_2) = E_2 \cup E_3.$$

At each vertex, exactly two edges belong to G_1, one from E_1 and one from E_2. Therefore, every vertex has degree 2 in G_1. Similarly, every vertex has degree 2 in G_2.

Thus both G_1 and G_2 are 2-regular spanning subgraphs. Consequently, each of them is a disjoint union of cycles.

Moreover, the cycles are even. Indeed, in a cycle of G_1, the edge colors must alternate between c_1 and c_2, so the cycle has even length. Similarly, the colors on every cycle of G_2 alternate between c_2 and c_3, and hence those cycles are also even.

Because G is planar, every cycle divides the plane into an interior region and an exterior region.

We now use the cycles of G_1 and G_2 to construct a 4-region coloring of G.

Let R be any region of G. Assign to R the ordered pair

$$(a_1, a_2),$$

where $a_1, a_2 \in \{0,1\}$ are defined as follows:

$$a_1 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_1, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_1, \end{cases}$$

and

$$a_2 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_2, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_2. \end{cases}$$

Thus every region receives one of the four possible colors

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1).$$

It remains to prove that this region coloring is proper.

Let R_1 and R_2 be two adjacent regions of G, and let e be their common boundary edge.

Every edge of G belongs to at least one of the subgraphs G_1 and G_2. More precisely,

$$e \in E_1 \Rightarrow e \in G_1,$$

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$$e \in E_2 \Rightarrow e \in G_1 \cap G_2,$$

and

$$e \in E_3 \Rightarrow e \in G_2.$$

Suppose first that $e \in G_1$. Since G_1 is a disjoint union of cycles, the edge e lies on exactly one cycle C of G_1.

The two regions R_1 and R_2 lie on opposite sides of the edge e. Hence one of them lies inside the cycle C and the other lies outside C.

For every other cycle C' of G_1, the regions R_1 and R_2 lie on the same side of C'. Therefore, when we pass from R_1 to R_2 across the edge e, the number of cycles of G_1 containing the region changes by exactly one.

Hence the parity changes. Therefore, the first coordinates of the colors of R_1 and R_2 are different:

$$a_1(R_1) \neq a_1(R_2).$$

Similarly, if $e \in G_2$, then e lies on a cycle of G_2, and the same argument shows that

$$a_2(R_1) \neq a_2(R_2).$$

Thus we have three cases.

If $e \in E_1$, then $e \in G_1$, so the first coordinates of the colors of R_1 and R_2 are different.

If $e \in E_3$, then $e \in G_2$, so the second coordinates are different.

If $e \in E_2$, then e belongs to both G_1 and G_2, so both coordinates are different.

Therefore, in every case, two adjacent regions receive different colors. Hence the constructed 4-region coloring is proper.

Thus G is 4-region colorable.

Therefore, a cubic bridgeless planar graph G is 4-region colorable if and only if it is 3-edge colorable. ■

1.4 Heawood Map-Colouring Theorem

Let S_p denote the orientable surface of genus p. Topologically, S_p may be regarded as a sphere with p handles attached. The chromatic number of S_p, denoted by $\chi(S_p)$, is defined to be the maximum chromatic number among all graphs that can be embedded on S_p. In particular, S_0 is the sphere, and the Four Colour Theorem asserts that $\chi(S_0) = 4$.

Definition 1.4.1 (Genus of a Graph)**.** The *genus* of a graph G, denoted by $p(G)$, is the minimum non-negative integer p such that G can be embedded on the orientable surface S_p.

Thus $p(G)$ is the minimum number of handles that must be added to a sphere in order to obtain a surface on which G can be embedded. In particular, $p(G) = 0$ if and only if G is planar.

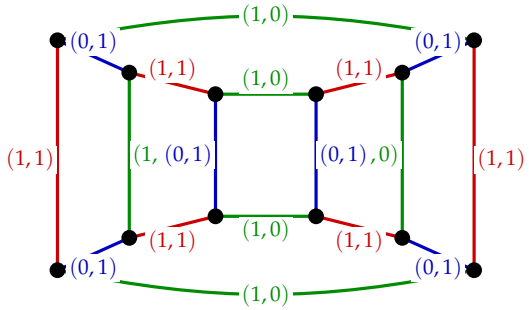
If a graph is embedded on S_p with n vertices, m edges, and f faces, then Euler's formula takes the form

$$n - m + f = 2 - 2p.$$

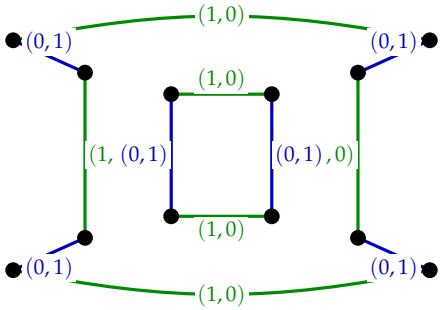
Theorem 1.4.2 (Genus of the Complete Graph)**.** *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

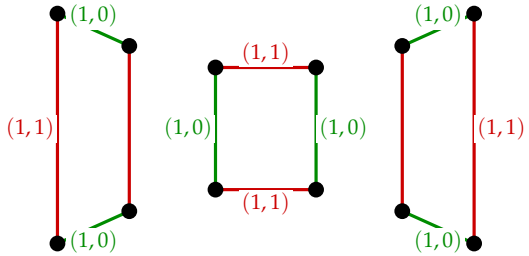
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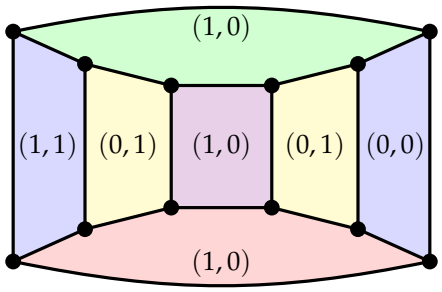
(a) 3-edge coloring of G



(b) G_1: edges colored $(0,1)$ and $(1,0)$



(c) G_2: edges colored $(1,0)$ and $(1,1)$



(d) 4-region coloring of G

Figure 1.9: Tait's Theorem: $\chi'(G) = 3 \Longrightarrow G$ is 4-region colorable.

Figure 1.10: Tait's Theorem

Proof. Let K_n be embedded on an orientable surface S_p of genus p. Let m and f denote the numbers of edges and faces, respectively. Since K_n has n vertices, we have $m = n(n-1)/2$.

Every face boundary contains at least three edges, and each edge is incident with two faces. Hence $3f \leq 2m$, so that $f \leq 2m/3$.

By Euler's formula, $n - m + f = 2 - 2p$. Therefore

$$2 - 2p = n - m + f \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Using $m = n(n-1)/2$, we get

$$2 - 2p \leq n - \frac{n(n-1)}{6}.$$

On simplification, this gives $12p \geq (n-3)(n-4)$. Hence $p \geq (n-3)(n-4)/12$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

For the reverse inequality, Ringel and Youngs proved that K_n can be embedded on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining the two inequalities, we obtain

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

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Detailed proof of the Ringel-Youngs Theorem is as follows.

Theorem 1.4.3 (Ringel–Youngs). *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Proof. Let K_n be embedded on an orientable surface S_p of genus p. Since K_n has n vertices and $m = n(n-1)/2$ edges, let f denote the number of faces of the embedding.

Every face boundary contains at least three edges, and every edge is incident with two faces. Hence $3f \leq 2m$. By Euler's formula, $n - m + f = 2 - 2p$, and therefore

$$2 - 2p \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Substituting $m = n(n-1)/2$ and simplifying, we obtain $12p \geq (n-3)(n-4)$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

It remains to prove that this lower bound is attainable. This is the main part of the Ringel–Youngs theorem.

For an embedding of a graph, define its *face excess* by

$$\varepsilon = \sum_F (|F| - 3),$$

where $|F|$ denotes the length of the boundary of the face F. Since the sum of the lengths of all face boundaries is $2m$, we have $3f + \varepsilon = 2m$. Consequently, $f = (2m - \varepsilon)/3$. Substitution in Euler's formula gives

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12}.$$

Thus, to obtain an embedding of minimum genus, it is sufficient to construct an embedding of K_n for which

$$\varepsilon = \frac{1}{2}\left(12\left\lceil \frac{(n-3)(n-4)}{12} \right\rceil - (n-3)(n-4)\right).$$

The value of ε depends only on the residue class of n modulo 12. The required values are

$n \pmod{12}$	0	1	2	3	4	5	6	7	8	9	10	11
ε	0	3	5	0	0	5	3	0	2	3	3	2.

Ringel and Youngs constructed embeddings having exactly these face excesses by means of *current graphs*. We describe the construction.

A current graph is an oriented graph whose directed edges are labelled by elements of a cyclic group $\mathbb{Z}_q$. If an edge receives the current a in one direction, it receives the current $-a$ in the opposite direction. At each ordinary vertex the currents are chosen to satisfy Kirchhoff's law: the sum of the currents entering the vertex is 0 in $\mathbb{Z}_q$.

The boundary of a face of the current graph is read successively, and the sequence of currents so obtained is called its *log*. The current graphs used by Ringel and Youngs are chosen so that every nonzero element of the current group occurs in the log in the required way. From such a log a rotation system is obtained as follows.

If

$$(a_1, a_2, \ldots, a_r)$$

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is the log at the vertex 0, then the rotation at a vertex $i \in \mathbb{Z}_q$ is obtained by adding i to every numerical entry:

$$i: \quad (i + a_1, i + a_2, \ldots, i + a_r),$$

where the additions are taken modulo q. Thus the single current graph generates simultaneously the rotations at all the numbered vertices.

Because every nonzero group element occurs in the log, every difference $j - i$, with $i \neq j$, occurs. Hence the derived graph contains every edge joining two numbered vertices. In other words, the numbered vertices induce a complete graph.

At an ordinary vertex of degree three, Kirchhoff's law implies that the corresponding faces in the derived embedding are triangles. Thus, when there are no exceptional vertices, the derived rotation system gives a triangular embedding of a complete graph.

This immediately settles the residue classes $n \equiv 0, 3, 4, 7 \pmod{12}$. In these cases $(n-3)(n-4)$ is divisible by 12, so that $\varepsilon = 0$, and Ringel–Youngs current graphs give triangular embeddings of K_n. For such an embedding, $3f = 2m$, and hence

$$p = \frac{(n-3)(n-4)}{12}.$$

For the remaining residue classes a triangular embedding of K_n is arithmetically impossible, because $(n-3)(n-4)$ is not divisible by 12. Ringel and Youngs therefore introduced certain exceptional vertices, called *vortices*, into the current graph. A vortex does not satisfy the ordinary Kirchhoff condition and produces one or more large faces in the derived embedding. New vertices are inserted into these faces and joined to the vertices occurring on their boundaries.

The resulting derived graph is either K_n itself or a graph of the form $K_n - L$, where L is a small graph consisting of the few edges between the newly introduced vertices that have not yet been obtained. The missing edges of L are then inserted by local modifications of the embedding. These modifications consist of diagonal changes in quadrilateral faces and, when necessary, the addition of handles. Each added handle is used to insert the required missing adjacencies without creating crossings.

The constructions are arranged so that the resulting embeddings have precisely the following face excesses:

$n \pmod{12}$	ε
$0, 3, 4, 7$	0
$8, 11$	2
$1, 6, 9, 10$	3
$2, 5$	5.

Consequently, in every residue class the constructed embedding satisfies

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12} = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Thus, for every $n \geq 3$, Ringel and Youngs obtained an embedding of K_n on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining this with the previously established lower bound gives

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

The following inequality is due to Heawood [113].

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Theorem 1.4.4. *The chromatic number of the orientable surface of positive genus p has the upper bound*

$$\chi(S_p) \leq \frac{7+\sqrt{1+48p}}{2}, \quad p > 0.$$

Proof. Let G be an (n, m) graph embedded on S_p. Since any graph can be augmented to a triangulation of the same genus by adding edges without reducing χ, we assume G to be a triangulation. Let d' be the average degree of the vertices of G. Then n, m and f (the number of regions) are related by the equations

$$d'n = 2m = 3f.$$

Solving for m and f in terms of n and using Euler's equation $n - m + f = 2 - 2p$, we get

$$d' = \frac{12(p-1)}{n} + 6. \tag{1.2}$$

As $d' \leq n - 1$, this gives $n - 1 \geq \frac{12(p-1)}{n} + 6$. Solving for n and taking the positive square root, we obtain $n \geq \frac{7+\sqrt{1+48p}}{2}$.

Let $H(p) = \frac{7+\sqrt{1+48p}}{2}$. Then we show that $H(p)$ colours are sufficient to colour the vertices of G. If $n = H(p)$, obviously we have sufficient colours. In case $n > H(p)$, we have $\frac{1}{n} < \frac{1}{H(p)}$ then equation (1.2) gives obtain

$$d' < \frac{12(p-1)}{H(p)} + 6 \tag{1.3}$$

Now,

$$\begin{aligned} 2H(p) &= 7 + \sqrt{1+48p} \\ \text{or } (2H(p)-7)^2 &= 1 + 48p \\ \text{or } 4H^2(p) - 28H(p) + 49 &= 1 + 48p \\ \text{or } H(p)\,[H(p) - 7] &= 12(p-1) \\ \text{or } H(p) - 7 &= \frac{12(p-1)}{H(p)} \\ \text{i.e. } H(p) - 1 &= \frac{12(p-1)}{H(p)} + 6 \end{aligned}$$

Hence, equation 1.3 gives, $d' < H(p) - 1$. Therefore, when $n > H(p)$, there is a vertex v of degree at most $H(p) - 2$. Identify v and any adjacent vertex by an elementary contraction to obtain a new graph G'. If $n' = n - 1 = H(p)$, then G' can be coloured in $H(p)$ colours. If $n' > H(p)$, repeat the argument and evidently we get an $H(p)$ colourable graph. It is then easy to see that the colouring of this graph induces a colouring of the preceding one in $H(p)$ colours, and so forth, so that G itself is $H(p)$-colourable. Hence the result follows.

■

Theorem 1.4.5 (Heawood Map-Colouring Theorem)**.** *For every positive integer p, the chromatic number of the orientable surface S_p of genus p is*

$$\chi(S_p) = \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor.$$

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Proof. By the upper-bound theorem, every graph G embeddable on S_p satisfies $\chi(G) \leq \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor$. Hence $\chi(S_p) \leq \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor$.

It remains to establish the reverse inequality. Recall that the orientable genus of the complete graph K_n is

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Let n be the largest integer such that K_n is embeddable on S_p. Then $p(K_n) \leq p$, whereas $p(K_{n+1}) > p$. Since p is an integer, the genus formula gives

$$\frac{(n-3)(n-4)}{12} \leq p < \frac{(n-2)(n-3)}{12}.$$

Thus $n^2 - 7n + 12 \leq 12p < n^2 - 5n + 6$. The first inequality yields $n \leq \frac{7+\sqrt{1+48p}}{2}$, while the second yields $n > \frac{5+\sqrt{1+48p}}{2}$. Therefore

$$\frac{5+\sqrt{1+48p}}{2} < n \leq \frac{7+\sqrt{1+48p}}{2}.$$

Since the two bounds differ by 1 and n is an integer,

$$n = \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor.$$

Now K_n is embeddable on S_p and $\chi(K_n) = n$. Hence $\chi(S_p) \geq n = \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor$.

Combining the two inequalities, we obtain

$$\chi(S_p) = \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor.$$

■

1.5 Uniquely Colorable Graphs

Definition 1.5.1 (Uniquely Colorable Graph)**.** A graph $G = (V, E)$ is said to be *uniquely k-colorable* if there is a unique partition of V into k independent sets. Equivalently, every proper k-coloring of G determines the same partition of V into color classes, apart from a permutation of the colors.

Theorem 1.5.2. *If G is uniquely k-colorable, then every vertex of G has degree at least $k-1$.*

Proof. Let c be the unique k-coloring of G, and suppose that some vertex v has degree less than $k-1$. Then at most $k-2$ colors occur among the neighbours of v. Hence, among the k available colors, there is a color different from $c(v)$ which is not used on any neighbour of v. Recoloring v with this color gives another proper k-coloring of G, and the resulting partition into color classes is different from the original one. This contradicts the uniqueness of the k-coloring. Therefore, $d(v) \geq k-1$ for every $v \in V(G)$. ■

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Theorem 1.5.3. *In a uniquely k-colorable graph G, the subgraph induced by the union of any two color classes is connected.*

Proof. Let C_i and C_j be two distinct color classes in the unique k-coloring of G, and let $H = G[C_i \cup C_j]$. Suppose that H is disconnected, and let H_1 be one of its components. Interchange the colors i and j on all vertices of H_1, leaving the colors of all other vertices unchanged. Since H_1 is a component of H, no edge of H joins a vertex of H_1 to a vertex of another component of H. Hence the interchange preserves proper coloring and produces another proper k-coloring of G inducing a different partition of the vertices into color classes. This contradicts the assumption that G is uniquely k-colorable. Therefore, $G[C_i \cup C_j]$ is connected. ■

Theorem 1.5.4. *A uniquely k-colorable graph is $(k-1)$-connected.*

Proof. Let G be uniquely k-colorable. If G is complete, the result is immediate. Suppose that G is not complete. Assume, to the contrary, that G is not $(k-1)$-connected. Then there exists a vertex cut X with $|X| \leq k-2$ such that $G - X$ is disconnected. Since at most $k-2$ vertices occur in X, at least two colors, say c_1 and c_2, are not used on any vertex of X. Let C_1 and C_2 denote the corresponding color classes. By the preceding theorem, the subgraph $G[C_1 \cup C_2]$ is connected. As no vertex of X has color c_1 or c_2, the entire subgraph $G[C_1 \cup C_2]$ lies in $G - X$ and therefore in a single component, say G_1. Let G_2 be another component of $G - X$, and choose $v \in V(G_2)$. Its color is different from c_1 and c_2. Consider the subgraph induced by the color of v together with c_1, and interchange these two colors on the component containing v. This produces another proper k-coloring of G with a different color-class partition, contradicting uniqueness. Hence G is $(k-1)$-connected. ■

Theorem 1.5.5. *Every planar graph is four-colorable if and only if every Hamiltonian planar graph is four-colorable.*

Proof. If every planar graph is four-colorable, then every Hamiltonian planar graph is four-colorable, since every Hamiltonian planar graph is planar. Conversely, suppose that every Hamiltonian planar graph is four-colorable. Using the standard fact that every planar graph can be embedded as a subgraph of a Hamiltonian planar graph, let G be an arbitrary planar graph and let G' be a Hamiltonian planar graph containing G as a subgraph. By hypothesis, G' is four-colorable. Restricting such a coloring to G gives a proper four-coloring of G. Hence every planar graph is four-colorable. ■

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CHAPTER
2

MATCHINGS AND FACTORS

Historical Note and Motivation

The theory of matchings began with **Julius Petersen's** 1891 study of regular graphs and evolved through **Philip Hall's** 1935 matching criteria for bipartite graphs. The field was revolutionized by **W.T. Tutte** in 1947, who characterized 1-factors in general graphs and later extended this to f-factors. These structural foundations were complemented by **Claude Berge's** algorithmic approach using augmenting paths and the **Erdős-Gallai** characterization of degree sequences. The study culminates in the factorization of K_n, a problem rooted in 19th-century tournament designs.
This chapter explores the fundamental tension between **local constraints** (vertex degrees) and **global structure** (factors). By mastering these topics, we learn how to determine when local demands can be universally satisfied, a concept essential for solving real-world assignment and scheduling problems. Theoretically, this section highlights the power of reduction, demonstrating how complex f-factor problems can be solved using the simpler mechanics of 1-factors.

2.1 Matchings

Definition 2.1.1 (Matching, Maximal, and Maximum)**.** A **matching** in a graph is a set of independent edges. That is, a subset M of the edge set E of a graph $G(V, E)$ is a matching if no two edges of M have a common vertex.

- A matching M is said to be **maximal** if there is no matching M' strictly containing M. That is, M is maximal if it cannot be enlarged (you cannot add another edge without breaking the matching property).
- A matching M is said to be **maximum** if it has the largest possible cardinality. That is, M is maximum if there is no matching M' such that $|M'| > |M|$.

Definition 2.1.2 (Saturated and Unsaturated Vertices)**.** Let M be a matching in a graph G.

- A vertex v in G is said to be M**-saturated** (or saturated by M) if there is an edge $e \in M$ incident with v.
- A vertex which is not incident with any edge of M is said to be M**-unsaturated**.

In other words, given a matching M in a graph G, the vertices belonging to the edges of M are M-saturated and the vertices not belonging to the edges of M are M-unsaturated.

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Consider the graph shown in Figure 2.1 below. Clearly, $M = \{v_1v_2, v_3v_7, v_4v_5\}$ is a matching. The vertices $v_1, v_2, v_3, v_4, v_5, v_7$ are M-saturated, but v_6 is M-unsaturated.

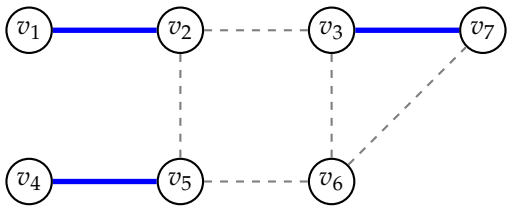


Figure 2.1: Visual representation of the matching M. The blue edges form the matching.

Definition 2.1.3 (Perfect Matching)**.** A matching M in a graph G is said to be a **perfect matching** if M saturates every vertex of G.

Definition 2.1.4 (Complete Matching in Bipartite Graphs)**.** If G is a bipartite graph with bipartition $V = V_1 \cup V_2$ and $|V_1| = n_1$, $|V_2| = n_2$, then a matching M of G saturating all the vertices in V_1 is called a **complete matching** (or V_1 matched into V_2).

Definition 2.1.5 (Alternating Path)**.** Let H be a subgraph of a graph G. An H**-alternating path (cycle)** in G is a path (cycle) whose edges are alternately in $E(G) \setminus E(H)$ and $E(H)$.

Specifically, if M is a matching in a graph G, then an M**-alternating path (cycle)** in G is a path (cycle) whose edges are alternately in $E(G) \setminus M$ and M. That is, in an M-alternating path, the edges alternate between M-edges and non-M-edges.

Definition 2.1.6 (Augmenting Path)**.** An M-alternating path whose end vertices are M-unsaturated is said to be an M**-augmenting path**.

Example 2.1.7. Examples to explain definitions given above.

- **Matching:** A set of edges without common vertices.
- **Maximal Matching:** A matching to which no more edges can be added.
- **Maximum Matching:** A matching with the largest possible number of edges.



Figure 2.2: **Maximal Matching**. Size = 1. We cannot add (A, B) or (C, D) because B and C are already busy.

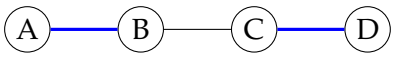


Figure 2.3: **Maximum Matching**. Size = 2. This is the largest possible matching for this graph.

Saturated vs. Unsaturated Vertices

A vertex is **Saturated** if it touches a matched edge (blue). Otherwise, it is **Unsaturated**.

Perfect Matching

A matching that saturates **every** vertex in the graph.

Complete Matching (Bipartite)

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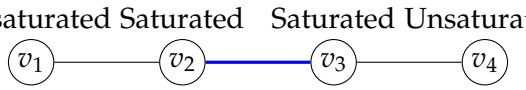


Figure 2.4: Vertices v_2, v_3 are saturated by the blue edge. v_1, v_4 are unsaturated.

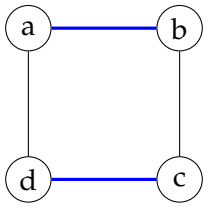


Figure 2.5: A Perfect Matching on C_4. All 4 vertices are covered.

In a bipartite graph $G = (V_1 \cup V_2, E)$, a matching is **complete from** V_1 **to** V_2 if every vertex in V_1 is saturated.

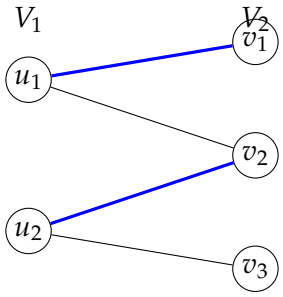


Figure 2.6: Complete Matching of V_1 into V_2. All vertices in V_1 are blue-connected. Vertex v_3 is left over, which is fine.

Alternating and Augmenting Paths

- **Alternating Path:** Edges alternate between Non-Matching and Matching.
- **Augmenting Path:** An alternating path that starts and ends at unsaturated vertices.

Theorem 2.1.8 (Berge). *A matching M of a graph G is maximum if and only if G contains no M-augmenting path.*

Proof. Suppose first that M is a maximum matching and that $P = v_1 v_2 \cdots v_k$ is an M-augmenting path. Then the edges of P alternate between edges outside M and edges in M, with the first and last edges outside M. Thus $v_2v_3, v_4v_5, \ldots, v_{k-2}v_{k-1} \in M$, while $v_1v_2, v_3v_4, \ldots, v_{k-1}v_k \notin M$.

Replacing the edges of M on P by the remaining edges of P produces a matching

$$M_1 = \big[M - \{v_2v_3, \ldots, v_{k-2}v_{k-1}\}\big] \cup \{v_1v_2, \ldots, v_{k-1}v_k\},$$

with $|M_1| = |M| + 1$, contradicting the maximality of M. Hence no M-augmenting path exists.

Conversely, suppose that G contains no M-augmenting path. If M were not maximum, there would exist a matching M' with $|M'| > |M|$. Let H be the spanning subgraph of G whose edge set is $M \triangle M'$.

Since each vertex is incident with at most one edge of M and at most one edge of M', every vertex of H has degree at most 2. Hence each component of H is an isolated vertex, a path, or a cycle, and its edges alternate between M and M'.

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Figure 2.7: **Alternating Path** $(1-2-3)$. Starts Unsat(1) $\rightarrow$ Sat(3).

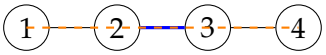


Figure 2.8: **Augmenting Path** $(1-2-3-4)$. Starts Unsat(1) $\rightarrow$ Ends Unsat(4).

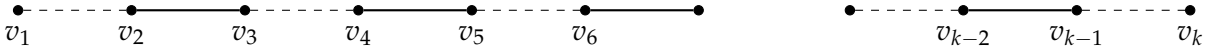


Figure 2.9:

Every cycle contains the same number of edges from M and M'. Since $|M'| > |M|$, some path component must contain one more edge from M' than from M. Such a path begins and ends with edges of M', and therefore its end vertices are unsaturated by M. Hence it is an M-augmenting path, a contradiction.

Therefore, M is maximum. ■

Theorem 2.1.9 (Hall's Marriage Theorem). *A bipartite graph $G = (X \cup Y, E)$ has a matching that covers all vertices in X if and only if for every $S \subseteq X$, $|N(S)| \geq |S|$.*

Proof. Assume a matching M covers all vertices in X. For any $S \subseteq X$, each vertex in S is matched by M to a distinct vertex in Y. Let $M(S)$ be this set of matched vertices in Y. Then $|M(S)| = |S|$. Since every vertex in S is connected to its matched partner, $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |M(S)| = |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction that a matching covering X exists. Suppose no such matching exists. Let M be a maximum matching in G. Since M does not cover X, there must be at least one unmatched vertex $u \in X$. By Berge's Lemma, since M is a maximum matching, there are no M-augmenting paths in G. Let A be the set of all vertices reachable from u by an M-alternating path. Let $X' = A \cap X$ and $Y' = A \cap Y$. Since u is unmatched and starts an M-alternating path, $u \in X'$. For any $y \in Y'$, it must be connected to some $x \in X'$ by an edge not in M. M must match this x to some $y' \in Y'$, otherwise we could extend the path to x to form an M-augmenting path from u to x, a contradiction (since x would be unmatched). Specifically, any vertex $x \in X'$ (other than u) must be matched by M to some $y \in Y'$. If x were unmatched, then an M-alternating path from u to x would be an M-augmenting path, a contradiction. Similarly, any vertex $y \in Y'$ must be matched by M to some $x \in X'$. If y were matched to $x \notin X'$, then combining the path from u to y (ending with an M-edge) with the edge (y, x) (which is in M) would form an M-alternating path from u to x ending with an M-edge. This would imply $x \in X'$, a contradiction. Therefore, all edges of M that are incident to A must connect a vertex in $X' \setminus \{u\}$ to a vertex in Y'. This implies that M provides a perfect matching

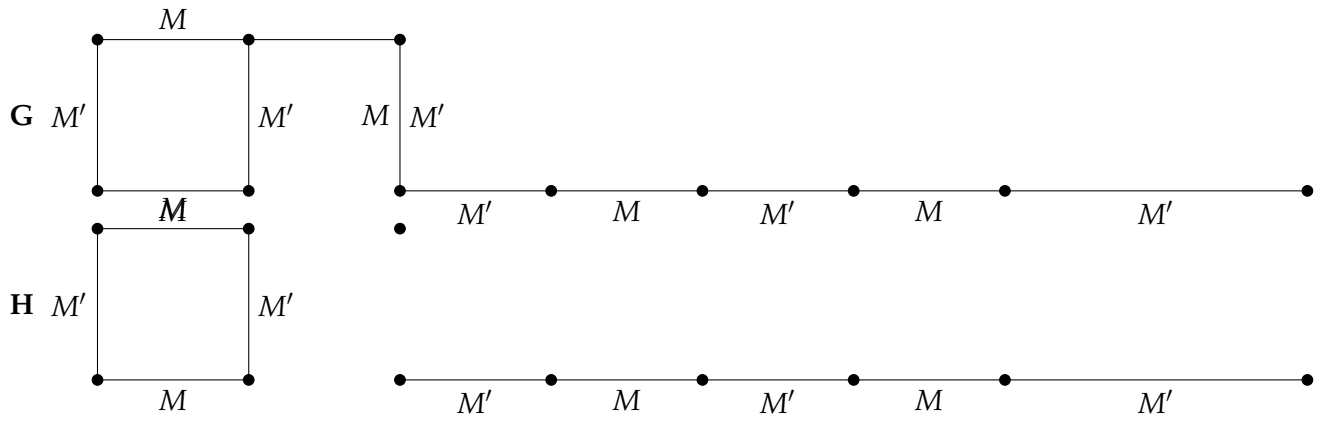


Figure 2.10:

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between $X' \setminus \{u\}$ and Y'. Thus, $|X'| - 1 = |Y'|$.

Now consider the set $S = X'$. By construction of A, all neighbors of X' must be in Y'. Thus $N(X') \subseteq Y'$. From our previous deduction, $|Y'| = |X'| - 1$. Therefore, $|N(X')| \leq |Y'| = |X'| - 1$. This means $|N(X')| < |X'|$, which contradicts the given condition that $|N(S)| \geq |S|$ for all $S \subseteq X$. Our initial assumption that no matching covering X exists must be false. Hence, such a matching exists. ■

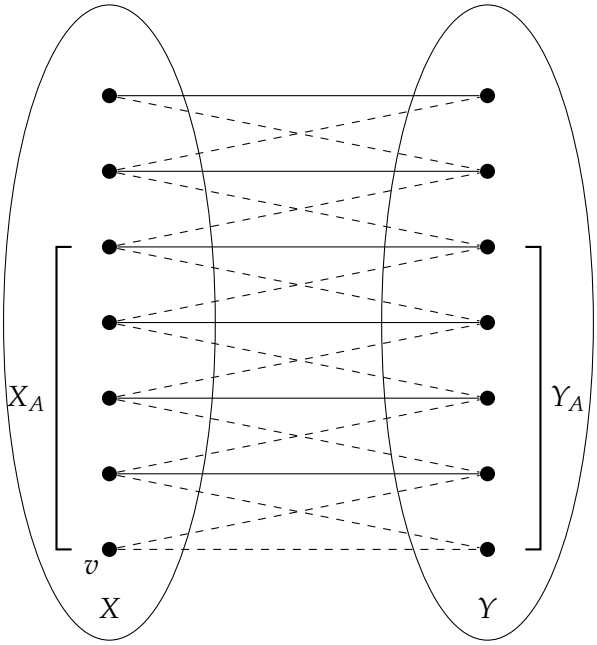


Figure 2.11: Bipartite graph illustrating sets $X_A \subseteq X$ and $Y_A \subseteq Y$ relative to a matching.

2nd Proof using Augmenting Paths. Assume a matching M covers all vertices in X. For any $S \subseteq X$, each $x \in S$ is matched to a distinct $y_x \in Y$. These y_x form the set $M(S)$. Clearly, $|M(S)| = |S|$ and $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction. Suppose no matching covers X. Let M be a maximum matching. Since it doesn't cover X, there's an unmatched $u \in X$. By the given theorem (Berge's Lemma), since M is maximum, there are no M-augmenting paths.

Let A be the set of vertices reachable from u by M-alternating paths. Let $X_A = A \cap X$ and $Y_A = A \cap Y$. (See Figure 2.11). Since there are no $M-$augmenting paths, u is the only vertex in A that is $M-$unsaturated. Furthermore, if $y \in Y_A$ is matched by M, its matched partner x' must be in X_A. (If $x' \notin X_A$, then extending the M-alternating path from u to y with the edge $(y, x') \in M$ would make x' reachable, implying $x' \in X_A$, a contradiction). Thus, M perfectly matches Y_A with $X_A \setminus \{u\}$. Therefore, $|Y_A| = |X_A| - 1$.

Now, let $S = X_A$. We claim $N(S) \subseteq Y_A$. Take any $x \in S$ and its neighbor $y \in N(x)$. If $y \notin Y_A$, then the edge (x, y) cannot be in M (because if it were, then x would be matched to y, but $x \in X_A$ and $y \notin Y_A$ implies x is not matched within A, a contradiction to M matching $X_A \setminus \{u\}$ to Y_A). If $(x, y) \notin M$, then combining the M-alternating path from u to x with (x, y) would make y reachable, implying $y \in Y_A$, a contradiction. Thus, $N(S) \subseteq Y_A$.

Combining these, $|N(S)| \leq |Y_A| = |X_A| - 1 = |S| - 1$. So, $|N(S)| < |S|$ for $S = X_A$. This contradicts our initial assumption. Therefore, a matching covering X must exist. ■

Theorem 2.1.10 (Hall's SDR Theorem). *Let $\mathcal{A} = \{A_i : 1 \leq i \leq r\}$ be a finite family of nonempty finite*

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sets. Then $\mathcal{A}$ has a system of distinct representatives if and only if, for every subset $I \subseteq \{1, 2, \ldots, r\}$,

$$\left|\bigcup_{i \in I} A_i\right| \geq |I|.$$

Equivalently, the union of any k members of $\mathcal{A}$ contains at least k elements, for $1 \leq k \leq r$.

Proof. Construct a bipartite graph $G = (X \cup Y, E)$ by taking $X = \{1, 2, \ldots, r\}$ and $Y = \bigcup_{i=1}^{r} A_i$, and joining $i \in X$ to $y \in Y$ precisely when $y \in A_i$.

A system of distinct representatives for $\mathcal{A}$ is exactly a matching in G that saturates every vertex of X. Indeed, choosing a representative $x_i \in A_i$ corresponds to choosing the edge ix_i, and the representatives are distinct precisely when the chosen edges form a matching.

By Hall's Marriage Theorem, such a matching exists if and only if $|N(S)| \geq |S|$ for every subset $S \subseteq X$. But, by the construction of G,

$$N(S) = \bigcup_{i \in S} A_i.$$

Hence Hall's condition is equivalent to

$$\left|\bigcup_{i \in S} A_i\right| \geq |S|$$

for every $S \subseteq X$.

Therefore, $\mathcal{A}$ has an SDR if and only if the union of any k members of $\mathcal{A}$ contains at least k elements. ■

Theorem 2.1.11. *Let G be a bipartite graph with bipartite sets V_1 and V_2 of orders n_1 and n_2, respectively, where $n_1 \leq n_2$. Suppose that G satisfies Hall's condition*

$$|S| \leq |N(S)|$$

for every subset $S \subseteq V_1$. Then G has at least

$$r(\delta, n_1) = \prod_{i=1}^{\min\{\delta, n_1\}} (\delta - i + 1)$$

complete matchings, where

$$\delta = \min\{d(v) : v \in V_1\}.$$

Proof. We proceed by induction on n_1.

If $n_1 = 1$, then $V_1 = \{v_1\}$ and $d(v_1) = \delta$. Each edge incident with v_1 gives a complete matching. Hence there are δ complete matchings. Since $r(\delta, 1) = \delta$, the result holds.

Assume that the result is true for bipartite graphs whose first partite set has $n_1 - 1$ vertices. Let $v_1 \in V_1$. Since $d(v_1) \geq \delta$, there are at least δ choices for a vertex $u_1 \in N(v_1)$.

For each such choice of u_1, consider the graph $G' = G - \{v_1, u_1\}$. Its first partite set is $V_1' = V_1 - \{v_1\}$ and has order $n_1 - 1$. On deleting u_1, a vertex of V_1' can lose at most one incident edge. Therefore the minimum degree of the vertices of V_1' in G' is at least $\delta - 1$.

By the induction hypothesis, G' has at least $r(\delta - 1, n_1 - 1)$ complete matchings. Each such matching, together with the edge v_1u_1, gives a complete matching of G. Since there are at least δ choices for u_1, the number of complete matchings of G is at least

$$\delta\, r(\delta - 1, n_1 - 1).$$

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If $\delta \geq n_1$, then $\delta - 1 \geq n_1 - 1$, and hence

$$\delta\, r(\delta - 1, n_1 - 1) = \delta \prod_{i=1}^{n_1-1} (\delta - i) = \delta(\delta - 1) \cdots (\delta - n_1 + 1) = r(\delta, n_1).$$

If $\delta < n_1$, then $\delta - 1 < n_1 - 1$, and therefore

$$\begin{aligned}
\delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - i) &= \delta \cdot \prod_{i=1}^{\delta-1} (\delta - i) \\
&= \delta\big((\delta - 1)(\delta - 2) \cdots 1\big) \\
&= \delta(\delta - 1)! \\
&= \delta! \\
&= \prod_{i=1}^{\delta} (\delta - i + 1).
\end{aligned}$$

Thus, in this case also, the number of complete matchings is at least $r(\delta, n_1)$.

Hence the result follows by induction. ■

Theorem 2.1.12. *Every k-regular bipartite graph with $k > 0$ has a perfect matching.*

Proof. Let $G = (X \cup Y, E)$ be a k-regular bipartite graph, where X and Y are the two partite sets.

Since every vertex has degree k, counting the edges from the side X gives

$$|E| = \sum_{x \in X} d(x) = k|X|.$$

Similarly, counting the same edges from the side Y gives

$$|E| = \sum_{y \in Y} d(y) = k|Y|.$$

Hence

$$k|X| = k|Y|.$$

Since $k > 0$, we obtain

$$|X| = |Y|.$$

We now verify Hall's condition. Let $S \subseteq X$, and let $N(S)$ denote the set of all vertices of Y adjacent to at least one vertex of S.

Every vertex of S has degree k, and all of its incident edges join a vertex of S to a vertex of $N(S)$. Therefore, the number of edges between S and $N(S)$ is

$$|E(S, N(S))| = k|S|.$$

On the other hand, each vertex of $N(S)$ has total degree k. Therefore, at most $k|N(S)|$ edges can join S to $N(S)$. Thus

$$|E(S, N(S))| \leq k|N(S)|.$$

Combining the two relations, we get

$$k|S| \leq k|N(S)|.$$

Since $k > 0$,

$$|S| \leq |N(S)|.$$

Thus Hall's condition holds for every subset $S \subseteq X$. By Hall's Marriage Theorem, G has a matching that saturates every vertex of X.

Finally, since $|X| = |Y|$, a matching that saturates all vertices of X also saturates all vertices of Y. Hence the matching is a perfect matching. ■

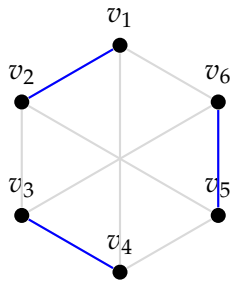
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2.2 Factors of Graphs

Definition 2.2.1 (Factor)**.** Let G be a graph. A **factor** F of G is a spanning subgraph of G. That is, $V(F) = V(G)$ and $E(F) \subseteq E(G)$.

Definition 2.2.2 ($\mathcal{F}$-Factor and H-Factor)**.** Let $\mathcal{F} = \{H_1, H_2, \ldots, H_k\}$ be a collection of graphs. A factor F is called an **$\mathcal{F}$-factor** if every connected component of F is isomorphic to some element of $\mathcal{F}$.

In the specific case where $\mathcal{F} = \{H\}$ (a singleton set), the factor F is called an **H-factor**. In this case, every component of the spanning subgraph is isomorphic to H.



K_2-Factor
The blue edges form a spanning subgraph
where every component is isomorphic to K_2.

Figure 2.12: Visualisation of an H-factor where $H = K_2$.

Degree Constraints: f-Factors and k-Factors

Let G_1 and G_2 be graphs. Recall the *Cartesian product* of G_1 and G_2, denoted by $G_1 \square G_2$, is the graph with vertex set

$$V(G_1 \square G_2) = V(G_1) \times V(G_2).$$

Two vertices (u_1, u_2) and (v_1, v_2) are adjacent in $G_1 \square G_2$ if and only if either $u_1 = v_1$ and $u_2 v_2 \in E(G_2)$, or $u_2 = v_2$ and $u_1 v_1 \in E(G_1)$.

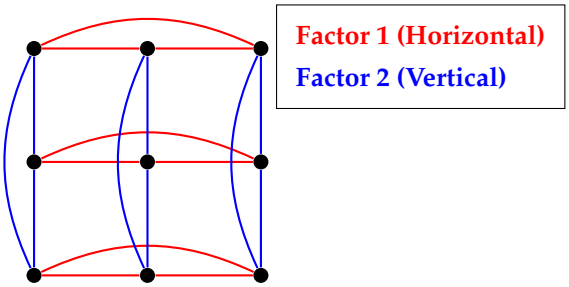
Example 2.2.3. Consider the graph $G = K_3 \square K_3$ (also known as the Rook's graph on a 3×3 grid). This graph has 9 vertices and is 4-regular. It admits a K_3-factorization consisting of two factors, F_1 and F_2.

- **Factor 1 (Red):** Three disjoint triangles formed horizontally.
- **Factor 2 (Blue):** Three disjoint triangles formed vertically.

Since $E(G) = E(F_1) \cup E(F_2)$ and $E(F_1) \cap E(F_2) = \emptyset$, this constitutes a factorization.

Definition 2.2.4 (Vertex Function)**.** A **vertex function** of a graph $G(V, E)$ is a mapping $f : V \to \mathbb{N}_0$, where $\mathbb{N}_0$ denotes the set of non-negative integers.

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Graph $G = F_1 \cup F_2$

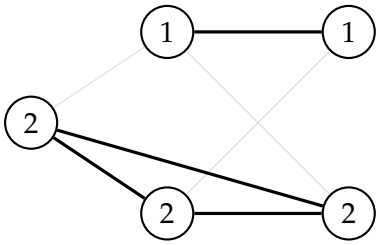
Figure 2.13: Visualisation of G decomposed into two K_3-factors.

Definition 2.2.5 (f-Factor)**.** A spanning subgraph F of a graph G is called an **f-factor** if the degree of every vertex v in F corresponds to the value assigned by the vertex function. That is:

$$\deg_F(v) = f(v) \quad \forall v \in V.$$

Definition 2.2.6 (k-Factor)**.** A **k-factor** is a special case of an f-factor where $f(v) = k$ for all $v \in V$. Thus, a k-factor is a k-regular spanning subgraph.

- A **1-factor** is a perfect matching (vertices have degree 1).
- A **2-factor** is a disjoint union of cycles covering all vertices.



Example of an f-factor
Numbers inside nodes represent the target degree $f(v)$. The thick edges form the subgraph satisfying these constraints.

Figure 2.14: An f-factor where some vertices require degree 1 and others degree 2.

Factorisation

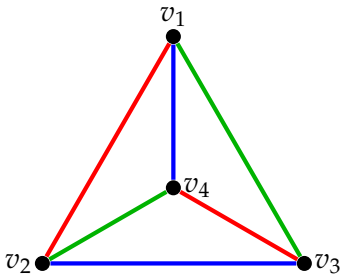
Definition 2.2.7 (Factorisation / Decomposition)**.** Let G be a graph and let $\{F_1, F_2, \ldots, F_q\}$ be a set of factors of G. These factors constitute a **factorisation** (or decomposition) of G if their edge sets form a partition of $E(G)$. We write:

$$G = F_1 \oplus F_2 \oplus \cdots \oplus F_q$$

This means every edge of G belongs to exactly one factor F_i.

Definition 2.2.8 (Isomorphic Factorisation)**.** If each factor F_i in a factorisation is isomorphic to the same graph F, the factorisation is called an **isomorphic factorisation** or an F**-decomposition** of G.

Definition 2.2.9 (k-Factorisation)**.** A factorisation is called a k**-factorisation** if every factor F_i is a k-factor. A graph admitting such a decomposition is said to be k**-factorable**.



Factor 1 — v_1v_2, v_3v_4
Factor 2 — v_1v_4, v_2v_3
Factor 3 — v_1v_3, v_2v_4

1-Factorisation of K_4

Figure 2.15: A 1-factorisation of the complete graph K_4. The edges are partitioned into three distinct perfect matchings (Red, Blue, Green).

Definition 2.2.10. A component of a graph is odd or even according to whether it has an odd or even number of vertices. The number of odd components of a graph G is denoted by $k_0(G)$.

Tutte's One-Factor Theorem: A graph $G = (V, E)$ has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. That is, $o(G - S) \leq |S|$ for all $S \subseteq V$.

Proof. A Video lecture available at
https://youtu.be/EW4oA9HZzgg?si=8QGKe7Sfpdem9Rg5
or visit YouTube via **https://aijaz.vercel.app/** ■

Theorem 2.2.11 (Tutte)**.** *A graph G has a 1-factor if and only if*

$$k_0(G - S) \leq |S|, \text{ for all } S \subseteq V. \tag{8.8.1}$$

Proof. Let G be a graph having 1-factor M. Let S be an arbitrary subset of V and let $O_1, O_2, \ldots, O_k$ be the odd components of $G - S$. For each i, the vertices in O_i can be adjacent only to other vertices in O_i and to vertices in S. Since G has a 1-factor (perfect matching), at least one vertex from each O_i is to be matched with a different vertex in S. Thus the number of vertices in S is at least k, the number of odd components. Hence $|S| \geq k$, so that $k_0(G - S) \leq |S|$ (Fig. T1).

Conversely, assume that the condition (8.8.1) holds. Let G have no 1-factor. Join pairs of non-adjacent vertices of G and continue this process till we get a maximal supergraph G^* of G having no 1-factor. Since joining two odd components by an edge results in an even component, therefore

$$k_0(G^* - S) \leq k_0(G - S). \tag{8.8.2}$$

Thus for G^* also, we have $k_0(G^* - S) \leq |S|$.

First, we note that G has even order. This is because the condition (8.8.1) must hold for $S = \phi$, which gives $k_0(G) = 0$. Therefore, in this case G has no odd component, so G itself is an even-ordered graph.

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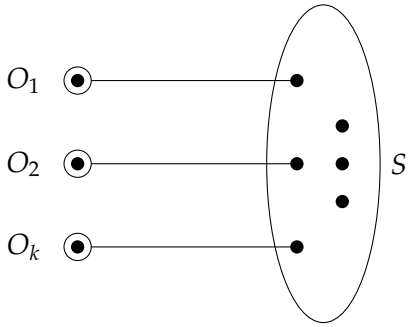


Figure T1

Also, for every pair of non-adjacent vertices u and v of G^*, $G^* + uv$ has a 1-factor, and any such 1-factor definitely contains the edge uv.

Now, let K be the set of those vertices of G^* which are having degree $n-1$. Clearly, $K \neq V$, since otherwise $G^* = K_n$ has a perfect matching.

Claim that each component of $G^* - K$ is complete. Assume, on the contrary, that in $G^* - K$, there exists a component G_1 which is not complete. Then in G_1, there are vertices x, y and z such that $xy, yz \in E(G^*)$ and $xz \notin E(G^*)$. Also, $y \in V(G_1)$ so $y \notin K$ and so $d(y|G^*) < n-1$ and thus there exists a vertex w of G^* with $yw \notin E(G^*)$. Evidently, $w \notin K$ (Fig. T2).

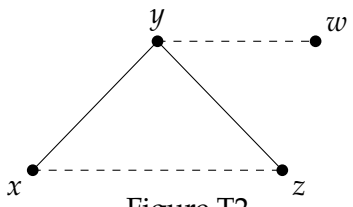
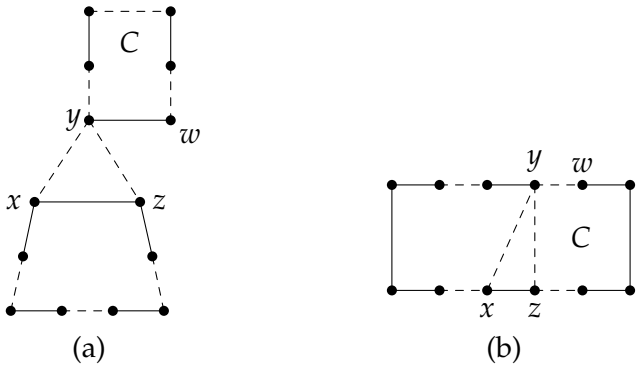


Figure T2

Now, by the choice of G^*, both $G^* + xz$ and $G^* + yw$ have 1-factors, say M_1 and M_2 respectively. M_1 must contain xz and M_2 must contain yw. Let H be the subgraph of $G^* + \{xz, yw\}$ induced by the edges that are in M_1 and M_2 but not in both (that is, in the symmetric difference of M_1 and M_2). Since M_1 and M_2 are 1-factors, each vertex of G^* is saturated by both M_1 and M_2, and H is a disjoint union of even cycles in which the edges alternate between M_1 and M_2. We have two cases to consider.

Case 1 Let xz and yw belong to different components of H, as shown in Figure T3 (a). If yw belongs to the cycle C, then the edge of M_1 in C together with the edges of M_2 not belonging to C form a 1-factor of G^*, contradicting the choice of G^*.



Ordinary lines correspond to the edges of M_1 and bold lines to the edges of M_2.

Figure T3

Case 2 Let xz and yw belong to the same component C of H. Since each component of H is a

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cycle, C is a cycle (Fig. T3 (b)). By symmetry of x and z, it can be assumed that the vertices x, y, w and z appear in that order on C. Then the edges of M_1 belonging to the $yw \ldots x$ section of C, together with the edge yz, and the edges of M_2 not in the $yw \ldots z$ section of C form a 1-factor of G^*, again contradicting the choice of G^*.

Thus each component of $G^* - K$ is complete.

By conditions (8.8.2), $k_0(G^* - K) \leq |K|$. Therefore one vertex of each of the odd components of $G^* - K$ is matched to a vertex of K. This is because each vertex of K is adjacent to every other vertex of G^*. Also the remaining vertices in each of the odd and even components of $G^* - K$ can be matched among themselves (Fig. T4). The total number of vertices thus matched is even. As $|V(G^*)|$ is even, the remaining vertices of K can be matched among themselves, thus giving a 1-factor of G^*, which is a contradiction.

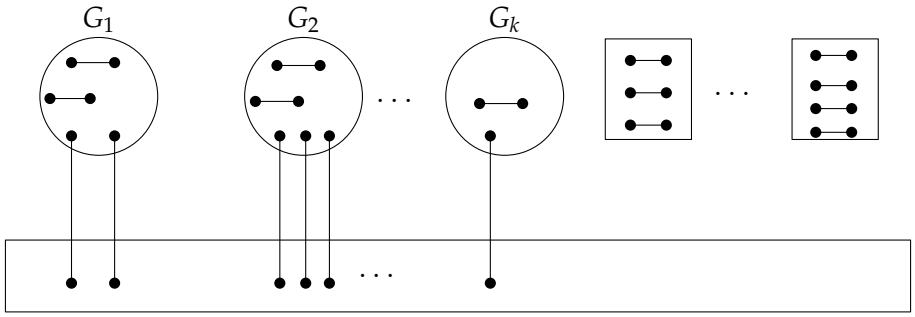


Figure T4

■

Remark 2.2.12. Among the various proofs of Theorem 8.8, one proof due to Anderson [4] uses Hall's theorem.

Tutte's characterisation of graphs admitting 1-factors has been used to obtain several results including those for regular graphs and many of these are reported in the review article by Akiyama and Kano [1]. We present some of these results. The first is due to Peterson [181].

Theorem 2.2.13 (Peterson). *Every cubic graph without cut edges has a 1-factor*

Proof. Let G be a cubic graph with vertex set $V(G)$. Let G have no cut edges and let $S \subseteq V$. Let $O_1, O_2, \ldots, O_r$ be the odd components in $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S. Then we have;

$$\sum_{v \in V(O_i)} \deg(v) = 3n_i = 2m_i + k_i.$$

This gives, $k_i = 3n_i - 2m_i$, $1 \leq i \leq r$. As n_i is odd, therefore, k_i must be odd. Moreover, since there are no cut edges in the graph, we have $k_i \neq 1$, that is, $k_i \geq 3$, that is, there are at least 3 edges from each O_i to S. Hence, we have

$$3r \leq \sum_{s \in S} \deg(s) = 3|S|.$$

Therefore,

$$k_o(G - S) = r \leq |S|$$

for each $S \subseteq V(G)$. The proof then follows by Tutte's theorem. ■

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Theorem 2.2.14 (Generalization of Theorem 2.2.13). *Let G be a $k-$regular graph having even number of vertices and is $(k-1)-$edge connected. Then G has a $1-$factor.*

Proof. Let G be a k-regular, $(k-1)$-edge-connected graph with even order and let S be any subset of V. Let $O_1, O_2, \ldots, O_r$ be the odd components of $G-S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)-$connected, we have

$$k_i \geq k-1, \tag{2.1}$$

Now, G is $k-$regular and vertices of any O_i are either adjacent within O_i or adjacent to some vertex of S, therefore,

$$\sum_{v \in V(O_i)} d(v) = kn_i = 2m_i + k_i \tag{2.2}$$

and vertices in S can be adjacent within S or adjacent to some vertex of O_i, $1 \leq i \leq r$ or may be adjacent to some other vetex of G, so

$$\sum_{v \in S} d(v) = k|S| \geq \sum_{i=1}^{r} k_i + 2|E(S)|. \tag{2.3}$$

Now, (2.2) gives $k_i = kn_i - 2m_i$, so that if k and $k)i$ have the same parity, therefore, $k_i \geq k-1$ implies $k_i \geq k$. Summing this over r odd components, we obtain by using (2.3)

$$rk \leq \sum_{i=1}^{r} k_i \leq k|S| + 2|E(S)| \leq k|S|.$$

Therefore, $k_0(G-S) = r \leq |S|$ and by Tutte's theorem, G has a 1-factor. ■

Definition 2.2.15 (Anti-factors). A subset $S \subseteq V$ of a graph $G(V,E)$ such that $k_0(G-S) > |S|$ is called an antifactor set. An antifactor set S such that no proper subset of S is an antifactor set is called a minimal antifactor set.

Theorem 2.2.16. *Let G be a $k-$regular graph of even order that is also $(k-1)-$edge connected. If G_1 is obtianed from G by deleting any $k-1$ edges, then G_1 has a $1-$factor.*

Proof. Let G be a k-regular, $(k-1)$-edge-connected graph with even order. Let $K \subseteq V(G)$ and let $V(G_1) = V(G) - K$ and let S be any subset of V. Let $O_1, O_2, \ldots, O_r$ be the odd components of $G-S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)-$connected, we have a_i =number of edges from K to S
b_i =number of edges within O_i
m_i =number of edges from O_i to S

$$a_i + m_i \geq k-1, \text{ then } \sum_{i=1}^{r} a_i + \sum_{i=1}^{r} m_i \geq k$$

Morevover, $\sum_{i=1}^{r} a_i + \frac{1}{2}\sum_{i=1}^{r} b_i \leq k-1$, which gives, $2\sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i \leq 2(k-1)$.

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Also, $\sum_{i=1}^{r} m_i \leq k|S|$. Adding, we get,

$$2\sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i + \sum_{i=1}^{r} m_i \leq 2(k-1) + k|S| = k(|S|+2) - 2.$$

Finally,

$$\begin{aligned} kr &\leq \sum_{i=1}^{r} a_i + \sum_{i=1}^{r} m_i \\ &\leq 2\sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i + \sum_{i=1}^{r} m_i \\ &\leq k(|S|+2) - 2 \\ &< k(|S|+2). \end{aligned}$$

Thus, $r < |S| + 2$, a contradiction. ■

Corollary 2.2.17. *A $(k-1)$-regular simple graph on $2k$ vertices has a 1-factor.*

Proof. Assume G is a $(k-1)$-regular simple graph with $2k$ vertices having no 1-factor. Then G has an antifactor set S and by Lemma 8.4, $k_0(G-S) \geq |S|+2$. Therefore,

$$|S| + (|S|+2) \leq 2k \implies |S| \leq k-1.$$

Let $|S| = k - r$. Then $r \neq 1$. For if $r = 1$, then $|S| = k-1$ and therefore $k_0(G-S) = k+1$. Thus each odd component of $G - S$ is a singleton and therefore each such vertex is adjacent to all the $k-1$ vertices of S, as G is $(k-1)$-regular. This implies that every vertex of S is of degree at least $k+1$, a contradiction.

Thus, $|S| = k - r$, with $2 \leq r \leq k-1$.

If G_1 is any component of $G - S$ and $v \in V(G_1)$, then v can be adjacent to at most $|S|$ vertices of S. Since G is $(k-1)$-regular, v is adjacent to at least $(k-1) - (k-r) = r-1$ vertices of G_1. So, $|V(G_1)| \geq r$.

Counting the vertices of S and the components, we obtain:

$$(|S|+2)r + |S| \leq 2k$$

Substituting $|S| = k - r$:

$$(k - r + 2)r + (k - r) \leq 2k$$

Expanding and simplifying:

$$kr - r^2 + 2r + k - r \leq 2k$$

$$kr - r^2 + r - k \leq 0$$

Multiplying by -1 and factoring:

$$r^2 - r(k+1) + k \geq 0$$

$$(r-1)(r-k) \geq 0$$

Since $2 \leq r \leq k-1$, the term $(r-1)$ is positive and $(r-k)$ is negative. Their product is negative, which violates the condition $(r-1)(r-k) \geq 0$. ■

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Definition 2.2.18. A *vertex function* for a general graph $G(V, E)$ is a function $f : V \to \mathbb{N}_0$ such that $0 \leq f(v) \leq d(v)$, for every $v \in V$.

For any function $f : V \to \mathbb{N}_0$ and for any $U \subseteq V$, we set

$$f(U) = \sum_{v \in U} f(v).$$

Definition 2.2.19. Three pair-wise disjoint (possibly empty) subsets S, T and U of the vertex set V of a general graph G, such that $V = S \cup T \cup U$, is called a *decomposition* $D = (S, T, U)$ of G. The components, say $C_1, C_2, \ldots, C_k$, of the vertex-induced subgraph $\langle U \rangle$ of G are called the components of U. For each C_i, let

$$h_i = f(C_i) + q_G[T, C_i],$$

where f is a given vertex function of G, and $q_G[T, C_i]$ denotes the number of edges of G, whose one end is in T and the other end is in C_i. Then C_i is said to be an odd or even component of U, depending upon h_i being odd or even, and the number of odd components of U is denoted by $k'_0(D, f)$.

The *f-deficiency* of D for the given G is defined as

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Construction 2.2.20 (Construction of Graph G' for $f-$factor Theorem)**.** Let G be a general graph with the vertex function f. This defines for each vertex v_i, its degree $d_i = d(v_i|G)$, $f_i = f(v_i)$ and $k_i = d_i - f_i$. For each vertex v_i of G, take two subsets X_i and Y_i of vertices of G', where $|X_i| = d_i$ and $|Y_i| = k_i$. Let the vertices of Y_i be labelled $y_j^i, 1 \leq j \leq k_i$, arbitrarily. Now, label the vertices of X_i in the following manner.

Label the edges of G arbitrarily as $1, 2, \ldots, m$. Since $|X_i|$ = number of link edges l_i incident with v_i plus twice the number of loops ℓ_i^0 incident with v_i, a vertex $x_j^i \in X_i$ is taken corresponding to each link edge j incident with v_i and two vertices $x_h^i, x_{h'}^i$ are taken corresponding to each loop h incident at v_i. As for edges, if link j joins v_i and v_k, then $x_j^i x_j^k$ is made an edge. Corresponding to loop h incident with x_h^i, the edge $x_h^i x_{h'}^i$ is taken. In addition, all the edges of the complete bipartite subgraph with vertex partition $\{X_i, Y_i\}$ are taken for each i. These constitute the graph G'. The induced subgraph $< X_i \cup Y_i >$ of G' is called the *star graph* at i and is denoted by $St(i)$. For multi graph G, it is a complete bipartite graph, but for general graphs it will contain edges in $< X_i >$ which we call *loop edges*. The other edges of $St(i)$ are called *star edges* and the remaining edges of G' are called as *link edges*.

Illustration: This construction is illustrated in Figure 2.16. In the general graph G, $d(v_1) = 5$, $d(v_2) = 3$ and $d(v_3) = 2$. Let $f = (3, 2, 1)$ be the vertex function, so that $f(v_1) = 3$, $f(v_2) = 2$ and $f(v_3) = 1$. Therefore, $k_1 = 2$, $k_2 = 1$ and $k_3 = 1$. Here $|X_1| = 5$, $|X_2| = 3$ and $|X_3| = 2$, and $|Y_1| = 2$, $|Y_2| = 1$ and $|Y_3| = 1$.

Label the edges of G as 1, 2, 3, 4 and 5 as shown. Then $Y_1 = \{y_1^1, y_2^1\}$, $Y_2 = \{y_1^2\}$ and $Y_3 = \{y_1^3\}$, and we have $X_1 = \{x_1^1, x_{1'}^1, x_2^1, x_3^1, x_4^1\}$, $X_2 = \{x_2^2, x_3^2, x_5^2\}$ and $X_3 = \{x_4^3, x_5^3\}$. The vertex set of G' is $X_i \cup Y_i$, $1 \leq i \leq 3$. The edges of G' are $x_1^1 x_{1'}^1$, $x_2^1 x_2^2$, $x_3^1 x_3^2$, $x_4^1 x_4^3$, $x_5^2 x_5^3$ (shown by dotted lines) and the star edges (shown by bold lines). The dotted lines in G form an f-factor with $f = (3, 2, 1)$.

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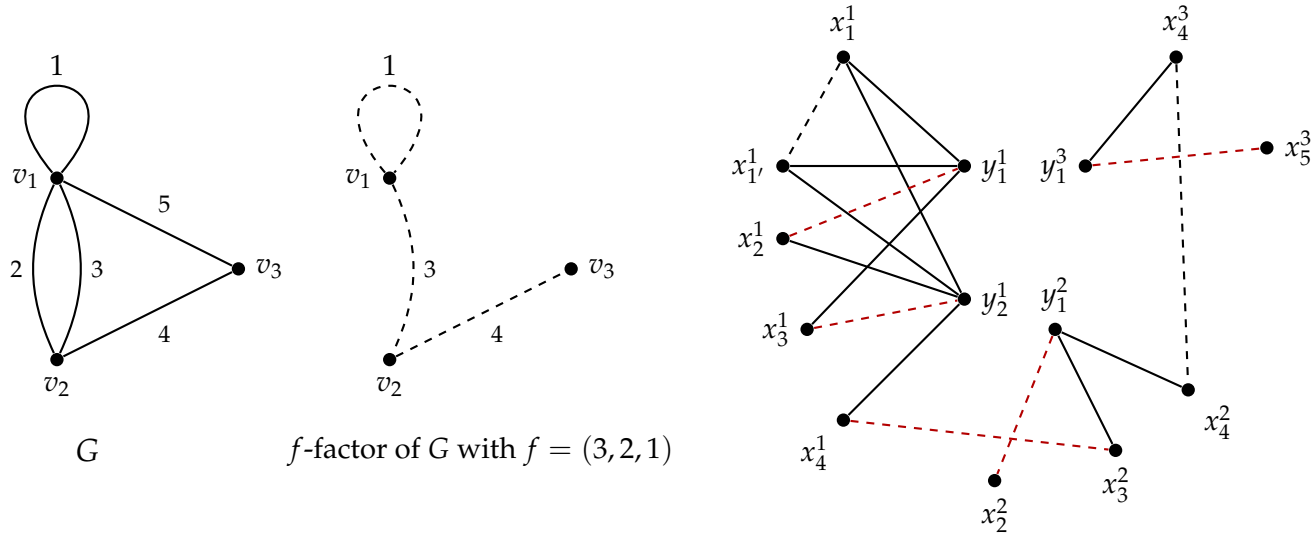


Figure 2.16: Dark dotted edges in G' represent edges induced by $f-$factor of G for a $1-$factor of G' and red dotted edges are additional edges required to produce a $1-$factor of G'.

Example 2.2.21 (Construction of G' for a Cycle C_4)**.** Let G be a cycle C_4 with vertices v_1, v_2, v_3, v_4. Let $f(v_i) = 1$ for all i. Thus, for every vertex v_i:

$$|X_i| = d(v_i) = 2 \quad \text{and} \quad |Y_i| = d(v_i) - f(v_i) = 1.$$

The construction is visualized below (Figure 2.17)The graph G' consists of 4 "corner" gadgets. Finding a 1-factor in G' corresponds to finding a 1-factor in G.

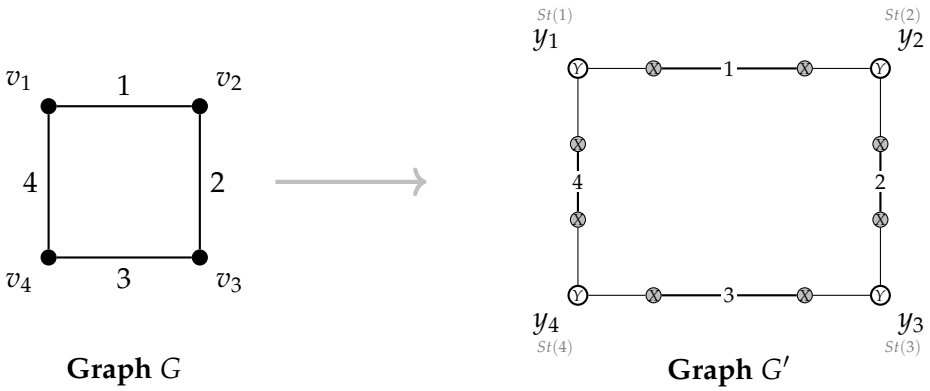


Figure 2.17: Transformation of C_4 into G' with $f(v) = 1$.

Theorem 2.2.22. *A general graph G has an f-factor if and only if G' has a 1-factor*

Proof. We prove the necessary and sufficient conditions separately.

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(⇒) Necessity Assume G has an f-factor F with link edges L and loop edges L^0. Let g_i denote the number of link edges and f_i^0 the number of loop edges of F incident at vertex v_i. Consequently, the degree in the factor is $f_i = g_i + 2f_i^0$.

Recall that for the constructed graph G', the size of the sets X_i and Y_i are defined as:

$$|X_i| = d_i = l_i + 2l_i^0 \quad \text{and} \quad |Y_i| = k_i = d_i - f_i,$$

where l_i is the number of link edges incident with v_i and l_i^0 is the number of loops incident with v_i in the original graph G. We can expand the expression for $|Y_i|$ as follows:

$$\begin{aligned} |Y_i| &= (l_i + 2l_i^0) - (g_i + 2f_i^0) \\ &= (l_i - g_i) + 2(l_i^0 - f_i^0). \end{aligned}$$

Let L' and $(L^0)'$ be the set of link and loop edges in G' corresponding to L and L^0. The matching consisting of these edges keeps all vertices of Y_i unsaturated. Furthermore, it leaves unsaturated a set of vertices in X_i with cardinality:

$$(l_i - g_i) + 2(l_i^0 - f_i^0) = k_i = |Y_i|.$$

We can now match the vertices of Y_i with these remaining unsaturated vertices of X_i using the star edges $St(i)$. Let S' be this set of edges. Clearly, the union $L' \cup (L^0)' \cup S'$ constitutes a 1-factor in G'.

(⇐) Sufficiency Conversely, assume G' has a 1-factor H. Since the vertices in Y_i can only be matched to vertices in X_i (by the construction of the star graphs), there are exactly

$$d_i - |Y_i| = d_i - (d_i - f_i) = f_i$$

vertices of X_i that are *not* matched by H to vertices in Y_i. Let us denote this set of vertices as X_i^*. Note that $f_i = g_i + 2f_i^0$.

Clearly, $2f_i^0$ vertices from X_i^* correspond to loop edges of the induced subgraph $\langle X_i \rangle$ and are thus matched in pairs amongst themselves. The remaining g_i vertices in X_i^* are matched by the link edges of H to vertices in X_j belonging to other star graphs $St(j)$ of G'.

If F is the set of edges in G corresponding to these loop edges and link edges selected by H, then the degree of v_i in the subgraph induced by F is:

$$d_{<F>}(v_i) = g_i + 2f_i^0 = f_i.$$

Thus, F is an f-factor of G. ■

Theorem 2.2.23 ($f-$factor theorem)**.** *A general graph G has an $f-$factor if and only if it has no decomposition D with $\delta(D, f) > 0$.*

OR

If f is a vertex function for a graph G, then either G has an $f-$factor or an $f-$barrier but not both.

Proof. Refer to "Introduction to Graph Theory" by S Pirzada. ■

Definition 2.2.24. Suppose $D = (S, T, U)$ is a decomposition of V and $x \in S$, $y \in T$. The decomposition $D_1 = (S \setminus \{x\}, T, U \cup \{x\})$ is said to be obtained by an **x-transfer from S**, and the decomposition $D_2 = (S, T \setminus \{y\}, U \cup \{y\})$ is said to be obtained by a **y-transfer from T**.

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Lemma 2.2.25. *Let $D = (S, T, U)$ be a decomposition of $V(G)$. An x-transfer from S does not reduce the deficiency of D if $f(x) = d(x)$ or $d(x) - 1$, and a y-transfer from T does not reduce the deficiency of D if $f(y) = 0$ or 1.*

Proof. Let $D' = (S - \{x\}, T, U \cup \{x\})$ be obtained from D by an x-transfer. Then

$$\delta(D', f) - \delta(D, f) \geq f(x) - d(x).$$

Thus the difference is at least 0 when $f(x) = d(x)$, and at least -1 when $f(x) = d(x) - 1$. Since the deficiencies of any two decompositions have the same parity, the latter difference cannot be -1. Hence

$$\delta(D', f) \geq \delta(D, f).$$

Similarly, if $D'' = (S, T - \{y\}, U \cup \{y\})$ is obtained by a y-transfer, then

$$\delta(D'', f) - \delta(D, f) \geq -f(y).$$

If $f(y) = 0$, this is nonnegative, while if $f(y) = 1$, it is at least -1. Again parity excludes the value -1. Therefore

$$\delta(D'', f) \geq \delta(D, f).$$

Hence neither transfer reduces the deficiency under the stated hypotheses. ■

Theorem 2.2.26. *The f-factor theorem implies the 1-factor theorem of Tutte.*

Proof. Take $f(v) = 1$ for every $v \in V(G)$. Then an f-factor is exactly a 1-factor.

Assume that $k_0(G - S) \leq |S|$ for every $S \subseteq V(G)$. Suppose, to the contrary, that G has no 1-factor. By the f-factor theorem, there exists a decomposition $D = (S, T, U)$ with positive deficiency.

Since $f(y) = 1$ for every $y \in T$, Lemma 2.2.25 allows us to transfer every vertex of T into U without decreasing the deficiency. Hence we obtain the decomposition

$$(S, \varnothing, V(G) \setminus S)$$

with positive deficiency.

For this decomposition, since $f(S) = |S|$, we have

$$\delta(D, 1) = k_0(G - S) - |S|.$$

Thus $k_0(G - S) > |S|$, contrary to the assumed condition. Therefore G has a 1-factor. Hence the f-factor theorem implies Tutte's 1-factor theorem. ■

Theorem 2.2.27 (Erdős–Gallai Theorem)**.** Let $d_1 \geq d_2 \geq \cdots \geq d_n \geq 0$ be a nonincreasing sequence of integers. Then $(d_1, d_2, \ldots, d_n)$ is graphical if and only if $\sum_{i=1}^{n} d_i$ is even and, for every $k = 1, 2, \ldots, n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min\{d_i, k\}.$$

Corollary 2.2.28. *The Erdős–Gallai theorem on degree sequences follows from the f-factor theorem.*

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Proof. Let $d_1 \geq d_2 \geq \cdots \geq d_n \geq 0$ and suppose that $\sum_{i=1}^{n} d_i \equiv 0 \pmod 2$. To obtain this condition, we observe that d is graphic if and only if the complete graph K_n has a $d-$factor. Let $V(K_n) = \{v_1, \ldots, v_n\}$ and define $f(v_i) = d_i$ for $1 \leq i \leq n$. Then $(d_1, \ldots, d_n)$ is graphic if and only if K_n has an f-factor.

By the f-factor theorem, K_n has an f-factor if and only if $\delta(D, f) \leq 0$ for every decomposition $D = (S, T, U)$ of $V(K_n)$, where

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Let $r = |T|$. Since every vertex of K_n has degree $n-1$, we have $d(T) = r(n-1)$. Also, every vertex of S is adjacent to every vertex of T, so $q[S, T] = |T||S| = r|S|$. Hence

$$\begin{aligned}\delta(D, f) &= k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T] \\ &= k'_0(D, f) - f(S) + f(T) - r(n-1) + r|T| \\ &= k'_0(D, f) + f(T) - f(S) - r(n - 1 - |S|)\end{aligned}$$

Now $K_n[U]$ is complete. Thus, if $U \neq \varnothing$, it has only one component, and consequently $k'_0(D, f) \in \{0, 1\}$. Moreover, $f(V(K_n)) = \sum_{i=1}^{n} d_i$ is even, and therefore $\delta(D, f)$ is even because $\delta(D, f)$ and $\sum f(V(G))$ always have the same parity. Hence the condition $\delta(D, f) \leq 0$ is equivalent to

$$f(T) \leq f(S) + r(n - 1 - |S|).$$

Since $U = V(K_n) \setminus (S \cup T)$, we have $n - 1 - |S| = (r - 1) + |U|$. Therefore,

$$f(T) \leq r(r - 1) + f(S) + r|U|.$$

For fixed T, the sets S and U partition $V(K_n) \setminus T$. For each $v \notin T$, placing v in S contributes $f(v)$, whereas placing v in U contributes r. Hence the minimum possible value of $f(S) + r|U|$ is $\sum_{v \notin T} \min\{f(v), r\}$. Thus the f-factor condition is equivalent to

$$f(T) \leq r(r - 1) + \sum_{v \notin T} \min\{f(v), r\}$$

for every $T \subseteq V(K_n)$ with $|T| = r$.

Taking $T = \{v_1, \ldots, v_r\}$ gives $f(T) = \sum_{i=1}^{r} d_i$ and $\sum_{v \notin T} \min\{f(v), r\} = \sum_{i=r+1}^{n} \min\{d_i, r\}$. Hence

$$\sum_{i=1}^{r} d_i \leq r(r - 1) + \sum_{i=r+1}^{n} \min\{d_i, r\},$$

which is the Erdős–Gallai inequality.

■

Theorem 2.2.29 (k-factor Theorem). *Let $d = [d_i]_1^n$ be a graphical sequence. If the sequence $d' = [d_i - k]_1^n$ is also graphical, then there exists a realization of d that contains a k-factor.*

Proof. Let G_1 be a realization of $d = [d_i]_1^n$ and let G_2 be a realization of $d' = [d_i - k]_1^n$, both on the same vertex set $V = \{v_1, \ldots, v_n\}$. Thus $d_{G_1}(v_i) = d_i$ and $d_{G_2}(v_i) = d_i - k$ for every i. Choose G_1 and G_2 so that $|E(G_1) \cap E(G_2)|$ is as large as possible. Put $E_1 = E(G_1)$ and $E_2 = E(G_2)$. We claim that $E_2 \subseteq E_1$. Once this is proved, the graph with edge set $E_1 \setminus E_2$ has degree $d_i - (d_i - k) = k$ at every vertex, and hence is a k-factor of G_1.

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We first record a switching observation. Suppose that $v_iv_j \in E_2 \setminus E_1$ and $v_iv_r, v_jv_s \in E_1 \setminus E_2$. Then $v_rv_s \in E_1 \setminus E_2$. Indeed, if $v_rv_s \in E_2$, then in G_2 we may replace the edges v_iv_j and v_rv_s by v_iv_r and v_jv_s. This preserves every vertex degree and increases $|E_1 \cap E_2|$, a contradiction. If $v_rv_s \notin E_1$, then in G_1 we may replace v_iv_r and v_jv_s by v_iv_j and v_rv_s. Again all degrees are preserved and the number of common edges increases. Hence $v_rv_s \in E_1 \setminus E_2$.

Now suppose, to the contrary, that $E_2 \setminus E_1 \neq \varnothing$. For each vertex u, let $t(u) = d_{E_2 \setminus E_1}(u)$. Since $d_{G_1}(u) - d_{G_2}(u) = k$, we have $d_{E_1 \setminus E_2}(u) = t(u) + k$. Choose a vertex v_1 for which $t(v_1)$ is maximum, and write $t = t(v_1) > 0$. Choose v_j such that $v_1v_j \in E_2 \setminus E_1$. Since $t(v_j) \geq 1$, we have $d_{E_1 \setminus E_2}(v_j) = t(v_j) + k \geq k + 1$. Hence there exists a vertex w such that $v_jw \in E_1 \setminus E_2$.

Let $A = N_{E_1 \setminus E_2}(v_1)$. Since $d_{E_1 \setminus E_2}(v_1) = t + k$, we have $|A| = t + k$. First suppose that $w \notin A$. For every $a \in A$, the switching observation applied to $v_1v_j \in E_2 \setminus E_1$ and $v_1a, v_jw \in E_1 \setminus E_2$ gives $aw \in E_1 \setminus E_2$. Thus w is incident with the $t + k$ edges joining it to the vertices of A, as well as the edge v_jw. Therefore $d_{E_1 \setminus E_2}(w) \geq t + k + 1$, and hence $t(w) \geq t + 1$, contradicting the maximality of t.

Now suppose that $w \in A$. For every $a \in A \setminus \{w\}$, the same switching observation gives $aw \in E_1 \setminus E_2$. In addition, both v_1w and v_jw belong to $E_1 \setminus E_2$. Hence $d_{E_1 \setminus E_2}(w) \geq (t + k - 1) + 2 = t + k + 1$, so again $t(w) \geq t + 1$, contradicting the maximality of t.

Therefore $E_2 \setminus E_1 = \varnothing$, and hence $E_2 \subseteq E_1$. Consequently, the graph $H = (V, E_1 \setminus E_2)$ satisfies $d_H(v_i) = d_i - (d_i - k) = k$ for every i. Hence H is a k-factor of G_1. ■

2.3 Factorization of Complete Graphs

Definition 2.3.1. A **factorization** of a graph G is a decomposition of the edges of G into edge-disjoint spanning subgraphs, known as factors. If the graph is decomposed into k-factors (where every vertex has degree k), it is called a **k-factorization**.

For the complete graph K_n, the type of factorization possible depends on the parity of n.

Case 1: n is Even (1-Factorization)

Consider a complete graph K_{2n}. It has $2n$ vertices and is $(2n-1)$-regular.

- A 1-factor is a perfect matching (regular of degree 1).
- Since the total degree of each vertex is $2n - 1$, K_{2n} can be decomposed into exactly $2n - 1$ perfect matchings.

Case 2: n is Odd (2-Factorization)

Consider a complete graph K_{2n+1}. It has $2n + 1$ vertices and is $2n$-regular.

- A 2-factor is a spanning subgraph where every vertex has degree 2 (a disjoint union of cycles).
- In the context of K_n, we are often interested in decomposing it into Hamiltonian cycles.
- Since the total degree is $2n$, K_{2n+1} can be decomposed into n Hamiltonian cycles.

Theorem 2.3.2. *For every positive integer n, the complete graph K_{2n} is 1-factorable.*

Proof. Let the vertex set be $V = \{0, 1, \ldots, 2n-2, 2n-1\}$. We designate the vertex $2n - 1$ as the fixed center and treat the remaining vertices $\{0, 1, \ldots, 2n - 2\}$ as elements of $\mathbb{Z}_{2n-1}$ (the integers modulo $2n - 1$).

We construct $2n - 1$ factors, denoted $F_0, F_1, \ldots, F_{2n-2}$. For each $i \in \{0, \ldots, 2n - 2\}$, define the set of edges F_i as:

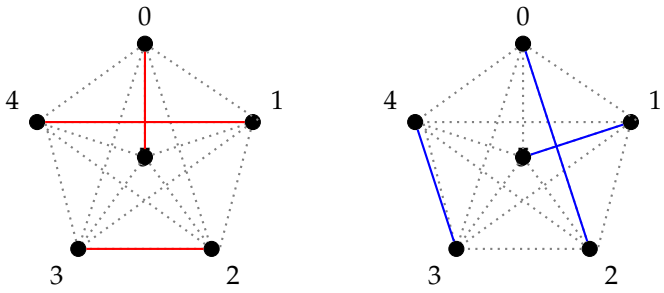
$$F_i = \{\{2n-1, i\}\} \cup \{\{i-k, i+k\} \pmod{2n-1} \mid k = 1, \ldots, n-1\}.$$

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1. F_i **is a 1-factor:** The fixed vertex $2n-1$ is matched to vertex i. The remaining $2n-2$ vertices form the polygon and are paired distinctly. The relation $i-k \equiv i+k \pmod{2n-1}$ implies $2k \equiv 0$, which is impossible for $1 \leq k \leq n-1$ since $2n-1$ is odd. Thus, every vertex has degree exactly 1 in F_i.
2. Decomposition:

- Every edge incident to the center $2n-1$ is of the form $\{2n-1, x\}$ and appears uniquely in factor F_x.
- For any edge $\{u, v\}$ not incident to $2n-1$, the sum of the indices modulo $2n-1$ is invariant for a fixed factor. Specifically, $u+v \equiv (i-k)+(i+k) \equiv 2i \pmod{2n-1}$. Since $2n-1$ is odd, $\gcd(2, 2n-1) = 1$, meaning 2 is invertible. Thus, $i \equiv (u+v) \cdot 2^{-1} \pmod{2n-1}$ is uniquely determined.

Therefore, every edge of K_{2n} belongs to exactly one factor, constituting a 1-factorization. ■



Factor F_0 **Factor** F_1

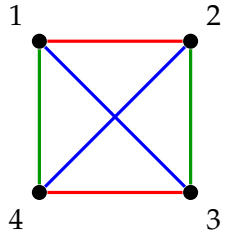
Figure 2.18: 1-Factorization of K_6 ($n = 3$). The solid colored lines represent the specific 1-factor, while the dotted lines represent the remaining edges of the complete graph.

Theorem 2.3.3. *For every positive integer n, the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Example 2.3.4 (1-Factorization of K_4)**.** Let $n = 4$. The graph K_4 has 4 vertices and degree 3. It can be decomposed into $4-1 = 3$ perfect matchings (1-factors).

The Factors:

- F_1 **(Red)**: Matches $\{(1,2),(3,4)\}$
- F_2 **(Blue)**: Matches $\{(1,3),(2,4)\}$
- F_3 **(Green)**: Matches $\{(1,4),(2,3)\}$



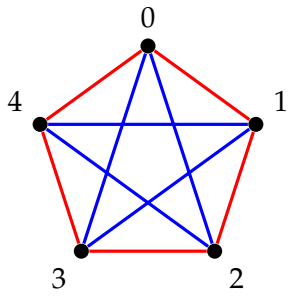
Decomposition of K_4

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Example 2.3.5 (2-Factorization of K_5)**.** Let $n = 5$. The graph K_5 has 5 vertices and degree 4. It can be decomposed into $(5-1)/2 = 2$ Hamiltonian cycles.

The Factors:

- H_1 **(Red)**: The outer pentagon cycle $\{(0,1),(1,2),(2,3),(3,4),(4,0)\}$.
- H_2 **(Blue)**: The inner star cycle $\{(0,2),(2,4),(4,1),(1,3),(3,0)\}$.



Decomposition of K_5

Theorem 2.3.6. *For every positive integer n, the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Proof. Let the vertex set be $V = \{0, 1, \ldots, 2n\}$. We designate the vertex $2n$ as the fixed center and treat the remaining set $\{0, \ldots, 2n-1\}$ as elements of $\mathbb{Z}_{2n}$.

We construct n Hamiltonian cycles $C_0, C_1, \ldots, C_{n-1}$ as follows:

1. The Base Cycle C_0: Define the cycle C_0 by the sequence of vertices:

$$C_0 = (2n,\ 0,\ 1,\ 2n-1,\ 2,\ 2n-2,\ \ldots,\ n,\ 2n).$$

The path within $\mathbb{Z}_{2n}$ (excluding the fixed vertex $2n$) zigzags through the vertices. The lengths of the edges in this path correspond to the distances $1, 2, \ldots, n$ in the modular arithmetic of $\mathbb{Z}_{2n}$.

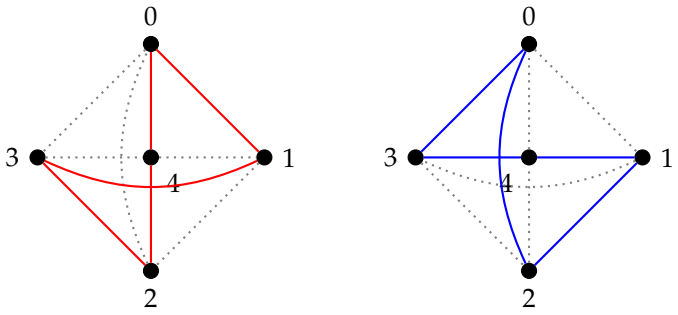
2. The General Cycles: For each $i \in \{0, \ldots, n-1\}$, let C_i be the cycle obtained by adding $i \pmod{2n}$ to every vertex in $\{0, \ldots, 2n-1\}$, while keeping the vertex $2n$ fixed.

3. Verification:

- **Edges incident to $2n$:** In cycle C_i, the vertex $2n$ connects to i and $n+i$. As i ranges from 0 to $n-1$, these pairs $\{i, n+i\}$ are disjoint and cover all vertices in $\mathbb{Z}_{2n}$. Thus, every edge incident to $2n$ appears exactly once.
- **Edges within $\mathbb{Z}_{2n}$:** The base path in C_0 contains exactly one edge of each "length" class $d \in \{1, \ldots, n\}$. By the properties of cyclic groups, rotating a set of edges that contains one representative of each length generates all possible edges within the complete graph induced by $\mathbb{Z}_{2n}$.

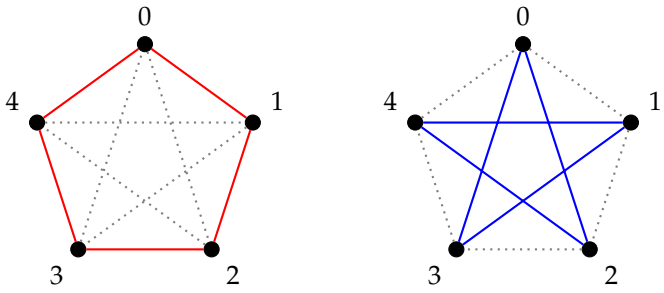
Thus, the collection $\{C_0, \ldots, C_{n-1}\}$ forms a 2-factorization of K_{2n+1}. ■

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Cycle C_0 **Cycle** C_1

Figure 2.19: Decomposition of K_5. We curve the diagonal edges (0-2 and 1-3) so they do not overlap with the central vertex 4. All 10 edges of K_5 are present in both figures (either solid or dotted).



Factor 1: Pentagon **Factor 2: Star**

Figure 2.20: Decomposition of K_5 into 2 Hamiltonian Cycles. The dotted lines in each figure represent the edges belonging to the other factor.

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CHAPTER

3

EDGE GRAPHS AND ECCENTRICITY SEQUENCES

The construction now commonly called the *line graph* has appeared in the literature under several names, including *edge graph*, *interchange graph*, *derivative graph*, and *adjoint graph*. We use the term *edge graph* and the standard notation $L(G)$. The central idea is to replace each edge of a graph by a vertex and to record incidences between edges as adjacencies between the new vertices.

3.1 Edge Graphs

Definition 3.1.1 (Edge Graph)**.** Let $G = (V, E)$ be a finite simple graph. The **edge graph** (or **line graph**) of G, denoted by $L(G)$, is the graph whose vertex set is $E(G)$, with two distinct vertices adjacent precisely when their corresponding edges in G have a common end-vertex. Thus,

$$V(L(G)) = E(G), \qquad e \sim_{L(G)} f \iff e \neq f \text{ and } e \cap f \neq \varnothing.$$

A graph F is called an **edge graph** if $F \cong L(H)$ for some graph H. Such a graph H is called a *root graph* of F. Figure 3.1 illustrates the construction.

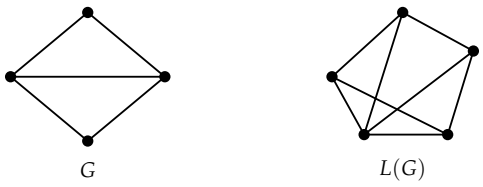


Figure 3.1: The graph G and its edge graph $L(G)$.

Throughout this section all graphs are finite and simple, unless stated otherwise. Isolated vertices of G do not appear in $L(G)$; consequently, statements that recover G from $L(G)$ require an assumption excluding isolated vertices. The following elementary properties will be used repeatedly.

1. If G has at least one edge and no isolated vertices, then G is connected if and only if $L(G)$ is connected. Indeed, a trail of consecutive edges in G corresponds to a path in $L(G)$.
2. If H is a subgraph of G, then $L(H)$ is a subgraph of $L(G)$.
3. For any $v \in V(G)$, the edges incident with v correspond to a clique of order $d_G(v)$ in $L(G)$. This clique need not be maximal (for example, consider a triangle in G).
4. If $e = uv \in E(G)$, then the neighbours of e in $L(G)$ correspond to the other edges incident with u or v. Because G is simple, these sets are disjoint. Hence $d_{L(G)}(e) = d_G(u) + d_G(v) - 2$.
5. The *iterated edge graphs* are defined by $L^0(G) = G$ and $L^r(G) = L(L^{r-1}(G))$ for $r \geq 1$.

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The following result determines the number of edges in an edge graph.

Theorem 3.1.2. *The number of edges m' in $L(G)$ when G has degree sequence $[d_i]_1^n$ is given by*

$$m' = \frac{1}{2}\left(\sum_{i=1}^{n} d_i^2\right) - m.$$

Proof. Let $[d_i]_1^n$ be the degree sequence of the graph G and let $L(G)$, the edge graph of G, have m' edges. As the degree of the vertex v_i in G is d_i, there are d_i edges incident on v_i. From these d_i edges, any two are adjacent at v_i in G. Hence the number of edges contributed by v_i to $L(G)$ is $\binom{d_i}{2}$. Thus

$$m' = \sum_{i=1}^{n} \binom{d_i}{2} = \sum_{i=1}^{n} \frac{d_i(d_i-1)}{2} = \frac{1}{2}\sum_{i=1}^{n}(d_i^2 - d_i) = \frac{1}{2}\sum_{i=1}^{n} d_i^2 - \frac{1}{2}\sum_{i=1}^{n} d_i = \frac{1}{2}\left(\sum_{i=1}^{n} d_i^2\right) - m.$$

■

The following observation is immediate.

Theorem 3.1.3. *For a graph G with at least one edge and no isolated vertices, $L(G)$ is a path if and only if G is a path.*

Proof. If $G = P_n$ with $n \geq 2$, its $n-1$ edges can be arranged consecutively; two of them share an endpoint exactly when they are consecutive. Hence $L(P_n) = P_{n-1}$ (where P_1 denotes the one-vertex path).

Conversely, suppose $L(G)$ is a path. Since a path is connected, the first observation above shows that G is connected. If some vertex of G had degree at least 3, three of its incident edges would give a triangle in $L(G)$, which is impossible. Thus $\Delta(G) \leq 2$. A finite connected graph of maximum degree at most 2 is a path or a cycle (including the single-edge case). A cycle C_n has $L(C_n) \cong C_n$, which is not a path. Therefore G must be a path. ■

We now have the following stronger result.

Theorem 3.1.4. *A finite connected simple graph G is isomorphic to $L(G)$ if and only if G is a cycle.*

Proof. Let G be a connected graph with n vertices, m edges and with degree sequence $[d_i]_1^n$. Let $L(G)$ be the edge graph of G. The number of vertices in $L(G)$ is m. The number of edges m' in $L(G)$ is given by

$$m' = \frac{1}{2}\left(\sum_{i=1}^{n} d_i^2\right) - m.$$

If $G = C_n$, its edges occur in a cyclic order, and consecutive edges are precisely the adjacent pairs. Thus $L(C_n) \cong C_n$.

Conversely, let $G \cong L(G)$. Then G and $L(G)$ have the same number of vertices and edges. So $n = m$ and

$$m = \frac{1}{2}\left(\sum_{i=1}^{n} d_i^2\right) - m.$$

Therefore $n = m$ and $\sum_{i=1}^{n} d_i^2 = 4m$. Using the handshaking identity $\sum_{i=1}^{n} d_i = 2m$, we obtain

$$\mathrm{Var}\{[d_i]\} = \frac{1}{n}\sum_{i=1}^{n} d_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} d_i\right)^2 = \frac{4m}{m} - \frac{4m^2}{m^2} = 0.$$

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Therefore the d_i's are equal and G is regular of degree d, say. So $nd = 2m$ implies that $d = 2m/n = 2$. Thus G is a 2-regular connected graph, that is, C_n. ■

We next examine how graph isomorphisms behave under the edge-graph construction.

Theorem 3.1.5. *If the graphs G_1 and G_2 are isomorphic, then $L(G_1)$ and $L(G_2)$ are also isomorphic.*

Proof. Assume (ϕ, θ) to be an isomorphism of G_1 onto G_2. Then θ is a bijection of $E(G_1)$ onto $E(G_2)$. We show that θ is an isomorphism of $L(G_1)$ to $L(G_2)$ by showing that θ preserves adjacency. Let e_i and e_j be two adjacent vertices of $L(G_1)$. So there exists a vertex v of G_1 incident to both e_i and e_j, and therefore $\phi(v)$ is a vertex incident to both $\theta(e_i)$ and $\theta(e_j)$. Thus $\theta(e_i)$ and $\theta(e_j)$ are adjacent vertices in $L(G_2)$.

Let $\theta(e_i)$ and $\theta(e_j)$ be adjacent vertices in $L(G_2)$. Then they are adjacent edges in G_2 and therefore there exists a vertex v' of G_2 incident to both $\theta(e_i)$ and $\theta(e_j)$. Then $\phi^{-1}(v')$ is a vertex of G_1 incident to both e_i and e_j, and thus e_i and e_j are adjacent vertices of $L(G_1)$. Therefore e_i and e_j are adjacent vertices of $L(G_1)$ if and only if $\theta(e_i)$ and $\theta(e_j)$ are adjacent vertices of $L(G_2)$. Hence θ is an isomorphism of $L(G_1)$ onto $L(G_2)$. ■

The converse fails in one remarkable case: $L(K_3) \cong K_3 \cong L(K_{1,3})$, although $K_3 \ncong K_{1,3}$. Whitney's theorem states that this is the only exceptional pair among connected simple graphs. The result expresses the near-uniqueness of the root graph of a connected edge graph.

Theorem 3.1.6 (Whitney)**.** *Let G and G' be connected graphs with isomorphic edge graphs. Then G and G' are isomorphic unless one is K_3 and the other is $K_{1,3}$.*

Proof. First suppose that $n(G)$ and $n(G')$ are less than or equal to 4. A necessary condition for $L(G)$ and $L(G')$ to be isomorphic is that $m(G) = m(G')$. The only nonisomorphic connected graphs on at most four vertices are those shown in Figure 3.2.

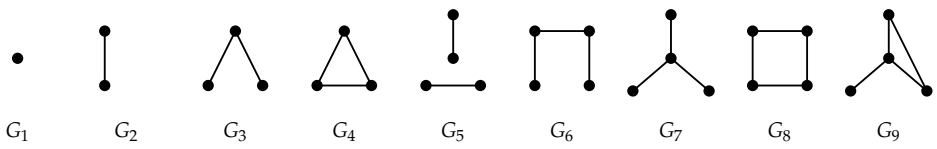


Figure 3.2: The connected graphs used in the proof of Whitney's theorem.

In Figure 3.2, graphs G_4, G_5 and G_6 are the three graphs having three edges each. We know that G_4 and G_6 have isomorphic edge graphs, namely K_3. The edge graph of G_5 is a path of length 2 and hence $L(G_5)$ cannot be isomorphic to $L(G_4)$ or $L(G_6)$. Further, G_7 and G_8 are the only two graphs in the list having four edges each. Clearly, $L(G_8) \cong G_8$ and $L(G_7)$ is isomorphic to G_9. Thus the edge graphs of G_7 and G_8 are not isomorphic. No two of the remaining graphs have the same number of edges. Hence the only non-isomorphic graphs with at most four vertices having isomorphic edge graphs are G_4 and G_6.

Now suppose that either G or G', say G, has at least five vertices and that $L(G)$ and $L(G')$ are isomorphic under an isomorphism ϕ_1. So ϕ_1 is a bijection from the edge set of G onto the edge set of G'.

We now prove that ϕ_1 transforms an induced $K_{1,3}$ subgraph of G onto a $K_{1,3}$ subgraph of G'. Let $e_1 = uv_1, e_2 = uv_2$ and $e_3 = uv_3$ be the edges of an induced $K_{1,3}$ subgraph of G. As G has at least five vertices and is connected, there exists an edge e adjacent to only one or all three of edges e_1, e_2 and e_3, as illustrated in Figure 3.3.

Clearly, $\phi_1(e_1), \phi_1(e_2)$ and $\phi_1(e_3)$ form either a $K_{1,3}$ subgraph or a triangle in G'. If $\phi_1(e_1), \phi_1(e_2)$ and $\phi_1(e_3)$ form a triangle in G', $\phi_1(e)$ can be adjacent to precisely two of $\phi_1(e_1), \phi_1(e_2)$ and $\phi_1(e_3)$, whereas $\phi_1(e)$ must be adjacent to only one or all the three. This contradiction shows that $\{\phi_1(e_1), \phi_1(e_2), \phi_1(e_3)\}$

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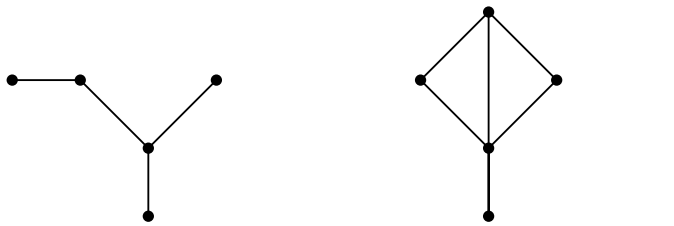


Figure 3.3: The configurations used in the proof of Whitney's theorem.

is not a triangle in G' and therefore forms a $K_{1,3}$ in G'. It is clear that a similar result holds as well for ϕ_1^{-1}, since it is an isomorphism on $L(G')$ onto $L(G)$.

Let $S(u)$ denote the star subgraph of G formed by the edges of G incident at a vertex u of G. We show that ϕ_1 maps $S(u)$ onto the unique star subgraph $S(u')$ of G'.

(i) First suppose the degree of u is at least 2. Let f_1 and f_2 be any two edges incident at u. The edges $\phi_1(f_1)$ and $\phi_1(f_2)$ of G' have an end vertex u' in common. If f is any other edge of G incident with u, then $\phi_1(f)$ is incident with u', and conversely, for every edge f' of G' incident with u', $\phi_1^{-1}(f')$ is incident with u. Thus $S(u)$ in G is mapped to $S(u')$ in G'.

(ii) Let the degree of u be 1 and $e = uv$ be the unique edge incident with u. As G is connected and $n(G) \geq 5$, degree of v must be at least 2 in G, and therefore, by (i), $S(v)$ is mapped to a star $S(v')$ in G'. Also $\phi_1(uv) = u'v'$ for some $u' \in V(G')$. If the degree of u' is greater than 1, by (i), the star at u' in G' is transformed by ϕ_1^{-1} either to the star at u in G or to the star at v in G. But as the star at v in G is mapped to the star at v' in G' by ϕ_1, ϕ_1^{-1} should map the star at u' to the star at u only. As ϕ_1^{-1} is one-one, this means that $\deg u \geq 2$, a contradiction. Therefore $\deg u' = 1$ and so $S(u)$ in G is mapped to $S(u')$ in G'.

Define $\phi : V(G) \to V(G')$ by setting $\phi(u) = u'$ if $\phi_1(S(u)) = S(u')$. Since $S(u) = S(v)$ only when $u = v$ ($G \neq K_2, G' \neq K_2$), ϕ is one-one. ϕ is also onto since, for v' in G', $\phi_1^{-1}(S(v')) = S(v)$ for some $v \in V(G)$, and by the definition of ϕ, $\phi(v) = v'$. Finally, if uv is an edge of G, then $\phi_1(uv)$ belongs to both $S(u')$ and $S(v')$, where $\phi_1(S(u)) = S(u')$ and $\phi_1(S(v)) = S(v')$. This means $u'v'$ is an edge of G'. But $u' = \phi(u)$ and $v' = \phi(v)$. Consequently, $\phi(u)\phi(v)$ is an edge of G'. If u and v are nonadjacent in G, $\phi(u)\phi(v)$ must be nonadjacent in G'. Otherwise, $\phi(u)\phi(v)$ belongs to both $S(\phi(u))$ and $S(\phi(v))$ and hence $\phi_1^{-1}(\phi(u)\phi(v)) = uv \in E(G)$, a contradiction. Thus G and G' are isomorphic under ϕ. ■

Not every graph is an edge graph. The following simple obstruction is fundamental to the structural characterizations developed below.

Lemma 3.1.7. *The star $K_{1,3}$ is not an edge graph.*

Proof. Suppose $K_{1,3} \cong L(H)$. Let x be its vertex of degree 3 and let a, b, c be the three pairwise nonadjacent neighbours of x. The vertices x, a, b, c correspond to four edges e_x, e_a, e_b, e_c of H. Each of e_a, e_b, e_c meets e_x, but no two of them meet one another. Since e_x has only two endpoints, at least two of e_a, e_b, e_c must meet e_x at the same endpoint. Those two edges would then be adjacent in $L(H)$, a contradiction. Hence $K_{1,3}$ is not an edge graph.

For comparison, Figure 3.4 shows some small possible root graphs with four edges.

Two examples are $L(C_4) \cong C_4$ and the line graph of a triangle with a pendant edge, which is isomorphic to $K_4 - e$. See Figure 3.5.

Figure 3.6 gives three further examples: P_5 has edge graph P_4; a tree with degree sequence $(3, 2, 1, 1, 1)$ has a triangle with a pendant vertex as its edge graph; and $K_{1,4}$ has edge graph K_4.

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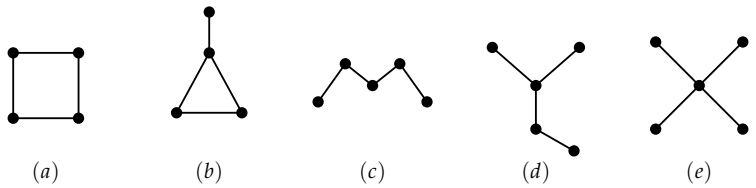


Figure 3.4: Examples of connected graphs with four edges.

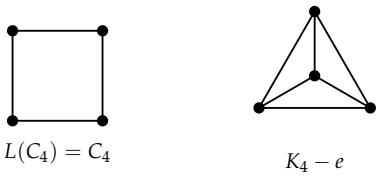


Figure 3.5: The edge graphs of the first two possibilities in Figure 3.4.

The diagrams provide further examples of edge graphs, but the preceding incidence argument establishes the assertion without requiring an exhaustive classification of four-edge graphs. ■

The first characterization, due to Krausz, translates the question of whether a graph is a line graph into a partition problem involving cliques. A *clique* is a set of pairwise adjacent vertices; here an *edge partition into cliques* means that every edge belongs to exactly one chosen clique.

Theorem 3.1.8 (Krausz). *A finite simple graph G is an edge graph if and only if its edge set can be partitioned into the edge sets of complete subgraphs in such a way that each vertex of G belongs to at most two of these complete subgraphs.*

Proof. **Necessity.** Suppose $G = L(H)$ for a finite simple graph H. For every vertex $u \in V(H)$, collect the vertices of G corresponding to edges of H incident with u. This set is a clique C_u. Every edge of G represents two distinct edges of H sharing a vertex, and because H is simple, their common vertex is unique. Therefore each edge of G belongs to exactly one clique C_u. Moreover, a vertex of G represents an edge uv of H, so it belongs to at most the two cliques C_u and C_v. Discard cliques of size less than 2, which contribute no edges. The required partition follows.

Sufficiency. Conversely, suppose the required clique partition is $S_1, \dots, S_k$. We first treat the case when G has no isolated vertices. Every vertex of G belongs to one or two of the chosen cliques. For each vertex v_j belonging to exactly one clique, adjoin a new singleton set $\{v_j\}$. Consider the family

$$S = \{S_1, S_2, \dots, S_k, \{v_1\}, \{v_2\}, \dots, \{v_\ell\}\},$$

and construct a graph H with one vertex for each member of S. For each vertex $x \in V(G)$, join the two vertices of H corresponding to the two sets of S containing x. These are distinct because the sets have been introduced separately. No two vertices $x, y \in V(G)$ can determine the same edge of H: otherwise x and y would lie together in two chosen cliques, and the edge xy of G would appear twice in the partition. Thus H is simple.

The edges of H correspond bijectively to the vertices of G. Two edges of H share an endpoint exactly when their corresponding vertices lie together in one of the selected cliques, which is equivalent to adjacency in G. Consequently $L(H) \cong G$. If G has isolated vertices, add one disjoint copy of K_2 to H for each such vertex; this produces the required isolated vertices in $L(H)$. ■

Krausz's theorem gives a useful constructive characterization. Van Rooij and Wilf developed a different description in terms of small induced subgraphs and the parity of triangles. We introduce the necessary terminology.

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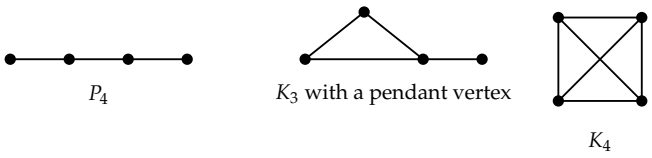


Figure 3.6: Edge graphs of the three tree possibilities in Figure 3.4.

Definition 3.1.9. For $X \subseteq V(G)$, the **induced subgraph** $G[X]$ contains every edge of G whose endpoints are both in X. A triangle T in G is called **odd** if some vertex $v \in V(G) \setminus V(T)$ has an odd number of neighbours in $V(T)$. Otherwise T is called **even**. Equivalently, T is even when every vertex outside T has either 0 or 2 neighbours in T. A **double triangle** is an induced subgraph isomorphic to $K_4 - e$; it consists of two triangles sharing one edge.

Theorem 3.1.10 (Van Rooij–Wilf). *A finite simple graph G is an edge graph if and only if G contains no induced $K_{1,3}$ and no induced double triangle in which both constituent triangles are odd.*

Proof. **Necessity.** Let $G = L(H)$. The graph G contains no induced $K_{1,3}$: an induced subgraph of a line graph is itself the line graph of the subgraph formed by the corresponding edges, whereas $K_{1,3}$ is not a line graph. Now we observe that the vertices of a double triangle in G correspond to the edges of a $K_{1,3} + e$ in H. In particular, one of these double triangles in G is generated by a triangle in H. Obviously a triangle in G generated by a triangle in H is even, since an edge incident to a triangle in H intersects exactly two edges of the triangle in H (Figure 3.7).

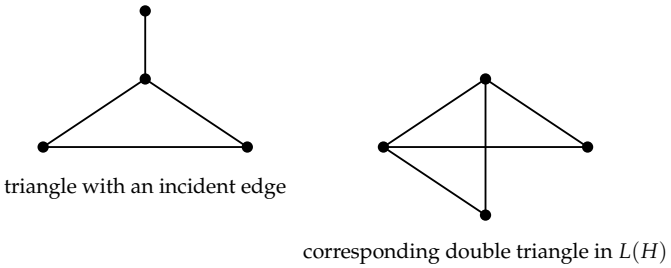


Figure 3.7: A triangle in an edge graph generated by a triangle in the original graph.

Sufficiency. Let G not contain an induced subgraph $K_{1,3}$ and let no double triangle of G have two odd triangles. Assume G is connected, for otherwise, we apply the construction to each component. In case G is $K_{1,3}$-free and has a double triangle with both triangles even, then G is one of the graphs given in Figure 3.8.

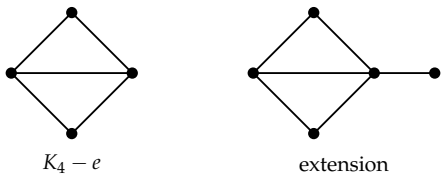


Figure 3.8: The possibilities when a $K_{1,3}$-free graph has a double triangle with both triangles even.

Thus we consider the case when every double triangle of G has exactly one odd triangle. To prove the result, it suffices by Theorem 3.1.8, to partition $E(G)$ into cliques that cover each vertex at most twice. Now, let $S_1, S_2, \ldots, S_k$ be the maximal cliques of G that are not even triangles and let $T_1, T_2, \ldots, T_\ell$

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be the edges that belong to one even triangle and no odd triangle. We claim that $\mathcal{B} = \{S_i\} \cup \{T_j\}$ partitions $E(G)$ into cliques using each vertex at most twice.

Now, every edge appears in a maximal clique, but every triangle in a clique with more than three vertices is odd. Therefore T_j is not in any clique S_i. Also S_i and $S_{i'}$ have no common edge, because G has no double triangles with both triangles odd. Thus the cliques in $\mathcal{B}$ are edge-disjoint. If $e \in E(G)$, then e belongs to some S_i unless the only maximal clique containing e is an even triangle. In this case e is a T_j, since the double triangles do not have both triangles even (Figure 3.9).

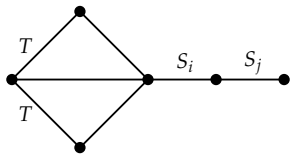


Figure 3.9: The clique configuration used in the proof of Theorem 3.1.10.

We now show that each $v \in G$ appears at most twice in $\mathcal{B}$. Assume v belongs to $B_1, B_2, B_3 \in \mathcal{B}$. Edge-disjointness implies that v has neighbours x, y, z with each belonging to only one of B_1, B_2, B_3. Since G has no induced $K_{1,3}$, assume that xy is an edge. Now by edge-disjointness, the triangle vxy does not belong to a member of $\mathcal{B}$. Therefore vxy is an even triangle. Thus z has exactly one other edge to vxy, say zx, while zy is not an edge. But now the same argument shows that zvx is an even triangle and we have a double triangle with both triangles even. This contradicts our supposition and hence each $v \in G$ appears at most twice in $\mathcal{B}$. ■

3.1.1 Beineke's Characterisation

The characterisations of Krausz and van Rooij–Wilf describe line graphs in terms of clique partitions and the structure of their induced subgraphs, respectively. A particularly useful consequence is that the class of line graphs admits a characterisation by a *finite list of forbidden induced subgraphs*.

Recall that a graph H is a *forbidden induced subgraph* for line graphs if no line graph contains an induced copy of H. Beineke's theorem states that exactly nine minimal obstructions are needed. The first is the claw $K_{1,3}$; the remaining eight arise from configurations involving two triangles with a common edge.

Theorem 3.1.11 (Beineke). *A graph G is the edge graph of a simple graph if and only if G contains no induced subgraph isomorphic to any of the nine graphs $G_1, \dots, G_9$ displayed in Figure 3.10.*

Proof. **Necessity.** Suppose $G = L(H)$ for a simple graph H. By Krausz's theorem (Theorem 3.1.8), the edges of G can be partitioned into cliques so that each vertex belongs to at most two cliques. None of the nine graphs in Figure 3.10 has such a partition. Moreover, an induced subgraph of $L(H)$ on a set $F \subseteq E(H)$ is precisely $L(H[F])$, where $H[F]$ is the edge-induced subgraph of H. Hence no line graph contains any of $G_1, \dots, G_9$ as an induced subgraph. Figure 3.10.

Sufficiency. Suppose that G has none of $G_1, \dots, G_9$ as an induced subgraph. In particular, G is *claw-free*, since $G_1 = K_{1,3}$. By the van Rooij–Wilf characterisation (Theorem 3.1.10), it is enough to prove that no induced diamond has two odd triangles.

Suppose the contrary. Let $D = G[\{a, b, c, d\}]$ be a diamond with $ab, ac, bc, ad, bd \in E(G)$ and $cd \notin E(G)$. Its two triangles abc and abd are assumed odd. A *witness* to the oddness of a triangle means a vertex outside that triangle adjacent to exactly one or all three of its vertices. Notice that c and d do not themselves witness oddness of the opposite triangle: each is adjacent to the two vertices a, b.

Case 1: One vertex witnesses both triangles. Let $x \notin D$ be adjacent to an odd number of vertices in each of abc and abd. Put $s = |N(x) \cap \{a, b\}|$. Both $s + \mathbf{1}_{xc \in E(G)}$ and $s + \mathbf{1}_{xd \in E(G)}$ are odd. Therefore

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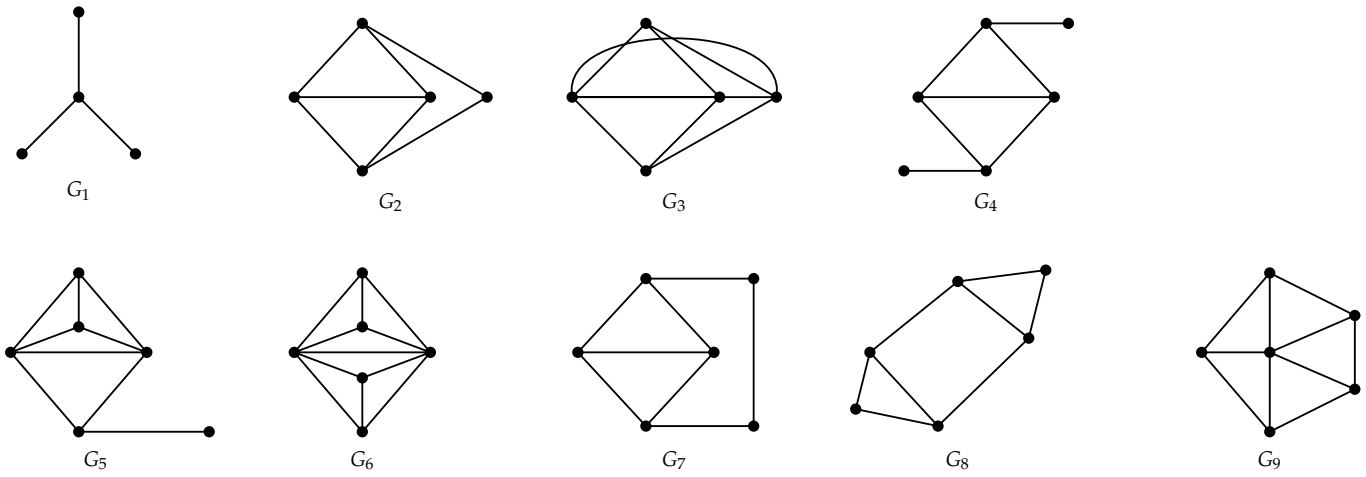


Figure 3.10: The nine forbidden induced subgraphs in Beineke's characterisation of line graphs.

there are only three possibilities:

(i) If $s = 1$, then $xc, xd \notin E(G)$. If, for example, $xa \in E(G)$, then a is adjacent to the pairwise nonadjacent vertices c, d, x. Thus $\{a, c, d, x\}$ induces the claw G_1.

(ii) If $s = 0$, then $xc, xd \in E(G)$; the diamond together with x induces G_2.

(iii) If $s = 2$, then $xc, xd \in E(G)$; hence x is adjacent to all four diamond vertices and these five vertices induce G_3.

All three possibilities contradict our assumption. The three configurations are displayed in Figure 3.11.

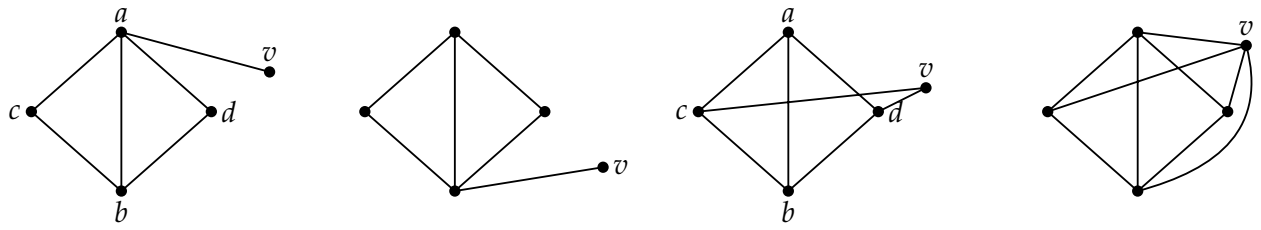


Figure 3.11: The configurations arising in Case 1 of the proof of Beineke's theorem.

Case 2: The triangles have no common witness. Choose $u, v \notin D$ so that u witnesses abc and v witnesses abd. They are distinct, and, by the present assumption, u has an even number of neighbours in abd while v has an even number in abc. Thus

$$|N(u) \cap \{a, b, c\}| \in \{1, 3\}, \quad |N(u) \cap \{a, b, d\}| \in \{0, 2\},$$
$$|N(v) \cap \{a, b, d\}| \in \{1, 3\}, \quad |N(v) \cap \{a, b, c\}| \in \{0, 2\}.$$

The following two observations make the analysis short. First, a witness adjacent to a or b cannot be adjacent to neither c nor d: otherwise that shared diamond vertex, its two nonadjacent diamond neighbours, and the witness form an induced claw. Second, no witness can be adjacent to both c and d: the parity conditions would make it adjacent to neither or both of a, b, and then it would be a common witness (Case 1). These facts explain all forced additional edges below.

Case 2.1: Each witness sees exactly one vertex of its triangle. By swapping a, b and simultaneously swapping c, d and u, v where appropriate, it suffices to consider four pairs of initial edges. Any further adjacency to D is determined by parity and claw-freeness. The complete possibilities are recorded in Figure 3.12.

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(i) $uc, vd \in E(G)$. Neither witness sees another vertex of D. If uv is an edge, the six vertices induce G_4; otherwise they induce G_7.

(ii) $ub, vd \in E(G)$. To avoid the claw centred at b, we must have $ud \in E(G)$; parity gives $vc \notin E(G)$. If $uv \notin E(G)$, the four vertices $\{a, d, u, v\}$ induce G_1 (with centre d). If $uv \in E(G)$, all six vertices induce G_8.

(iii) $ub, va \in E(G)$. Claw-freeness forces $ud, vc \in E(G)$. If $uv \notin E(G)$, the six vertices induce G_8; if $uv \in E(G)$, they induce G_2 (up to isomorphism).

(iv) $ub, vb \in E(G)$. Again $ud, vc \in E(G)$. If $uv \notin E(G)$, $\{a, b, u, v\}$ induces the claw G_1; if $uv \in E(G)$, the six vertices induce G_9.

In every possibility we obtain a forbidden induced subgraph.

Case 2.2: One witness sees three vertices, the other one. Interchanging the two triangles if necessary, assume $ua, ub, uc \in E(G)$. Since u is not odd for abd, $ud \notin E(G)$. By symmetry, the sole neighbour of v in abd is either d or b.

(i) If $vd \in E(G)$, parity forces $vc \notin E(G)$. The six vertices induce G_5 when uv is absent; when uv is present, the induced subgraph on $\{a, b, c, u, v\}$ is G_2.

(ii) If $vb \in E(G)$, parity forces $vc \in E(G)$. If uv is absent, $\{b, d, u, v\}$ induces a claw (centre b). If uv is present, $\{a, b, c, u, v\}$ induces G_3.

These four configurations appear in Figure 3.13.

Case 2.3: Both witnesses see three vertices. We have $ua, ub, uc, va, vb, vd \in E(G)$. As neither witness is odd for the other triangle, $ud, vc \notin E(G)$. If uv is absent, the six vertices induce G_6. If uv is present, $\{a, b, c, u, v\}$ induces G_3. See Figure 3.13.

Thus each case yields one of the nine forbidden induced subgraphs, a contradiction. No diamond of G can have both triangles odd. Together with the absence of an induced claw, the van Rooij–Wilf characterisation implies that G is a line graph. ■

> *Remark* 3.1.12. The preceding sufficiency argument uses the finite case classification underlying Beineke's theorem; it is a proof outline rather than an enumeration of every adjacency pattern. The forbidden-subgraph criterion is especially useful because it reduces recognition of line graphs to the exclusion of nine explicitly specified graphs.

3.1.2 Edge Graphs of Trees

A *block* of a graph is a maximal connected subgraph with no cut vertex. In particular, a bridge together with its endvertices constitutes a block. A *block graph* is a graph in which every block is complete. The following theorem characterises the line graphs of trees in terms of their block structure.

Theorem 3.1.13 (Chartrand). *A graph G is the edge graph of a tree if and only if G is a connected block graph in which every cut vertex belongs to exactly two blocks.*

Proof. **Necessity.** Suppose $G = L(T)$, where T is a tree. The edges incident with a vertex u of T form a clique in $L(T)$. Since T has no cycles, these cliques (when they contain at least two vertices) are precisely the nontrivial blocks of G. Thus each block of G is complete. As T is connected, so is G (including the case $T = K_2$, for which $G = K_1$).

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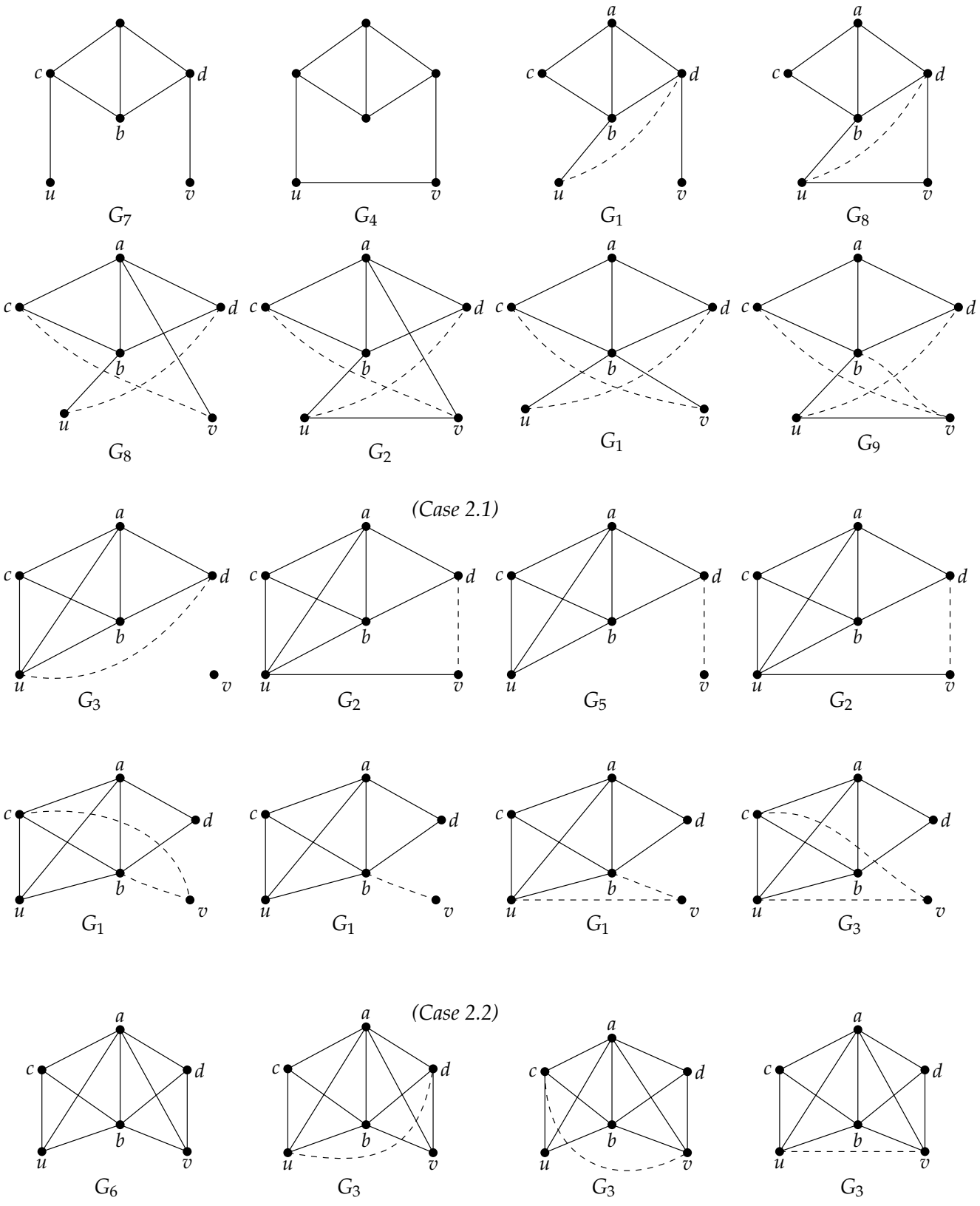


Figure 3.13: Bieneke's Theorem

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Let $e = uv$ be an edge of T. The corresponding vertex of G is a cut vertex if and only if both u and v have degree at least two in T. In this case, it belongs to exactly two blocks: the cliques formed by the edges incident with u and v. Hence G has the required properties.

Sufficiency. Conversely, let G be a connected block graph in which every cut vertex belongs to exactly two blocks. We construct a tree T such that $L(T) \cong G$.

First suppose $G = K_1$. Taking $T = K_2$ gives the required result. Otherwise, introduce a vertex b_B for each block B of G. For every cut vertex x of G lying in blocks B and C, join b_B to b_C by an edge e_x. For each non-cut vertex y of G, lying in its unique block B, introduce a new leaf ℓ_y and add the edge $e_y = b_B\ell_y$.

The block–cut-vertex incidence graph of a connected graph is a tree. Because every cut vertex of G lies in precisely two blocks, replacing each cut-vertex node of that incidence tree by an edge produces a tree on the block vertices b_B. Attaching the leaves ℓ_y preserves the tree property, so the resulting graph T is a tree.

The assignment $x \mapsto e_x$ is a bijection from $V(G)$ onto $E(T)$. Two distinct vertices x, y of a block graph are adjacent precisely when they lie in a common block. By construction, this happens exactly when the edges e_x and e_y of T share an endvertex. Therefore adjacency is preserved in both directions, and $G \cong L(T)$. ■

The construction in the proof provides a direct method for recovering a tree from its edge graph: represent each block by a vertex, replace each cut vertex shared by two blocks with an edge between their representatives, and attach a leaf for each non-cut vertex.



(a) Block graph G

(b) Tree T

Figure 3.14: A block graph formed by two triangles sharing one vertex, and a tree T whose edge graph is isomorphic to it.

The edge graphs of complete graphs were independently characterised by Chang and Hoffman; the edge graphs of complete bipartite graphs were characterised by Moon and Hoffman.

3.2 Edge Graphs and Traversability

The relationship between traversability in a graph and Hamiltonicity in its edge graph is particularly useful. Throughout this section, graphs are finite and simple, and a Hamiltonian graph is understood to contain a Hamiltonian cycle (of length at least three). We write $L^0(G) = G$ and $L^{r+1}(G) = L(L^r(G))$.

Theorem 3.2.1. *If G is an Eulerian graph with at least three edges, then $L(G)$ is both Eulerian and Hamiltonian.*

Proof. Let $e_1, e_2, \ldots, e_m$ be the edges encountered, in order, on an Eulerian circuit of G. Every two consecutive edges, including e_m and e_1, share an endvertex. Consequently,

$$e_1 e_2 \cdots e_m e_1$$

is a Hamiltonian cycle of $L(G)$, where the edges of G are regarded as vertices of $L(G)$.

For any edge $e = uv$ of G, the degree of the corresponding vertex of $L(G)$ is

$$d_{L(G)}(e) = d_G(u) + d_G(v) - 2.$$

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As G is Eulerian, both degrees on the right are even, so every vertex of $L(G)$ has even degree. Moreover, $L(G)$ is connected because G is connected and has an edge. Thus $L(G)$ is Eulerian. ■

The converse fails, even for connected graphs. Indeed, $L(K_{1,3}) \cong K_3$ is both Eulerian and Hamiltonian, whereas the three leaves of $K_{1,3}$ have odd degree.

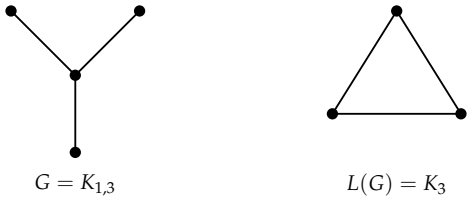


Figure 3.15: The edge graph can be Eulerian and Hamiltonian although the original graph is not Eulerian.

Definition 3.2.2 (Dominating Closed Trail)**.** A *dominating closed trail* of G is a closed trail W such that every edge of G has at least one endvertex on W. We also allow a single vertex as a closed trail of length zero; it dominates G precisely when it is incident with every edge of G. Equivalently, G has a connected Eulerian subgraph (possibly a single vertex) whose vertex set meets every edge of G.

It is essential here that the edges of the trail are not repeated. A closed *walk* may repeat edges and is not an appropriate replacement in the characterization below: every connected graph has a closed walk visiting all its vertices.

For example, the four edges of the cycle $v_1v_2v_3v_4v_1$ in Figure 3.16 form a dominating closed trail: the two edges attached to the cycle each have an endvertex on it.

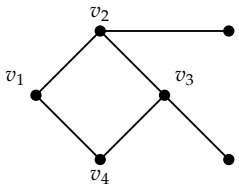


Figure 3.16: The cycle $v_1v_2v_3v_4v_1$ is a dominating closed trail.

Theorem 3.2.3 (Harary–Nash-Williams)**.** *Let G be a graph with at least three edges. Then $L(G)$ is Hamiltonian if and only if G has a dominating closed trail.*

Proof. Suppose first that G has a dominating closed trail W. If W consists of a single vertex v, all edges of G are incident with v, so $L(G)$ is a complete graph and hence is Hamiltonian. Otherwise, write the edges of W, in their cyclic order, as $e_1, e_2, \ldots, e_k$. Each edge not belonging to W has an endpoint on W. Assign each such edge to one of its endpoints on W. At each occurrence of a vertex v between two successive edges of W, insert all edges assigned to v between these two edges. All edges in the resulting cyclic sequence are distinct, and consecutive edges share an endpoint. The sequence therefore gives a Hamiltonian cycle in $L(G)$.

Conversely, let $e_1, e_2, \ldots, e_m$ be the cyclic ordering of the edges of G arising from a Hamiltonian cycle in $L(G)$. For each i choose a common endpoint x_i of e_i and e_{i+1}, with indices taken modulo m. Retain precisely those edges e_i for which $x_{i-1} \neq x_i$. Every retained edge joins x_{i-1} to x_i. If there is at least one retained edge, the retained edges, read in their cyclic order, form a closed trail: whenever

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intervening edges are omitted, all their corresponding transition vertices coincide. Each omitted edge has an endpoint at that common transition vertex, which lies on the retained trail. Hence the trail dominates every edge of G.

If no edge is retained, then $x_{i-1} = x_i$ for every i; thus all transition vertices equal one vertex v. Every e_i is incident with v, and the single-vertex trail v dominates G. ■

Corollary 3.2.4. *If G is Hamiltonian, then $L(G)$ is Hamiltonian.*

Proof. A Hamiltonian cycle of G meets every vertex and hence dominates all edges of G. Apply Theorem 3.2.3. ■

The converse is false: the cycle with two pendant edges shown in Figure 3.17 is not Hamiltonian, whereas its cycle is a dominating closed trail, so its edge graph is Hamiltonian.

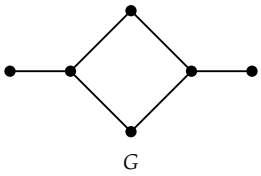


Figure 3.17: A non-Hamiltonian graph with a Hamiltonian edge graph.

Lemma 3.2.5. *If G is connected, has at least one edge, and every edge of G lies in a triangle, then G has a dominating closed trail.*

Proof. Among the connected Eulerian subgraphs of G, allowing a single vertex, choose one, H, with as many vertices as possible. Such a subgraph exists because G contains a triangle. Suppose there is an edge uv with $u, v \notin V(H)$. As G is connected, a path from H to uv contains an edge xy with $x \in V(H)$ and $y \notin V(H)$. Choose a triangle xyz containing xy.

If $z \notin V(H)$, the union of H with the triangle xyz is a larger connected Eulerian subgraph, a contradiction. If $z \in V(H)$, adjoining the two edges xy and yz to H would make x and z odd. Instead, if $xz \in E(H)$, replace H by the symmetric difference of its edges with the triangle; this remains connected because an edge of an Eulerian graph is not a bridge. If $xz \notin E(H)$, add the entire triangle. In either case the resulting connected even subgraph contains y, again a contradiction. Therefore every edge has an endpoint in $V(H)$, and an Eulerian circuit of H is dominating. ■

Corollary 3.2.6. *If G is connected, has at least one edge, and every edge of G belongs to a triangle, then $L(G)$ is Hamiltonian.*

Proof. By Lemma 3.2.5, G has a dominating closed trail. The assertion follows from Theorem 3.2.3. ■

Theorem 3.2.7 (Chartrand–Wall). *If G is connected and $\delta(G) \geq 3$, then $L^2(G)$ is Hamiltonian.*

Proof. Two adjacent vertices of $L(G)$ represent two edges of G sharing a vertex v. Since $d_G(v) \geq 3$, a third edge at v gives a triangle in $L(G)$ containing the given edge. Thus every edge of the connected graph $L(G)$ lies in a triangle. Apply Corollary 3.2.6 to $L(G)$. ■

Theorem 3.2.8 (Nebeský). *If G is connected and has at least three vertices, then $L(G^2)$ is Hamiltonian.*

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Proof. Recall that distinct vertices are adjacent in G^2 if their distance in G is at most two. We show that every edge uv of G^2 belongs to a triangle. If $d_G(u,v) = 2$, take the middle vertex w of a shortest u–v path; then u, v, w form a triangle in G^2. If $uv \in E(G)$, choose a third vertex w which is adjacent in G to u or v. Such a vertex exists because G is connected and has at least three vertices. Then $uw, vw \in E(G^2)$, so uvw is a triangle in G^2. Since G^2 is connected, Corollary 3.2.6 now applies. ■

Theorem 3.2.9. *Let G be a connected graph, and let e_1, e_2 be distinct edges such that $G - \{e_1, e_2\}$ is connected. Then G has a spanning walk whose first edge is e_1 and whose last edge is e_2.*

Proof. Write $e_1 = ab$ and $e_2 = cd$, and put $H = G - \{e_1, e_2\}$. Begin at a and traverse e_1 to b. Because H is connected, there exists a walk in H beginning at b, visiting every vertex of H, and ending at c. Append this walk, and finally traverse e_2 from c to d. The resulting walk visits every vertex of G and has the prescribed initial and terminal edges. ■

Remark 3.2.10. In the preceding theorem, "walk" is meant literally: vertices and edges may be repeated. The statement does not generally remain true if "walk" is replaced by "trail". Notice also that the hypothesis about every edge belonging to a triangle, sometimes included in statements of this result, is unnecessary.

Theorem 3.2.11 (Jaeger). *If G is a 4-edge-connected graph, then $L(G)$ is Hamiltonian. In particular, this holds for every 4-vertex-connected graph.*

Proof. The spanning-tree packing theorem implies that a 4-edge-connected graph has two edge-disjoint spanning trees T_1 and T_2. Let S be the set of odd-degree vertices of T_1; the handshaking lemma shows that $|S|$ is even.

We use the following elementary parity property of a tree: for every even set S of its vertices, there is a unique set $F \subseteq E(T_2)$ such that the vertices of odd degree in $(V(G), F)$ are exactly the vertices of S. Indeed, delete an edge f of T_2 and include f in F exactly when one of the two resulting components contains an odd number of vertices of S. The parity of the degree at each vertex then has the required value.

Since T_1 and T_2 are edge-disjoint, the subgraph

$$H = (V(G), E(T_1) \cup F)$$

is connected and spanning, and each of its vertices has even degree. Thus H has an Eulerian circuit, which is necessarily a dominating closed trail of G. By Theorem 3.2.3, $L(G)$ is Hamiltonian. ■

We next distinguish two constructions involving line graphs. Let $S_r(G)$ denote the graph obtained by inserting exactly r new degree-two vertices in every edge of G, with $S_0(G) = G$. Define

$$L_r(G) = L(S_{r-1}(G)), \qquad r \geq 1.$$

Thus $L_1(G) = L(G)$, whereas $L_2(G) = L(S_1(G))$ need not be isomorphic to the iterated line graph $L^2(G)$. (See Figure 3.18

Theorem 3.2.12 (Harary–Nash-Williams). *If G is Hamiltonian, then $L_2(G)$ is Hamiltonian. Conversely, if $L_2(G)$ is Hamiltonian, then $L(G)$ is Hamiltonian.*

Proof. Let C be a Hamiltonian cycle of G. Subdividing each edge of C once gives a cycle $S_1(C)$ in $S_1(G)$. Every edge of $S_1(G)$ is incident with an original vertex of G, and every such vertex belongs

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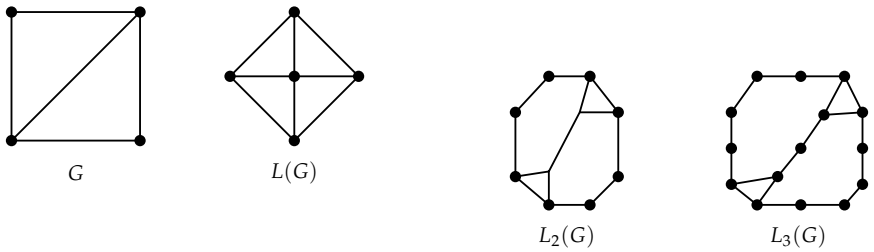


Figure 3.18: Successive subdivisions of a graph.

to $S_1(C)$. Thus $S_1(C)$ is a dominating closed trail of $S_1(G)$, and $L_2(G) = L(S_1(G))$ is Hamiltonian by Theorem 3.2.3.

For the converse, let $H = S_1(G)$ and let W be a dominating closed trail of H. If W is a single vertex, H is a star; for a simple graph G with at least three edges this cannot occur when every edge has been subdivided. Hence W contains an edge. Each subdividing vertex of H has degree two, so whenever W passes through one, it uses both incident edges. Consequently W is obtained by subdividing every edge of a closed trail W' in G. If an original edge uv of G does not lie on W', its two half-edges in H can be dominated only if u and v, respectively, lie on W'. In particular, W' dominates every edge of G. Thus $L(G)$ is Hamiltonian by Theorem 3.2.3. ■

Corollary 3.2.13. *Let G be a connected graph with at least three edges. Then G is Eulerian if and only if $L_3(G)$ is Hamiltonian.*

Proof. If G is Eulerian, subdividing each edge twice preserves every vertex degree and connectedness, so $S_2(G)$ is Eulerian. Theorem 3.2.1 shows that $L_3(G) = L(S_2(G))$ is Hamiltonian.

Conversely, assume $L_3(G)$ is Hamiltonian. By Theorem 3.2.3, $S_2(G)$ has a dominating closed trail W. Consider an original edge uv replaced by the path $u - x - y - v$, where x, y are its two new degree-two vertices. Since the middle edge xy must be dominated, at least one of x, y lies on W. Because a nontrivial closed trail uses both edges incident with every degree-two vertex it visits, it must traverse the entire path $u - x - y - v$. Therefore W traverses all the edges of $S_2(G)$, so this subdivision graph is Eulerian. Its original vertices have the same degrees as in G; hence all degrees in G are even. As G is connected, G is Eulerian. ■

Theorem 3.2.14 (Chartrand). *Let G be a connected graph of order $n \geq 3$ that is not a path. Then $L^k(G)$ is Hamiltonian for every integer $k \geq n - 3$.*

Proof. If $\Delta(G) = 2$, then, since G is connected and is not a path, G must be a cycle. Consequently, $L(G) \cong G$, and hence $L^k(G)$ is Hamiltonian for every $k \geq 0$.

Now suppose that $\Delta(G) \geq 3$. Recall that the *Hamiltonian index* $h(G)$ is the least nonnegative integer r for which $L^r(G)$ is Hamiltonian.

By the theorem of Saražin, every connected graph G that is not a path satisfies

$$h(G) \leq |V(G)| - \Delta(G).$$

Since $|V(G)| = n$ and $\Delta(G) \geq 3$, we obtain

$$h(G) \leq n - \Delta(G) \leq n - 3.$$

Thus $L^{n-3}(G)$ is Hamiltonian.

Finally, recall that the edge graph of a Hamiltonian graph is Hamiltonian. Therefore, once an iterated edge graph of G is Hamiltonian, every subsequent iterated edge graph is also Hamiltonian. It follows that $L^k(G)$ is Hamiltonian for every integer $k \geq n - 3$. ■

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Remark 3.2.15. This is Chartrand's classical bound for the Hamiltonian index of a connected graph. Its proof requires a separate iterative line-graph argument and does not follow directly from the preceding subdivision results; it should not be replaced by an unjustified one-line appeal to those results. A full proof may be inserted here after the necessary auxiliary lemmas on iterated line graphs have been established.

3.3 Total Graphs

A total graph combines vertex adjacency, edge adjacency, and vertex–edge incidence within a single graph. Throughout this section, graphs are finite and simple.

Definition 3.3.1 (Total Graph)**.** Let $G = (V, E)$ be a graph. The *total graph* of G, denoted by $T(G)$, has vertex set $V(T(G)) = V \cup E$, where each edge of G is regarded as a new vertex. Two distinct vertices of $T(G)$ are adjacent if and only if one of the following holds:

(i) they are vertices of G that are adjacent in G;

(ii) one is a vertex of G and the other an edge of G incident with it;

(iii) they are edges of G having a common end vertex.

Thus $T(G)$ records the three natural relationships among the vertices and edges of G. Figure 3.19 illustrates the construction for $G = K_3$. The filled vertices represent vertices of G, while the unfilled vertices represent its edges.

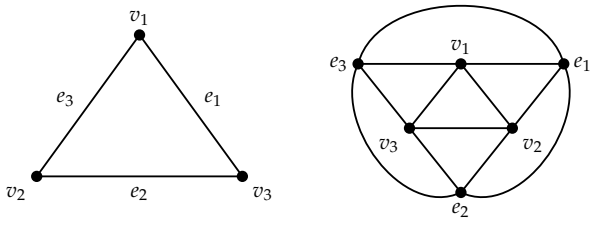


Figure 3.19: A graph $G = K_3$ and its total graph $T(G)$.

Remark 3.3.2. The subgraphs of $T(G)$ induced by V and by E are G and $L(G)$, respectively. Moreover, the edges joining V to E form the incidence graph of G, which is isomorphic to its subdivision graph $S(G)$. Indeed, the vertex of $T(G)$ corresponding to an edge uv is joined to u and v, giving the two edges $u(uv)$ and $(uv)v$ of $S(G)$.

3.4 Eccentricity Sequences and Sets

Recall that the *eccentricity* of a vertex v in a connected graph G is $e_G(v) = \max\{d_G(v, u) : u \in V(G)\}$. The radius and diameter are $r(G) = \min_{v \in V(G)} e_G(v)$ and $\operatorname{diam}(G) = \max_{v \in V(G)} e_G(v)$, respectively. Unless otherwise stated, the graphs in this section are finite, connected, and have at least two vertices.

Definition 3.4.1 (Eccentricity Sequence and Eccentricity Set)**.** Let G have vertices $v_1, \ldots, v_n$. The

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nondecreasing arrangement

$$e_1 \leq e_2 \leq \cdots \leq e_n$$

of the eccentricities $e_G(v_1), \ldots, e_G(v_n)$ is called the *eccentricity sequence* of G. The set of its distinct entries is the *eccentricity set* of G. A sequence or a set is said to be *realizable* if it is the eccentricity sequence or eccentricity set of some graph, respectively.

If the eccentricity set is $\{a_1, \ldots, a_t\}$, where $a_1 < \cdots < a_t$, and a_j occurs n_j times, the sequence may be written in the compact form

$$[a_1^{n_1}, a_2^{n_2}, \ldots, a_t^{n_t}].$$

Here the exponents indicate multiplicities, not powers. The sequence $(n_1, \ldots, n_t)$ is called the *eccentricity frequency sequence*. In particular, $a_1 = r(G)$ and $a_t = \operatorname{diam}(G)$. The familiar inequalities

$$r(G) \leq \operatorname{diam}(G) \leq 2r(G)$$

therefore give the necessary condition $a_1 \leq a_t \leq 2a_1$.

Theorem 3.4.2. *If uv is an edge of a connected graph G, then*

$$|e_G(u) - e_G(v)| \leq 1.$$

Proof. For every vertex w, the triangle inequality and $d_G(u, v) = 1$ give $d_G(u, w) \leq 1 + d_G(v, w) \leq 1 + e_G(v)$. Taking the maximum over all w yields $e_G(u) \leq 1 + e_G(v)$. Interchanging u and v gives $e_G(v) \leq 1 + e_G(u)$. Combining the two inequalities proves the assertion. ■

Thus adjacent vertices have equal eccentricities or eccentricities that differ by exactly one. Along any path, eccentricity cannot jump over an integer value.

Corollary 3.4.3. *Let $u_0u_1 \cdots u_m$ be a path in a connected graph G, and suppose $e_G(u_0) < e_G(u_m)$. For every integer k satisfying $e_G(u_0) < k < e_G(u_m)$, there exists j with $0 < j < m$ such that $e_G(u_j) = k$.*

Proof. Let j be the smallest index such that $e_G(u_j) \geq k$. Then $j \geq 1$, $e_G(u_{j-1}) < k$, and, by Theorem 3.4.2,

$$e_G(u_j) \leq e_G(u_{j-1}) + 1 \leq k.$$

Hence $e_G(u_j) = k$. Since $k < e_G(u_m)$, necessarily $j < m$. ■

The following necessary conditions are due to Lesniak.

Theorem 3.4.4 (Lesniak). *Let $[e_1, \ldots, e_n]$ be the nondecreasing eccentricity sequence of a connected graph G of order $n \geq 2$. Then:*

(i) $2e_1 \leq n$;

(ii) $e_n \leq \min\{n - 1, 2e_1\}$;

(iii) each integer k with $e_1 < k \leq e_n$ occurs at least twice in the sequence.

Proof. **(i)** Choose a central vertex c of G, so that $e_G(c) = e_1 = r$. Let T be a breadth-first spanning tree rooted at c. Distances from c are preserved, and therefore $e_T(c) = r$. Since distances in T are no smaller than in G, $r(T) \geq r(G) = r$; on the other hand, $r(T) \leq e_T(c) = r$. Thus $r(T) = r$. If d is the diameter of

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T, the standard center property of trees gives $r = \lceil d/2 \rceil$, so $d \geq 2r - 1$. A diametral path contains $d + 1$ vertices. Consequently,

$$n \geq d + 1 \geq 2r = 2e_1.$$

(ii) Every shortest path in an n-vertex connected graph has length at most $n - 1$, so $e_n = \operatorname{diam}(G) \leq n - 1$. Moreover, if c is a central vertex and u, v are arbitrary, then $d_G(u, v) \leq d_G(u, c) + d_G(c, v) \leq 2r$. Taking the maximum gives $e_n \leq 2e_1$.

(iii) Write $r = e_1$ and $d = e_n$. By Corollary 3.4.3, every integer k with $r < k \leq d$ is the eccentricity of some vertex w (take a path from a central to a peripheral vertex; the endpoint covers $k = d$). We show that it is also the eccentricity of a second vertex. Choose a vertex z with $d_G(w, z) = e_G(w) = k$, and a central vertex c. Since $d_G(c, z) \leq r < k$, a shortest c–z path does not contain w. Furthermore, $e_G(c) = r < k$, while $e_G(z) \geq d_G(z, w) = k$. Applying the intermediate-value argument along this path gives a vertex x with $e_G(x) = k$. As w is not on the path, $x \neq w$, as required. ■

For trees, Lesniak's conditions lead to an exact characterization. Recall that a tree has either one central vertex or two adjacent central vertices, and its radius is the ceiling of half its diameter.

Theorem 3.4.5 (Lesniak: Eccentricity Sequences of Trees). *Let $[e_1, \ldots, e_n]$ be a nondecreasing sequence of positive integers, where $n \geq 3$. It is the eccentricity sequence of a tree if and only if the following conditions hold:*

(i) Every integer k satisfying $e_1 < k \leq e_n$ appears at least twice;

(ii) either $e_n = 2e_1$ and $e_1 < e_2$, or $e_n = 2e_1 - 1$ and $e_1 = e_2 < e_3$.

Proof. **Necessity.** Let T be a tree of radius r and diameter d. Condition (i) follows from Theorem 3.4.4. If $d = 2r$, the tree has a unique central vertex and exactly one occurrence of the minimum eccentricity r; hence $e_1 < e_2$. If $d = 2r - 1$, the tree has exactly two central vertices, both of eccentricity r, and all remaining vertices have greater eccentricity. Thus $e_1 = e_2 < e_3$. This proves (ii).

Sufficiency. Put $r = e_1$ and $d = e_n$. By (ii), either $d = 2r$ or $d = 2r - 1$. Begin with a path $P = x_0 x_1 \cdots x_d$ of length d. At its vertex x_i,

$$e_P(x_i) = \max\{i, d - i\}.$$

If $d = 2r$, the eccentricity r occurs once on P, and if $d = 2r - 1$, it occurs twice. Every integer k with $r < k \leq d$ occurs exactly twice on P.

Let m_k be the required multiplicity of k in the given sequence. Condition (i) ensures $m_k \geq 2$ for $r < k \leq d$. For each such k, choose an internal vertex $x_{i(k)}$ of P with $e_P(x_{i(k)}) = k - 1$ and attach $m_k - 2$ new pendant vertices to it. This is possible because $k - 1 \leq d - 1$. Denote the resulting tree by T.

We verify that its eccentricities are precisely those prescribed. If x_i is an old vertex and y a new leaf attached to x_j, then $1 \leq j \leq d - 1$ and

$$d_T(x_i, y) = |i - j| + 1 \leq \max\{i, d - i\} = e_P(x_i).$$

Thus no old vertex acquires a larger eccentricity, and every old vertex retains its eccentricity because $P \subseteq T$.

Next, let y be a new leaf attached to x_j and let $k = e_P(x_j) + 1$. One endpoint of P is at distance k from y, so $e_T(y) \geq k$. For an old vertex x_i,

$$d_T(y, x_i) = 1 + |i - j| \leq 1 + e_P(x_j) = k.$$

For any other new leaf z attached to an internal vertex x_h, we also have

$$d_T(y, z) = 2 + |j - h| \leq 1 + \max\{j, d - j\} = k.$$

Hence $e_T(y) = k$. The tree therefore has precisely m_k vertices of eccentricity k for every k, and its nondecreasing eccentricity sequence is the given sequence. ■

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Example 3.4.6 (Constructing a Tree from an Eccentricity Sequence)**.** Construct a tree whose eccentricity sequence is

$$[4^2, 5^4, 6^3, 7^4].$$

Here the radius is 4 and the diameter is 7. Begin with the path $P = u_0u_1 \cdots u_7$. Its eccentricity sequence is $[4^2, 5^2, 6^2, 7^2]$, so we need two additional vertices of eccentricity 5, one of eccentricity 6, and two of eccentricity 7.

Attach leaves u_8 and u_9 to u_3 and u_4, respectively. Both have eccentricity 5. Next, attach u_{10} to u_2, which has eccentricity 5 on the original path; the new leaf has eccentricity 6. Finally, attach u_{11} and u_{12} to u_1 and u_6, respectively. These two leaves have eccentricity 7. The resulting tree, shown in Figure 3.20, has the required sequence.

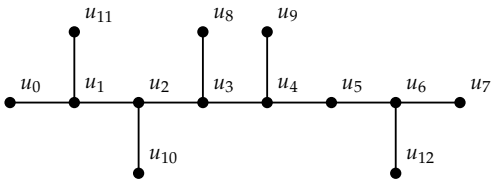


Figure 3.20: A tree realizing the eccentricity sequence $[4^2, 5^4, 6^3, 7^4]$.

Remark 3.4.7. The construction is not unique: different choices of attachment vertices can give nonisomorphic trees with the same eccentricity sequence.

3.4.1 Neighbourhoods

Let G be a connected graph and let $v \in V(G)$. For each integer $i \geq 0$, the *ith neighbourhood* of v is the set

$$N_i(v) = \{u \in V(G) : d_G(u, v) = i\}.$$

In particular, $N_0(v) = \{v\}$, while $N_1(v)$ is the usual *(open) neighbourhood* of v, denoted by $N(v)$. Thus, $N(v)$ consists precisely of the vertices adjacent to v.

Example 3.4.8. Consider the star with centre v and leaves $u_1, \ldots, u_6$ shown in Figure 3.21. Then

$$N(v) = N_1(v) = \{u_1, u_2, u_3, u_4, u_5, u_6\},$$

whereas $N_2(v) = \varnothing$. For each leaf u_i, $N(u_i) = \{v\}$ and $N_2(u_i) = \{u_j : j \neq i\}$.

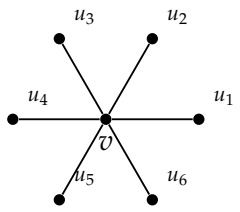


Figure 3.21: The neighbourhood of the central vertex of a star.

Two vertices having the same open neighbourhood are sometimes called *false twins*. Their distances to other vertices are closely related, as the next result shows.

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Lemma 3.4.9. *Let G be a connected graph. If $N(u) = N(v)$, then $e_G(u) = e_G(v)$.*

Proof. If $u = v$, there is nothing to prove. Suppose $u \neq v$. Since $N(u) = N(v)$, the vertices u and v cannot be adjacent: otherwise $u \in N(v) = N(u)$, which is impossible in a simple graph.

Interchanging u and v while fixing all other vertices preserves adjacency and therefore defines an automorphism of G. An automorphism preserves distances, so

$$d_G(u, x) = d_G(v, x) \qquad (x \in V(G) \setminus \{u, v\}).$$

Taking the maximum over all vertices, and noting that $d_G(u, v) = d_G(v, u)$, gives $e_G(u) = e_G(v)$. ■

For adjacent vertices, the corresponding condition involves removing each vertex from the neighbourhood of the other. Such vertices are sometimes called *true twins*.

Lemma 3.4.10. *Let u and v be adjacent vertices of a connected graph G. If*

$$N(u) \setminus \{v\} = N(v) \setminus \{u\},$$

then $e_G(u) = e_G(v)$.

Proof. The hypothesis says that every vertex other than u and v is adjacent to u if and only if it is adjacent to v. Since $uv \in E(G)$, interchanging u and v and fixing all remaining vertices is an automorphism of G. Hence u and v have the same eccentricity. ■

Multiplication of a vertex

We next describe two operations that replace a vertex by several vertices having identical connections to the rest of a graph.

Definition 3.4.11 (Replacement of a Vertex)**.** Let $v \in V(G)$ and let p be a positive integer. Delete v and introduce a set $X = \{x_1, \ldots, x_p\}$ of new vertices. Join every $x_i \in X$ to every vertex of $N_G(v)$, retaining all edges of $G - v$.

If X is made into a clique K_p, the resulting graph is said to be obtained by *linked multiplication* of v. If X is made into an independent set $\overline{K}_p$, the operation is called *multiplication* of v.

Thus the new vertices have the same neighbours outside X. They are pairwise adjacent in linked multiplication and pairwise nonadjacent in ordinary multiplication.

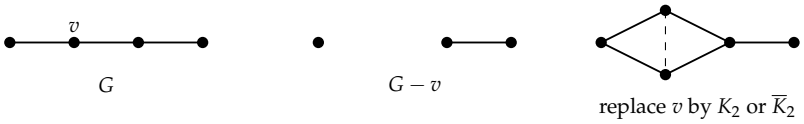


Figure 3.22: Replacing a vertex by two vertices. The dashed edge is included for K_2 and omitted for $\overline{K}_2$.

The effect of these operations on eccentricities is particularly useful in constructing graphs with prescribed eccentricity sequences.

Lemma 3.4.12. *Let G be a connected graph with diameter greater than 1, and let H be obtained by replacing $v \in V(G)$ by K_p, where $p \geq 1$. Then*

$$e_H(w) = e_G(w) \quad (w \in V(G) \setminus \{v\}), \qquad e_H(x) = e_G(v) \quad (x \in V(K_p)).$$

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Consequently, G and H have the same eccentricity set.

The same conclusion holds for replacement by $\overline{K}_p$ provided either $p = 1$ or $e_G(v) \geq 2$.

Proof. Write X for the set of vertices replacing v. Every path of G that passes through v can be represented in H by a path passing through any specified vertex of X. Conversely, a path in H between vertices outside X can be shortened, if necessary, by replacing any portion inside X by the single vertex v. Thus, for distinct $w, z \in V(G) \setminus \{v\}$ and $x \in X$,

$$d_H(w,z) = d_G(w,z), \qquad d_H(w,x) = d_G(w,v).$$

For $x, y \in X$, $x \neq y$, their distance is 1 if X induces K_p, and 2 if X induces $\overline{K}_p$ with $p \geq 2$. The latter assertion holds because G is connected and has more than one vertex, so v has a neighbour.

It follows that

$$e_H(w) = e_G(w) \qquad (w \in V(G) \setminus \{v\}),$$

and, when $p \geq 2$,

$$e_H(x) = \begin{cases} \max\{e_G(v), 1\}, & X \cong K_p, \\ \max\{e_G(v), 2\}, & X \cong \overline{K}_p. \end{cases}$$

Since $e_G(v) \geq 1$, linked multiplication preserves the eccentricity of v. Ordinary multiplication does so when $e_G(v) \geq 2$. For $p = 1$, the resulting graph is simply isomorphic to G. This proves the assertions. ■

Remark 3.4.13. The restriction on ordinary multiplication is necessary. For example, replacing the centre of $K_{1,3}$ by $\overline{K}_2$ produces $K_{2,3}$. The centre of $K_{1,3}$ has eccentricity 1, whereas each of its two replacements has eccentricity 2.

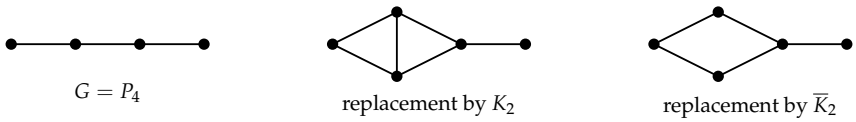


Figure 3.23: Both kinds of multiplication preserve the eccentricity set $\{2, 3\}$ of this example.

Eccentricity sequences and sets

For a connected graph, list its distinct eccentricities as $e_1 < e_2 < \cdots < e_k$, and let r_i denote the number of vertices of eccentricity e_i. Its eccentricity sequence may then be written as

$$[e_1^{r_1}, e_2^{r_2}, \ldots, e_k^{r_k}],$$

where the exponents indicate repetition rather than ordinary powers.

Theorem 3.4.14 (Lesniak). *Let $e_1 < \cdots < e_k$ be positive integers with $e_k > 1$, and let $r_1, \ldots, r_k$ and $s_1, \ldots, s_k$ be positive integers satisfying $s_i \leq r_i$ for all i. If*

$$[e_1^{s_1}, e_2^{s_2}, \ldots, e_k^{s_k}]$$

is the eccentricity sequence of a connected graph, then

$$[e_1^{r_1}, e_2^{r_2}, \ldots, e_k^{r_k}]$$

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is also an eccentricity sequence. In particular, a sequence with $e_k > 1$ is realisable if and only if it has a realisable subsequence containing at least one occurrence of each of its distinct eccentricities (the whole sequence is allowed as a subsequence).

Proof. Let G realise the shorter sequence. For each i, choose a vertex v_i whose eccentricity is e_i. Put

$$p_i = r_i - s_i + 1.$$

Replace v_1 by K_{p_1}. By Lemma 3.4.12, all other eccentricities remain unchanged, and v_1 is replaced by p_1 vertices of eccentricity e_1. Hence the frequency of e_1 increases by $p_1 - 1 = r_1 - s_1$.

Repeat this construction at the vertices corresponding to $v_2, \dots, v_k$. Each replacement preserves the eccentricity of every vertex already present, so after the ith step the frequency of e_i is r_i, while the frequencies of the remaining eccentricities have not yet been changed. After k steps, the resulting connected graph has precisely the required eccentricity sequence.

The final assertion follows by taking the subsequence to be the entire sequence in the necessity direction and by the construction above in the sufficiency direction. ■

Remark 3.4.15. The construction is generally not unique: different choices of vertices, as well as other graph constructions, may realise the same eccentricity sequence. Determining which realisable sequences contain no smaller realisable subsequence with the same distinct eccentricities is a separate problem.

We now turn from eccentricity sequences, which record the multiplicity of each eccentricity, to *eccentricity sets*, which record only the distinct values.

Theorem 3.4.16 (Behzad–Simpson). *A nonempty set $S = \{e_1, e_2, \dots, e_k\}$ of positive integers, where $e_1 < e_2 < \cdots < e_k$, is the eccentricity set of a connected graph if and only if*

$$e_{i+1} = e_i + 1 \quad (1 \le i < k), \qquad k \le e_1 + 1.$$

Proof. **Necessity.** Suppose that S is the eccentricity set of a connected graph G. Let $r = e_1$ and $D = e_k$ denote its radius and diameter. Along any path, the eccentricities of consecutive vertices differ by at most 1. A path from a central vertex of eccentricity r to a vertex of eccentricity D must therefore encounter every integer between r and D. Thus

$$S = \{r, r+1, \dots, D\}, \qquad k = D - r + 1.$$

The familiar radius–diameter inequality $D \le 2r$ now gives $k \le r + 1 = e_1 + 1$.

Sufficiency. Suppose that the two conditions hold. Write $r = e_1$ and $\ell = k - 1$. Then $0 \le \ell \le r$ and

$$S = \{r, r+1, \dots, r+\ell\}.$$

If $r = 1$ and $\ell = 0$, the graph K_2 has eccentricity set $\{1\}$. If $r = 1$ and $\ell = 1$, the path P_3 has eccentricity set $\{1, 2\}$.

Now let $r \ge 2$. Start with the cycle C_{2r} and choose a vertex x on it. Attach to x a path of length ℓ whose other vertices are new; when $\ell = 0$, take just the cycle. Denote the resulting graph by H.

For a vertex y on the cycle, put $t = d_{C_{2r}}(x, y)$. Its farthest vertex on the cycle is at distance r, and its farthest vertex on the attached path is at distance $t + \ell$. Consequently,

$$e_H(y) = \max\{r, t + \ell\}.$$

As t ranges from 0 to r, these eccentricities include every integer from r through $r + \ell$ and no others.

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Furthermore, let z_j be the vertex on the attached path at distance j from x, where $0 \le j \le \ell$. Its greatest distance to a cycle vertex is $r + j$, while its greatest distance to a path vertex is at most ℓ. Since $\ell \le r$, we obtain

$$e_H(z_j) = r + j.$$

Thus the vertices on the attached path also have eccentricities in S, and the eccentricity set of H is exactly

$$\{r, r+1, \dots, r+\ell\} = S.$$

This completes the construction and the proof. ■

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CHAPTER
4

SPECTRAL THEORY OF GRAPHS

INTRODUCTION

Spectral graph theory studies a graph through the eigenvalues of matrices associated with it. For a simple graph, the adjacency matrix records which vertices are adjacent, while the Laplacian matrix combines adjacency information with vertex degrees. Their spectra provide algebraic invariants that reveal important structural information about the graph.

The subject has strong connections with linear algebra, combinatorics, and mathematical chemistry. Early spectral ideas in chemistry arose from Hückel's work on molecular orbitals, while the mathematical development of spectral graph theory was advanced by Collatz and Sinogowitz in the 1950s. Graph energy, introduced by Ivan Gutman in 1978, gave another important spectral invariant motivated by chemical applications.

In this chapter we develop the adjacency spectrum and Laplacian spectrum of a graph, compute the spectra of several standard graph families, examine cospectral graphs, regular graphs, complements, and line graphs, and conclude with graph energy and Laplacian energy. The emphasis is on deriving the results directly from the underlying matrices so that the connection between graph structure and spectral information remains clear.

Video Resources

A detailed lecture series for this chapter is available at https://www.youtube.com/@Prof.DrAijaz/playlists. Additional study material is available at https://aijaz.vercel.app/.

4.1 Spectrum of Graphs

Let $G = (V, E)$ be a simple undirected graph with vertex set $V = \{v_1, \ldots, v_n\}$ and m edges. Spectral graph theory investigates how the structural properties of a graph are reflected in the eigenvalues of matrices associated with it. We begin with the adjacency matrix, which records the adjacency relation algebraically, and then determine the spectra of several familiar families of graphs.

Definition 4.1.1 (Adjacency Matrix)**.** The **adjacency matrix** of G is the $n \times n$ matrix $A(G) = [a_{ij}]$ defined by

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is adjacent to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Since G is simple, $a_{ii} = 0$ for every i. Since G is undirected, $a_{ij} = a_{ji}$, and therefore $A(G)$ is symmetric.

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Definition 4.1.2 (Spectrum)**.** The **adjacency spectrum** of G, denoted by $\mathrm{Spec}(G)$, is the multiset of eigenvalues of $A(G)$, each repeated according to its algebraic multiplicity. Equivalently, the eigenvalues are the roots of the characteristic polynomial $\phi(G,\lambda) = \det(\lambda I - A(G))$. As we shall see, $A(G)$ is real and symmetric, so its eigenvalues are real. We may therefore arrange them in nonincreasing order as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

Theorem 4.1.3 (Real Eigenvalues)**.** *Every eigenvalue of the adjacency matrix $A(G)$ is real.*

Proof. Write $A = A(G) = [a_{ij}]$. Because G is undirected, an edge joins v_i to v_j if and only if it joins v_j to v_i. Consequently $a_{ij} = a_{ji}$, so $A^T = A$. In addition, every entry of A is real.

Let $\lambda \in \mathbb{C}$ be an eigenvalue, and choose a nonzero (possibly complex) eigenvector x such that $Ax = \lambda x$. Taking the conjugate transpose gives $x^*A = \overline{\lambda}x^*$, because $A^* = A$. Multiplying $Ax = \lambda x$ on the left by x^* yields

$$x^*Ax = \lambda x^*x.$$

On the other hand, taking complex conjugates and using $A^* = A$ gives $\overline{x^*Ax} = x^*Ax$. Thus x^*Ax is real. Since $x \neq 0$, we have $x^*x = \sum_{j=1}^{n} |x_j|^2 > 0$, and therefore

$$\lambda = \frac{x^*Ax}{x^*x} \in \mathbb{R}.$$

Hence every adjacency eigenvalue is real. Furthermore, the spectral theorem for real symmetric matrices implies that A is orthogonally diagonalizable: there is an orthogonal matrix U and a real diagonal matrix D such that $A = UDU^T$. The columns of U form an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A. ■

Theorem 4.1.4 (Sum of Eigenvalues)**.** *If $\lambda_1, \ldots, \lambda_n$ are the adjacency eigenvalues of a simple graph G, then* $\sum_{i=1}^{n} \lambda_i = 0$.

Proof. Let $A = A(G)$. Since A is a real symmetric $n \times n$ matrix, the spectral theorem gives $A = UDU^T$, where U is orthogonal and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. By the cyclic property of trace,

$$\mathrm{tr}(A) = \mathrm{tr}(UDU^T) = \mathrm{tr}(DU^TU) = \mathrm{tr}(D) = \sum_{i=1}^{n} \lambda_i.$$

Recall that $\mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii}$ is the sum of the diagonal entries of A. The diagonal entry a_{ii} records whether there is a loop at the vertex v_i. A simple graph has no loops, and hence $a_{ii} = 0$ for every i. Therefore

$$\sum_{i=1}^{n} \lambda_i = \mathrm{tr}(A) = \sum_{i=1}^{n} 0 = 0,$$

as required. ■

Lemma 4.1.5. *Let $A = [a_{ij}]$ be the adjacency matrix of a simple graph G, and let $t(G)$ denote the number of triangles in G. Then:*

(A) the sum of all 2×2 principal minors of A is $-|E(G)|$;

(B) the sum of all 3×3 principal minors of A is $2t(G)$.

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Proof. **(A)** Fix distinct indices $i < j$. A principal minor is obtained by selecting the same set of row and column indices. The principal submatrix on $\{i, j\}$ is

$$A[\{i,j\}] = \begin{pmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{pmatrix} = \begin{pmatrix} 0 & a_{ij} \\ a_{ij} & 0 \end{pmatrix},$$

where we used the absence of loops and the symmetry of A. Its determinant is

$$\det A[\{i,j\}] = -a_{ij}^2 = -a_{ij},$$

since a_{ij} is either 0 or 1. Thus the determinant is -1 if v_i and v_j are adjacent, and 0 otherwise. Each edge of G corresponds to precisely one pair $i < j$. Summing over these pairs yields

$$\sum_{1 \le i < j \le n} \det A[\{i,j\}] = -\sum_{1 \le i < j \le n} a_{ij} = -|E(G)|.$$

(B) Fix three distinct indices $i < j < k$. Their principal submatrix is

$$A[\{i,j,k\}] = \begin{pmatrix} 0 & a_{ij} & a_{ik} \\ a_{ij} & 0 & a_{jk} \\ a_{ik} & a_{jk} & 0 \end{pmatrix}.$$

Expanding its determinant along the first row gives

$$\begin{aligned} \det A[\{i,j,k\}] &= -a_{ij}(0 \cdot 0 - a_{jk}a_{ik}) + a_{ik}(a_{ij}a_{jk} - 0 \cdot a_{ik}) \\ &= a_{ij}a_{jk}a_{ik} + a_{ik}a_{ij}a_{jk} \\ &= 2a_{ij}a_{jk}a_{ik}. \end{aligned}$$

The product $a_{ij}a_{jk}a_{ik}$ is 1 exactly when all three edges v_iv_j, v_jv_k and v_iv_k are present. This occurs precisely when these vertices form a triangle. For any other triple the product is 0. Every triangle has a unique set of three vertices and is counted once by the condition $i < j < k$. It follows that

$$\sum_{1 \le i < j < k \le n} \det A[\{i,j,k\}] = 2t(G).$$

This proves both assertions. ■

Theorem 4.1.6 (Closed Walks and Spectral Moments). *Let N_k be the total number of closed walks of length k in G. Then $N_k = \sum_{i=1}^{n} \lambda_i^k$. In particular, $\sum_{i=1}^{n} \lambda_i^2 = 2m$.*

Proof. Let $A = [a_{ij}]$ be the adjacency matrix and let $v_1, \ldots, v_n$ be the vertices of G. We first establish the elementary connection between powers of A and walks. For $k = 1$, the entry $(A)_{ij} = a_{ij}$ counts walks of length one from v_i to v_j: there is one such walk if the vertices are adjacent, and none otherwise.

Assume that $(A^k)_{ij}$ counts walks of length k from v_i to v_j. Matrix multiplication gives

$$(A^{k+1})_{ij} = \sum_{\ell=1}^{n} (A^k)_{i\ell} a_{\ell j}.$$

For a fixed ℓ, the product $(A^k)_{i\ell}a_{\ell j}$ counts walks of length k from v_i to v_ℓ which can be extended by the edge $v_\ell v_j$. Every walk of length $k+1$ has a unique penultimate vertex v_ℓ, so the sum counts all such walks once. By induction the interpretation holds for every positive integer k.

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In particular, $(A^k)_{ii}$ counts walks which start at v_i and return to v_i after k edges. Summing over all possible starting vertices, we obtain

$$N_k = \sum_{i=1}^{n} (A^k)_{ii} = \operatorname{tr}(A^k).$$

Now A is real symmetric, hence it admits an orthogonal diagonalization $A = UDU^T$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Consequently $A^k = UD^kU^T$, and the trace identity gives

$$N_k = \operatorname{tr}(A^k) = \operatorname{tr}(D^k) = \sum_{i=1}^{n} \lambda_i^k.$$

Finally, a closed walk of length two in a simple graph has the form u, v, u with $uv \in E(G)$. For each edge uv, there are exactly two such walks, namely u, v, u and v, u, v. Therefore $N_2 = 2|E(G)| = 2m$, and so $\sum_{i=1}^{n} \lambda_i^2 = 2m$. ■

Theorem 4.1.7 (Spectrum of Bipartite Graphs). *A graph G is bipartite if and only if its adjacency spectrum is symmetric about 0. In other words, λ is an eigenvalue of multiplicity r if and only if $-\lambda$ is an eigenvalue of multiplicity r.*

Proof. Suppose first that G is bipartite, with vertex partition $V(G) = V_1 \cup V_2$ and no edges within either part. Numbering the vertices of V_1 first gives

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}.$$

Let λ be an eigenvalue and write a corresponding nonzero eigenvector in block form $x = (u^T, v^T)^T$. Then

$$Ax = \lambda x \quad \Longleftrightarrow \quad Bv = \lambda u, \qquad B^T u = \lambda v.$$

Define $S = \operatorname{diag}(I_{|V_1|}, -I_{|V_2|})$. The matrix S is invertible, $S^{-1} = S$, and direct block multiplication shows that $SAS = -A$, equivalently $AS = -SA$. Applying this identity to the eigenvalue equation yields

$$A(Sx) = -S(Ax) = -\lambda Sx.$$

Thus Sx is a nonzero eigenvector for $-\lambda$. Moreover, S provides a bijection from the eigenspace of λ to that of $-\lambda$, because applying S again recovers x. The two eigenspaces consequently have equal dimension. Since the real symmetric matrix A is diagonalizable, algebraic and geometric multiplicities coincide. Hence λ and $-\lambda$ occur with the same multiplicity, proving spectral symmetry.

Conversely, assume that the adjacency spectrum is symmetric about zero, with multiplicities. For every nonnegative integer r, the eigenvalues can be paired as $\lambda, -\lambda$ (with any zero eigenvalues contributing zero), whence

$$\operatorname{tr}(A^{2r+1}) = \sum_{i=1}^{n} \lambda_i^{2r+1} = 0.$$

By the closed-walk interpretation of adjacency-matrix powers, this trace is the number of closed walks of length $2r+1$. It is a nonnegative integer. If an odd cycle of length $\ell = 2r+1$ existed, traversing that cycle from any one of its vertices would give a closed walk of length ℓ. Thus $\operatorname{tr}(A^\ell) \geq 1$, contradicting the equality above. Therefore G has no odd cycle. By the standard characterization of bipartite graphs, a graph is bipartite exactly when it has no odd cycle. Hence G is bipartite. ■

Theorem 4.1.8 (Spectrum of K_n). *For $n \geq 2$, $\operatorname{Spec}(K_n) = \{(n-1)^{(1)}, (-1)^{(n-1)}\}$.*

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Proof. Let J denote the $n \times n$ matrix in which every entry is 1, and let I be the identity matrix. In K_n, every pair of distinct vertices is adjacent, but there are no loops. Hence the adjacency matrix is $A = J - I$.

Set $\mathbf{1} = (1, \ldots, 1)^T$. Since the sum of the entries in each row of J is n, we have $J\mathbf{1} = n\mathbf{1}$. Therefore

$$A\mathbf{1} = (J - I)\mathbf{1} = (n-1)\mathbf{1},$$

so $n-1$ is an eigenvalue of A. Consider next the subspace

$$W = \left\{ x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0 \right\}.$$

For every $x \in W$, each coordinate of Jx equals $\sum_i x_i = 0$, and therefore

$$Ax = (J - I)x = -x.$$

Thus every nonzero vector in W is an eigenvector of A for eigenvalue -1. The single linear equation $\sum_i x_i = 0$ defines an $(n-1)$-dimensional subspace. Also $\mathbf{1} \notin W$, so

$$\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \oplus W.$$

We have therefore found a basis of n eigenvectors: one associated with $n-1$ and $n-1$ associated with -1. These account for all eigenvalues, including their multiplicities, as claimed. ∎

2nd Proof. Let A be the adjacency matrix of K_n. Since every pair of distinct vertices is adjacent, $A = J - I$, where J is the $n \times n$ matrix whose entries are all 1.

Since every row of J is identical and nonzero, $\operatorname{rank}(J) = 1$. Thus J has exactly one nonzero eigenvalue, say λ, while its remaining $n-1$ eigenvalues are zero. Since the sum of the eigenvalues of a matrix equals its trace, we have $\lambda = \operatorname{Tr}(J) = n$. Hence the eigenvalues of J are n with multiplicity 1 and 0 with multiplicity $n-1$.

Now $A = J - I$. If μ is an eigenvalue of J with corresponding eigenvector x, then

$$Ax = (J - I)x = Jx - x = (\mu - 1)x.$$

Thus the eigenvalues of A are obtained by subtracting 1 from the eigenvalues of J, without changing their multiplicities. Therefore A has eigenvalue $n-1$ with multiplicity 1 and eigenvalue -1 with multiplicity $n-1$. Hence

$$\operatorname{Spec}(K_n) = \{(n-1)^{(1)}, (-1)^{(n-1)}\}.$$

∎

Theorem 4.1.9 (Spectrum of $K_{r,s}$). *For $r, s \geq 1$, $\operatorname{Spec}(K_{r,s}) = \{(\sqrt{rs})^{(1)}, (-\sqrt{rs})^{(1)}, 0^{(r+s-2)}\}$.*

Proof. Let V_1 and V_2 be the two parts of $K_{r,s}$, having r and s vertices respectively. Place the vertices of V_1 first in the ordering. Since every vertex of one part is joined to every vertex of the other, and no two vertices of the same part are joined, the adjacency matrix has block form

$$A = \begin{pmatrix} 0 & J_{r\times s} \\ J_{s\times r} & 0 \end{pmatrix}.$$

Write $\mathbf{1}_r$ and $\mathbf{1}_s$ for all-ones column vectors of the indicated lengths. Consider the two-dimensional subspace

$$U = \left\{ \begin{pmatrix} a\mathbf{1}_r \\ b\mathbf{1}_s \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

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If $x = (a\mathbf{1}_r, b\mathbf{1}_s)^T$, then direct multiplication gives

$$Ax = \begin{pmatrix} sb\,\mathbf{1}_r \\ ra\,\mathbf{1}_s \end{pmatrix}.$$

Thus U is invariant under A, and with respect to its natural basis the restriction of A to U is represented by $\begin{pmatrix} 0 & s \\ r & 0 \end{pmatrix}$. Its characteristic polynomial is $\lambda^2 - rs$, whose roots are $\sqrt{rs}$ and $-\sqrt{rs}$. These are therefore two nonzero eigenvalues of A.

To find the remaining eigenvalues, consider

$$W = \left\{ \begin{pmatrix} u \\ v \end{pmatrix} : \sum_{i=1}^{r} u_i = 0,\ \sum_{j=1}^{s} v_j = 0 \right\}.$$

For $x = (u^T, v^T)^T \in W$, the upper block of Ax is $J_{r\times s}v = 0$ and the lower block is $J_{s\times r}u = 0$. Hence $Ax = 0$. The two independent sum conditions give $\dim W = (r-1) + (s-1) = r + s - 2$. Finally $U \cap W = \{0\}$ and $\dim U + \dim W = r + s$, so $\mathbb{R}^{r+s} = U \oplus W$. The $r + s$ independent eigenvectors thereby obtained account for the entire spectrum. Consequently the eigenvalues are $\sqrt{rs}$ and $-\sqrt{rs}$, each once, and 0 with multiplicity $r + s - 2$. ■

2nd Proof. Let A be the adjacency matrix of $K_{r,s}$. By arranging the vertices according to the two partite sets, we obtain

$$A = \begin{pmatrix} 0 & J_{r\times s} \\ J_{s\times r} & 0 \end{pmatrix},$$

where $J_{r\times s}$ and $J_{s\times r}$ are matrices whose entries are all 1.

Since all rows corresponding to the same partite set are identical, A has exactly two distinct nonzero rows. These rows are linearly independent because their nonzero entries occur in disjoint sets of columns. Hence $\operatorname{rank}(A) = 2$. Since A is real symmetric, it is diagonalizable, and therefore it has exactly two nonzero eigenvalues, say λ_1 and λ_2, while its remaining $r + s - 2$ eigenvalues are zero.

Since every diagonal entry of A is zero, the sum of its eigenvalues gives

$$\lambda_1 + \lambda_2 = \operatorname{Tr}(A) = 0.$$

Thus $\lambda_2 = -\lambda_1$.

Now the sum of the squares of the eigenvalues equals the trace of A^2. Since $K_{r,s}$ has rs edges, the number of closed walks of length 2 is $2rs$. Consequently,

$$\lambda_1^2 + \lambda_2^2 = \operatorname{Tr}(A^2) = 2rs.$$

Substituting $\lambda_2 = -\lambda_1$, we obtain $2\lambda_1^2 = 2rs$, which gives $\lambda_1^2 = rs$. Therefore the two nonzero eigenvalues are $\sqrt{rs}$ and $-\sqrt{rs}$, each with multiplicity 1, while 0 has multiplicity $r + s - 2$. Hence

$$\operatorname{Spec}(K_{r,s}) = \{(\sqrt{rs})^{(1)}, (-\sqrt{rs})^{(1)}, 0^{(r+s-2)}\}.$$

■

Definition 4.1.10 (Cyclic Permutation Matrix)**.** For $n \geq 2$, let Q_n be the $n \times n$ cyclic permutation matrix defined by $Q_n e_j = e_{j+1} \quad (1 \leq j < n), \qquad Q_n e_n = e_1$, where $e_1, \ldots, e_n$ are the standard basis vectors of $\mathbb{R}^n$.

Lemma 4.1.11. *For $n \geq 2$, the eigenvalues of Q_n are $1, \omega, \omega^2, \ldots, \omega^{n-1}, \qquad \omega = e^{2\pi i/n}$.*

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Proof. We work over $\mathbb{C}$, since the eigenvalues need not be real. Recall that $Q_n e_j = e_{j+1}$ for $1 \le j < n$ and $Q_n e_n = e_1$. Thus applying Q_n moves each standard basis vector forward by one position, returning to its starting position after n steps. Hence $Q_n^n = I$. If $Q_n x = \lambda x$ with $x \neq 0$, it follows that

$$x = Q_n^n x = \lambda^n x,$$

so $\lambda^n = 1$. Every eigenvalue must therefore be an nth root of unity.

We now show that *every* nth root of unity actually occurs. Fix a complex number λ with $\lambda^n = 1$, and define

$$x(\lambda) = (1, \lambda^{-1}, \lambda^{-2}, \dots, \lambda^{-(n-1)})^T.$$

Since the matrix shifts the coordinates forward cyclically,

$$Q_n x(\lambda) = \left(\lambda^{-(n-1)}, 1, \lambda^{-1}, \dots, \lambda^{-(n-2)}\right)^T.$$

The identity $\lambda^n = 1$ gives $\lambda^{-(n-1)} = \lambda$, and each of the other displayed coordinates equals λ times the corresponding coordinate of $x(\lambda)$. Hence $Q_n x(\lambda) = \lambda x(\lambda)$. The vector is nonzero, since its first coordinate is 1.

The equation $\lambda^n = 1$ has exactly n distinct complex roots, namely $1, \omega, \dots, \omega^{n-1}$ with $\omega = e^{2\pi i/n}$. We have exhibited a nonzero eigenvector for each one. As Q_n has order n, these are all its eigenvalues. In particular, its characteristic polynomial is $t^n - 1$. ■

2nd Proof. Since Q_n is the permutation matrix corresponding to a full cycle of length n, we have $Q_n^n = I$. If λ is an eigenvalue of Q_n with eigenvector $x \neq 0$, then $Q_n x = \lambda x$ implies $x = Q_n^n x = \lambda^n x$. Hence $\lambda^n = 1$, so every eigenvalue of Q_n is an nth root of unity.

Moreover, since the underlying permutation is a full cycle, the vectors $e_1, Q_n e_1, \dots, Q_n^{n-1} e_1$ form a basis of $\mathbb{C}^n$. Thus no nonzero polynomial of degree less than n annihilates Q_n. Since $Q_n^n = I$, its minimal polynomial is $t^n - 1$. As the minimal polynomial divides the characteristic polynomial and both have degree n, we obtain

$$\det(tI - Q_n) = t^n - 1.$$

Therefore the eigenvalues of Q_n are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where $\omega = e^{2\pi i/n}$. ■

Theorem 4.1.12 (Spectrum of a Cycle). *For $n \ge 3$, the eigenvalues of C_n are $\lambda_k = 2\cos\left(\frac{2\pi k}{n}\right)$, $k = 0, 1, \dots, n-1$.*

Proof. Number the vertices $0, 1, \dots, n-1$ cyclically and write $Q = Q_n$ for the cyclic permutation matrix from the preceding definition. Each vertex of C_n is adjacent to the vertex preceding it and to the vertex following it, with indices understood modulo n. Accordingly

$$A(C_n) = Q + Q^T.$$

A permutation matrix is orthogonal, so $Q^T = Q^{-1}$. By the preceding lemma the eigenvalues of Q are ω^k for $k = 0, \dots, n-1$, where $\omega = e^{2\pi i/n}$, and their corresponding eigenvectors form a basis of $\mathbb{C}^n$ because the eigenvalues are distinct.

Fix k and choose a nonzero vector x such that $Qx = \omega^k x$. Applying Q^{-1} to this equation gives $Q^{-1}x = \omega^{-k}x$. Hence

$$\begin{aligned} A(C_n)x &= (Q + Q^{-1})x \\ &= \omega^k x + \omega^{-k} x \\ &= (\omega^k + \omega^{-k})x. \end{aligned}$$

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Euler's identity $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ now yields

$$\omega^k + \omega^{-k} = e^{2\pi ik/n} + e^{-2\pi ik/n} = 2\cos\left(\frac{2\pi k}{n}\right).$$

Thus x is an eigenvector of the adjacency matrix for the stated eigenvalue. Since the n eigenvectors of Q form a basis, they also give n linearly independent eigenvectors of $A(C_n)$. Consequently the displayed list contains the entire spectrum, with repetitions reflecting multiplicities. ■

Alternative proof. Label the vertices of C_n by $0, 1, \ldots, n-1$, in cyclic order, and let $\omega = e^{2\pi i/n}$. For each $k \in \{0, \ldots, n-1\}$ define

$$x^{(k)} = (1, \omega^k, \omega^{2k}, \ldots, \omega^{(n-1)k})^T.$$

We will check directly that this is an eigenvector. For a vertex j, the only adjacent vertices are $j-1$ and $j+1$, with subscripts interpreted modulo n. Therefore the jth coordinate of $Ax^{(k)}$ is

$$\begin{aligned}(Ax^{(k)})_j &= \omega^{k(j-1)} + \omega^{k(j+1)} \\ &= \omega^{kj}(\omega^{-k} + \omega^k) \\ &= 2\cos\left(\frac{2\pi k}{n}\right)(x^{(k)})_j.\end{aligned}$$

The same calculation works at $j = 0$ and $j = n-1$ because $\omega^n = 1$. Thus

$$Ax^{(k)} = 2\cos\left(\frac{2\pi k}{n}\right)x^{(k)}.$$

To verify that no eigenvalues have been missed, we prove that the vectors $x^{(0)}, \ldots, x^{(n-1)}$ are linearly independent. Suppose $\sum_{k=0}^{n-1} c_k x^{(k)} = 0$. Its jth coordinate says $\sum_k c_k(\omega^j)^k = 0$ for every $j = 0, \ldots, n-1$. Hence the polynomial $p(t) = \sum_{k=0}^{n-1} c_k t^k$, of degree at most $n-1$, vanishes at the n distinct roots $1, \omega, \ldots, \omega^{n-1}$. A nonzero polynomial of degree at most $n-1$ cannot have n distinct roots. Thus $p = 0$ and every $c_k = 0$. We have produced n independent eigenvectors of an $n \times n$ matrix, so the eigenvalues listed above are complete. ■

Lemma 4.1.13. *Let $X = (x_1, \ldots, x_n)^T$ be an eigenvector of $A(P_n)$ corresponding to the eigenvalue λ. Then λ is also an eigenvalue of C_{2n+2}. Two linearly independent eigenvectors of $A(C_{2n+2})$ corresponding to λ are $Y = (x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1, 0)^T$ and $Z = (0, x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1)^T$.*

Proof. We first treat $n \geq 2$. Put $x_0 = x_{n+1} = 0$. Because P_n has consecutive edges $v_i v_{i+1}$, the equation $A(P_n)X = \lambda X$ is equivalent to the single recurrence

$$x_{i-1} + x_{i+1} = \lambda x_i \qquad (1 \leq i \leq n),$$

where the endpoint conditions are incorporated by $x_0 = x_{n+1} = 0$. In particular, $x_2 = \lambda x_1$ and $x_{n-1} = \lambda x_n$.

Let $N = 2n+2$. Number the vertices of C_N cyclically, so that the eigenvector equation for a vector $y = (y_1, \ldots, y_N)^T$ takes the form

$$y_{j-1} + y_{j+1} = \lambda y_j \qquad (1 \leq j \leq N),$$

where $y_0 = y_N$ and $y_{N+1} = y_1$. Now set

$$Y = (x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1, 0)^T.$$

For positions $j = 1, \ldots, n$, the cycle recurrence is exactly the path recurrence, because the coordinates adjacent to the end positions are the inserted zeros. At position $n+1$, the two neighboring coordinates

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are x_n and $-x_n$, and their sum is $0 = \lambda \cdot 0$. At positions $n+2, \ldots, 2n+1$, reversing the order and changing the signs transforms the same path recurrence into the required cycle recurrence. At the final position $2n+2$, the neighbors are $-x_1$ and x_1, which again sum to 0. This verifies every coordinate of

$$A(C_N)Y = \lambda Y.$$

Let S denote cyclic right shift of the coordinates. Shifting the entries of Y one step gives

$$Z = SY = (0, x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1)^T.$$

Since shifting a cycle merely relabels its vertices, $A(C_N)S = SA(C_N)$. It follows that

$$A(C_N)Z = A(C_N)SY = SA(C_N)Y = \lambda SY = \lambda Z.$$

Both vectors are nonzero. To prove they are linearly independent, observe that $x_1 \neq 0$: if $x_1 = 0$, the first recurrence forces $x_2 = 0$, and then successively every $x_i = 0$, contrary to $X \neq 0$. Now suppose $aY + bZ = 0$. Its first coordinate is $ax_1 = 0$, so $a = 0$. Its second coordinate is then $bx_1 = 0$, so $b = 0$. Hence Y and Z are independent.

When $n = 1$, the sole eigenvalue of the one-vertex path P_1 is 0. The prescribed vectors are $Y = (x_1, 0, -x_1, 0)^T$ and $Z = (0, x_1, 0, -x_1)^T$; direct inspection shows that both are independent eigenvectors of C_4 for eigenvalue 0. This completes the proof in all cases. ■

The preceding lemma explains how path eigenvalues can also be viewed within the spectrum of a larger cycle. The following direct argument gives the eigenvalues and their eigenvectors explicitly.

Theorem 4.1.14 (Spectrum of a Path). *The eigenvalues of P_n are $\lambda_k = 2\cos\left(\frac{k\pi}{n+1}\right)$, $\quad k = 1, 2, \ldots, n$.*

Proof. Label the vertices of P_n in order as $v_1, \ldots, v_n$. An eigenvector $x = (x_1, \ldots, x_n)^T$ with eigenvalue λ must satisfy $x_2 = \lambda x_1$ at the first endpoint, $x_{j-1} + x_{j+1} = \lambda x_j$ for $2 \leq j \leq n-1$, and $x_{n-1} = \lambda x_n$ at the last endpoint. It is convenient to extend the coordinates formally by setting $x_0 = x_{n+1} = 0$. Then all these equations can be written uniformly as

$$x_{j-1} + x_{j+1} = \lambda x_j \qquad (1 \leq j \leq n).$$

For each integer k with $1 \leq k \leq n$, put

$$\theta_k = \frac{k\pi}{n+1}, \qquad x^{(k)} = (\sin\theta_k, \sin 2\theta_k, \ldots, \sin n\theta_k)^T.$$

Note first that $0 < \theta_k < \pi$, and hence $\sin\theta_k > 0$. Thus $x^{(k)}$ is not the zero vector. Moreover, $\sin 0 = 0$ and

$$\sin((n+1)\theta_k) = \sin(k\pi) = 0,$$

so the proposed vector satisfies both endpoint conditions.

The trigonometric addition formulas give

$$\begin{aligned}\sin((j-1)\theta) + \sin((j+1)\theta) &= \sin(j\theta - \theta) + \sin(j\theta + \theta) \\ &= 2\sin(j\theta)\cos\theta.\end{aligned}$$

Consequently, for every $j = 1, \ldots, n$ (including the endpoints),

$$(A(P_n)x^{(k)})_j = \sin((j-1)\theta_k) + \sin((j+1)\theta_k) = 2\cos\theta_k\,\sin(j\theta_k).$$

This says precisely that

$$A(P_n)x^{(k)} = 2\cos\left(\frac{k\pi}{n+1}\right)x^{(k)}.$$

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Thus each displayed number is an eigenvalue. As k increases from 1 to n, the angles $k\pi/(n+1)$ are distinct and lie in $(0,\pi)$. The cosine function is strictly decreasing on this interval, so the resulting n eigenvalues are distinct. Eigenvectors belonging to distinct eigenvalues are linearly independent. We have therefore found n linearly independent eigenvectors of the $n \times n$ adjacency matrix. There can be no additional eigenvalues, and the stated formula describes the entire spectrum. ■

Lemma 4.1.15. *Let $X = (x_1, \dots, x_n)^T$ be an eigenvector of $A(P_n)$ corresponding to an eigenvalue λ. Then λ is also an eigenvalue of $A(C_{2n+2})$. Moreover, two linearly independent eigenvectors of $A(C_{2n+2})$ corresponding to λ are*

$$Y = (x_1, \dots, x_n, 0, -x_n, \dots, -x_1, 0)^T$$

and

$$Z = (0, x_1, \dots, x_n, 0, -x_n, \dots, -x_1)^T.$$

Proof. Let $A = A(P_n)$ and $X = (x_1, \dots, x_n)^T$ be an eigenvector corresponding to λ. Since

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \ddots & \vdots \\ 0 & 1 & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix},$$

the eigenvalue equation $AX = \lambda X$ gives

$$\begin{aligned} x_2 &= \lambda x_1, \\ x_{i-1} + x_{i+1} &= \lambda x_i, \qquad 2 \le i \le n-1, \\ x_{n-1} &= \lambda x_n. \end{aligned}$$

Equivalently, putting $x_0 = x_{n+1} = 0$, these equations become

$$x_{i-1} + x_{i+1} = \lambda x_i, \qquad 1 \le i \le n.$$

Now let $B = A(C_{2n+2})$ and consider

$$Y = (x_1, \dots, x_n, 0, -x_n, \dots, -x_1, 0)^T.$$

Since each vertex of C_{2n+2} is adjacent to its two cyclic neighbors, multiplication by B gives

$$BY = \begin{pmatrix} x_2 \\ x_1 + x_3 \\ \vdots \\ x_{n-1} \\ x_n - x_n \\ -x_{n-1} \\ \vdots \\ -(x_1 + x_3) \\ -x_2 \\ -x_1 + x_1 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \\ 0 \\ -\lambda x_n \\ \vdots \\ -\lambda x_2 \\ -\lambda x_1 \\ 0 \end{pmatrix} = \lambda Y.$$

Thus Y is an eigenvector of B corresponding to λ.

Similarly, for $Z = (0, x_1, \dots, x_n, 0, -x_n, \dots, -x_1)^T$, the same coordinate equations yield

$$BZ = \lambda Z.$$

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Hence Z is also an eigenvector corresponding to λ.

Finally, $x_1 \neq 0$, since otherwise the eigenvalue equations would successively imply $x_2 = \cdots = x_n = 0$, contradicting $X \neq 0$. If $aY + bZ = 0$, comparison of the first two coordinates gives $ax_1 = 0$ and $bx_1 = 0$. Hence $a = b = 0$, proving that Y and Z are linearly independent. ■

2nd Proof for Spectrum of Path. By the preceding lemma (Lemma 4.1.15), every eigenvalue λ of P_n is also an eigenvalue of C_{2n+2}, and its multiplicity in C_{2n+2} is at least 2, since the lemma provides two linearly independent eigenvectors corresponding to λ.

Now the eigenvalues of C_{2n+2} are

$$\mu_k = 2\cos\left(\frac{2\pi k}{2n+2}\right) = 2\cos\left(\frac{k\pi}{n+1}\right), \qquad k = 0, 1, \ldots, 2n+1.$$

Since $\mu_k = \mu_{2n+2-k}$, the eigenvalues corresponding to $k = 1, 2, \ldots, n$ each have multiplicity 2, whereas the eigenvalues 2 and -2, corresponding to $k = 0$ and $k = n+1$, respectively, each have multiplicity 1.

Consequently, every eigenvalue of P_n must belong to the set

$$\left\{2\cos\left(\frac{k\pi}{n+1}\right) : k = 1, 2, \ldots, n\right\}.$$

Moreover, the eigenvector equation $A(P_n)X = \lambda X$, with $X = (x_1, \ldots, x_n)^T$, gives $x_2 = \lambda x_1$ and $x_{i+1} = \lambda x_i - x_{i-1}$ for $2 \leq i \leq n-1$. Thus every eigenvector is uniquely determined by its first coordinate, so each eigenspace has dimension 1. Since $A(P_n)$ is real symmetric, it is diagonalizable, and hence all its eigenvalues are simple.

As $A(P_n)$ is an $n \times n$ matrix, it has exactly n distinct eigenvalues. Therefore all the n values listed above must occur, giving

$$\mathrm{Spec}(P_n) = \left\{2\cos\left(\frac{k\pi}{n+1}\right) : k = 1, 2, \ldots, n\right\}.$$

■

Example 4.1.16 (Spectrum of P_4)**.** For P_4, the path-spectrum formula gives

$$\mathrm{Spec}(P_4) = \left\{2\cos\frac{\pi}{5}, 2\cos\frac{2\pi}{5}, 2\cos\frac{3\pi}{5}, 2\cos\frac{4\pi}{5}\right\}.$$

Using the exact cosine values,

$$\mathrm{Spec}(P_4) = \left\{\frac{1+\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}, -\frac{\sqrt{5}-1}{2}, -\frac{1+\sqrt{5}}{2}\right\}.$$

The eigenvalues occur in opposite pairs, as expected from the bipartiteness of every path. This example illustrates the spectral symmetry established earlier.

Definition 4.1.17 (Determinant)**.** Let $A = [a_{ij}]$ be an $n \times n$ matrix. The *determinant* of A is

$$\det(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)},$$

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where S_n is the set of all permutations of $\{1, \ldots, n\}$ and

$$\operatorname{sgn}(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is even,} \\ -1, & \text{if } \sigma \text{ is odd.} \end{cases}$$

A useful observation. If $\sigma \in S_n$ has t disjoint cycles, counting fixed points as cycles of length one, then $\operatorname{sgn}(\sigma) = (-1)^{n-t}$. In particular, when A is the adjacency matrix of a simple graph, every permutation with a fixed point contributes zero: its product contains a diagonal entry $a_{ii} = 0$. Hence only permutations without fixed points need be considered in the determinant expansion.

Example 4.1.18. Use the permutation definition to evaluate

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Solution. There are exactly two permutations in S_2: the identity and the transposition (12). Their signs are 1 and -1, respectively. Thus

$$\begin{aligned} \det(A) &= \operatorname{sgn}(\mathrm{id}) a_{11}a_{22} + \operatorname{sgn}((12)) a_{12}a_{21} \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned}$$

■

Definition 4.1.19 (Elementary Spanning Subgraph)**.** An *elementary spanning subgraph* of a graph G is a spanning subgraph in which every connected component is either a single edge K_2 or a cycle of length at least three. An *elementary subgraph* is defined similarly, but need not be spanning.

The connection with determinants is immediate from the cycle decomposition of permutations: a transposition represents an edge, while a permutation cycle of length at least three represents a cycle of the graph. Because a permutation moves every vertex in a nonzero term of the adjacency determinant, these edges and cycles together span the graph.

Theorem 4.1.20 (Determinant Expansion)**.** *Let A be the adjacency matrix of a simple graph G of order n. Then*

$$\det(A) = \sum_{H} (-1)^{n - c_1(H) - c(H)} 2^{c(H)},$$

where the sum runs over all elementary spanning subgraphs H of G, $c_1(H)$ is the number of components isomorphic to K_2, and $c(H)$ is the number of cycle components. If no such subgraph exists, the sum is zero.

Proof. Write $A = [a_{ij}]$. By definition,

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Since every diagonal entry of A is zero, a permutation contributes a nonzero term only if it has no fixed points and every pair $v_i, v_{\sigma(i)}$ is an edge of G.

Consider the disjoint-cycle decomposition of such a permutation. A transposition (ij) gives the factor $a_{ij}a_{ji} = 1$ and thus corresponds to the edge v_iv_j. A cycle $(i_1\, i_2\, \ldots\, i_r)$ of length $r \geq 3$ gives

$$a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_r i_1} = 1,$$

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and therefore corresponds to the graph cycle $v_{i_1}v_{i_2}\cdots v_{i_r}v_{i_1}$. As the permutation cycles are disjoint and cover all n indices, they determine an elementary spanning subgraph H.

Conversely, any elementary spanning subgraph determines precisely $2^{c(H)}$ contributing permutations: its K_2 components give unique transpositions, while each longer cycle has two possible orientations. Each of these permutations has $c_1(H)+c(H)$ disjoint cycles, and hence sign $(-1)^{n-c_1(H)-c(H)}$. Summing the contributions of the permutations associated with each H proves the formula. ■

Simplified Proof. Let $A = A(G)$ be the adjacency matrix of a simple graph G of order n. By the Leibniz formula for determinants,

$$\det(A) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\prod_{i=1}^{n} a_{i,\sigma(i)},$$

where S_n is the symmetric group on n elements.

Since G is simple, $a_{ii}=0$, and $a_{ij}=1$ precisely when the vertices i and j are adjacent. Consequently, a term in the determinant expansion is nonzero only if σ has no fixed points and every pair of consecutive vertices in each cycle of σ is adjacent in G.

Every permutation admits a unique decomposition into disjoint cycles. A cycle of length two, say (ij), corresponds to an edge ij of G, while a cycle of length $k\geq 3$ corresponds to a cycle C_k in G. Thus, every nonzero term determines a spanning elementary subgraph H whose components are either isolated edges or cycles of length at least three.

Let $C_1(H)$ denote the number of K_2 components of H, and let $C(H)$ denote the number of its cycle components of length at least three. The corresponding permutation has exactly $C_1(H)+C(H)$ disjoint cycles. Since the sign of a permutation having D disjoint cycles is $(-1)^{n-D}$, we obtain

$$\operatorname{sgn}(\sigma) = (-1)^{n-C_1(H)-C(H)}.$$

We now determine the number of permutations corresponding to a fixed spanning elementary subgraph H. Each K_2 component determines a unique transposition, whereas each cycle component of length at least three determines exactly two cyclic permutations, corresponding to its two possible orientations. Since the components are vertex-disjoint, these choices are independent. Hence precisely $2^{C(H)}$ permutations correspond to H.

Moreover, every such permutation contributes a product equal to 1, because all the corresponding entries of A are equal to 1. Therefore, the total contribution of H to the determinant is

$$(-1)^{n-C_1(H)-C(H)}2^{C(H)}.$$

Summing over all spanning elementary subgraphs $H\in\mathcal{S}$ of G, we conclude that

$$\det(A) = \sum_{H\in\mathcal{S}} (-1)^{n-C_1(H)-C(H)}2^{C(H)},$$

as required. ■

Example 4.1.21. Determine $\det(A)$ for the graph in Figure 4.1 by listing its elementary spanning subgraphs.

Solution. Every elementary spanning subgraph on five vertices must consist either of a 5-cycle or of a triangle together with an independent edge. In the given graph, the only 5-cycle is 1 2 3 4 5 1. The only two triangle–edge combinations covering all vertices are $(3\,4\,5)$ together with 12, and $(1\,2\,3)$ together with 45. These give precisely the three subgraphs H_1, H_2, H_3 shown in the figure.

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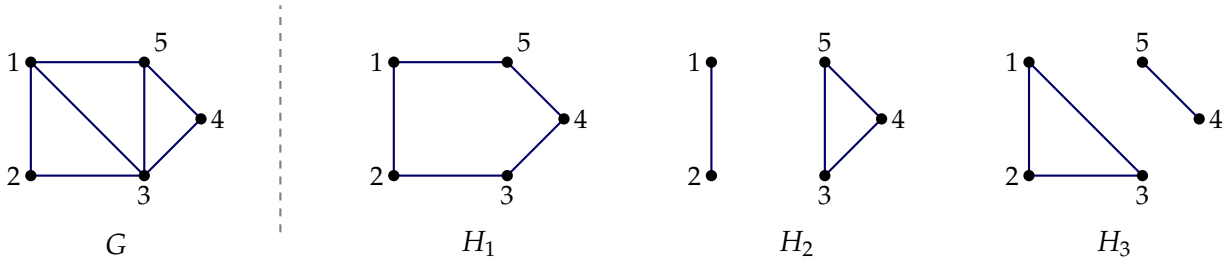


Figure 4.1: The elementary spanning subgraphs of G.

For H_1, we have $c_1(H_1) = 0$ and $c(H_1) = 1$, so its contribution is $(-1)^{5-1}2 = 2$. For each of H_2 and H_3, we have $c_1 = 1$ and $c = 1$, giving the contribution $(-1)^{5-2}2 = -2$. Therefore

$$\det(A) = 2 - 2 - 2 = \boxed{-2}.$$

■

The same argument gives a combinatorial formula for every coefficient of the characteristic polynomial. The essential idea is to apply the determinant expansion to the adjacency matrices of induced subgraphs.

Theorem 4.1.22 (Sachs' Theorem). *Let G be a simple graph of order n with characteristic polynomial*

$$\phi(G, \lambda) = \det(\lambda I - A(G)) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n.$$

Then, for $1 \leq i \leq n$,

$$a_i = \sum_{H \in \mathcal{H}_i} (-1)^{p(H)} 2^{c(H)},$$

where $\mathcal{H}_i$ denotes the family of all elementary subgraphs of G on exactly i vertices, that is, subgraphs whose connected components are either K_2 or cycles of length at least three. Here, $p(H)$ denotes the number of connected components of H, and $c(H)$ denotes the number of its cycle components.

Proof. Recall that the coefficient of λ^{n-i} in $\det(\lambda I - A)$ is $(-1)^i$ times the sum of the principal minors of A of order i. Consequently,

$$a_i = (-1)^i \sum_{\substack{S \subseteq V(G) \\ |S| = i}} \det A(G[S]).$$

Here $G[S]$ is the subgraph induced by S. Applying Theorem 4.1.20 to its adjacency matrix gives

$$\det A(G[S]) = \sum_{\substack{H \subseteq G[S] \\ H \text{ elementary and spanning}}} (-1)^{i-p(H)} 2^{c(H)}.$$

Every elementary subgraph of G on i vertices appears exactly once in this double sum, namely when $S = V(H)$. Therefore

$$\begin{aligned} a_i &= (-1)^i \sum_{H \in \mathcal{H}_i} (-1)^{i-p(H)} 2^{c(H)} \\ &= \sum_{H \in \mathcal{H}_i} (-1)^{p(H)} 2^{c(H)}, \end{aligned}$$

as required. ■

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Proof. Let $A = A(G)$. The coefficient a_i of the characteristic polynomial of G is given by

$$a_i = (-1)^i \sum_{|S|=i} \det A[S],$$

where $A[S]$ denotes the principal submatrix of A corresponding to a subset S of i vertices.

Expanding $\det A[S]$ over all permutations of S, we observe that permutations with fixed points contribute zero, since the diagonal entries of A vanish. Thus, every nonzero term corresponds to an elementary subgraph H whose components are either edges K_2 or cycles of length at least three.

Each edge component corresponds to a transposition, while each cycle component admits two possible orientations. Consequently, each elementary subgraph H contributes $2^{c(H)}$ terms, where $c(H)$ is the number of its cycle components.

Since the corresponding permutations have $p(H)$ disjoint cycles on i vertices, each has sign $(-1)^{i-p(H)}$. Combining this with the factor $(-1)^i$, we obtain

$$(-1)^i(-1)^{i-p(H)} = (-1)^{p(H)}.$$

Therefore, summing the contributions of all elementary subgraphs of order i yields

$$a_i = \sum_{H \in \mathcal{H}_i} (-1)^{p(H)} 2^{c(H)},$$

as required. ■

4.2 Isomorphism and Cospectral Graphs

Definition 4.2.1 (Cospectral Graphs)**.** Two graphs G_1 and G_2 are called *cospectral* if their adjacency matrices have the same eigenvalues, counted with multiplicities. Equivalently, their characteristic polynomials are equal: $\phi(G_1, \lambda) = \phi(G_2, \lambda)$.

An isomorphism merely relabels the vertices and therefore does not change the adjacency spectrum. Indeed, if $G_1 \cong G_2$, then their adjacency matrices satisfy $A(G_2) = P^T A(G_1) P$ for some permutation matrix P. Since $P^{-1} = P^T$, the two matrices are similar and consequently have the same characteristic polynomial. Thus *isomorphic graphs are cospectral*. The converse fails: eigenvalues do not always determine a graph up to isomorphism.

Example 4.2.2 (Non-isomorphic Cospectral Graphs)**.** The graphs $K_{1,4}$ and $C_4 \cup K_1$ (See Figure 4.2) are non-isomorphic but cospectral. Indeed, $\text{Spec}(K_{1,4}) = \text{Spec}(C_4 \cup K_1) = \{2, -2, 0, 0, 0\}$. They are clearly non-isomorphic because $K_{1,4}$ is connected whereas $C_4 \cup K_1$ is disconnected.

A connected example appears in Figure 4.3. Both graphs have characteristic polynomial

$$\phi(\lambda) = \lambda^6 - 7\lambda^4 - 4\lambda^3 + 7\lambda^2 + 4\lambda - 1.$$

They are not isomorphic: the graph on the left has a vertex of degree 5, whereas the maximum degree of the graph on the right is 3. Thus cospectral graphs need not have the same degree sequence.

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$K_{1,4}$ $\qquad$ $C_4 \cup K_1$

Figure 4.2: A pair of non-isomorphic cospectral graphs.



Figure 4.3: Two connected, non-isomorphic cospectral graphs on six vertices.

4.3 Regular Graphs

Definition 4.3.1 (k-regular Graph)**.** A graph G is **k-regular** if every vertex of G has degree k.

Theorem 4.3.2 (Spectral Properties of Regular Graphs)**.** *Let G be a k-regular graph of order n. Then:*

1. *k is an eigenvalue of $A(G)$ with eigenvector $\mathbf{1} = (1, \ldots, 1)^T$;*
2. *every eigenvalue λ of $A(G)$ satisfies $|\lambda| \leq k$;*
3. *the multiplicity of the eigenvalue k equals the number of connected components of G.*

In particular, k has multiplicity 1 if and only if G is connected.

Proof. Let A be the adjacency matrix of a k-regular graph G of order n. Since every vertex of G has degree k, each row and each column of A sums to k. Let $u = (1, 1, \ldots, 1)^t$. Then

$$Au = A \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ k \\ \vdots \\ k \end{pmatrix} = ku.$$

Thus, k is an eigenvalue of A with corresponding eigenvector u.

Now, let $x = (x_1, x_2, \ldots, x_n)^t$ be any eigenvector corresponding to k, so that $Ax = kx$. Choose j such that $|x_j| = \max\{|x_1|, \ldots, |x_n|\}$. If A_j denotes the jth row of A, then $A_j x = kx_j$, which gives

$$\sum_{v_i \sim v_j} x_i = kx_j.$$

Since G is k-regular, the sum on the left contains exactly k terms. Hence,

$$k|x_j| = \left| \sum_{v_i \sim v_j} x_i \right| \leq \sum_{v_i \sim v_j} |x_i| \leq k|x_j|.$$

Consequently, equality holds throughout. As x_j has maximum absolute value, all vertices adjacent to v_j must have the same eigenvector entry x_j. Since G is connected, repeating this argument along

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paths gives $x_1 = x_2 = \cdots = x_n = \alpha$. Therefore, $x = \alpha u$, showing that every eigenvector corresponding to k is a scalar multiple of u. Thus, the geometric multiplicity of k is one. Since A is a real symmetric matrix, it is diagonalizable, and hence the algebraic and geometric multiplicities coincide. Therefore, the algebraic multiplicity of k is also one.

Finally, let λ be any eigenvalue of A with corresponding nonzero eigenvector $y = (y_1, y_2, \ldots, y_n)^t$. Choose j such that $|y_j| = \max\{|y_1|, \ldots, |y_n|\}$. From $Ay = \lambda y$, we obtain

$$\lambda y_j = \sum_{v_i \sim v_j} y_i.$$

Taking absolute values and using the fact that v_j has exactly k neighbours, we get

$$|\lambda||y_j| = \left| \sum_{v_i \sim v_j} y_i \right| \leq \sum_{v_i \sim v_j} |y_i| \leq k|y_j|.$$

Since $y_j \neq 0$, it follows that $|\lambda| \leq k$, as required.

In general, let $G_1, \ldots, G_r$ be the connected components of G. After a suitable ordering of vertices, A is block diagonal with blocks $A(G_1), \ldots, A(G_r)$. Every component is k-regular, and the preceding argument shows that k has multiplicity 1 in each block. Therefore its multiplicity in A is r. As A is real symmetric, the algebraic and geometric multiplicities agree.

■

Example 4.3.3. The cycle C_n is 2-regular. Therefore 2 is an eigenvalue of C_n. Since C_n is connected, the eigenvalue 2 has multiplicity 1. This agrees with $\lambda_j = 2\cos\left(\frac{2\pi j}{n}\right), \quad j = 0, 1, \ldots, n-1$, for which the value 2 occurs only when $j = 0$.

4.3.1 Spectrum of the Complement

Recall that the complement $\overline{G}$ of a graph G has the same vertex set as G, with two distinct vertices adjacent in $\overline{G}$ exactly when they are not adjacent in G.

Theorem 4.3.4 (Spectrum of the Complement). *Let G be a k-regular graph of order n. Choose an orthonormal eigenbasis $x_1, x_2, \ldots, x_n$ of $A(G)$ such that $x_1 = \frac{1}{\sqrt{n}}\mathbf{1}$, with corresponding eigenvalues $k, \lambda_2, \ldots, \lambda_n$. Then* $\mathrm{Spec}(\overline{G}) = \{n-k-1, -\lambda_2 - 1, \ldots, -\lambda_n - 1\}$.

Proof. Let A and $\overline{A}$ be the adjacency matrices of G and its complement $\overline{G}$, respectively. Recall that for any two distinct vertices, exactly one of the graphs G and $\overline{G}$ contains the edge joining them. Therefore,

$$A + \overline{A} = J - I,$$

where J is the all-one matrix and I is the identity matrix. Hence,

$$\overline{A} = J - I - A.$$

Now let

$$x_1 = \frac{1}{\sqrt{n}}\mathbf{1},$$

where $\mathbf{1}$ denotes the all-one column vector of size n. Since every row of J sums to n, we have

$$Jx_1 = nx_1.$$

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Also, because G is k-regular, each row sum of A is k, so

$$Ax_1 = kx_1.$$

Using the relation $\overline{A} = J - I - A$, we obtain

$$\overline{A}x_1 = (J - I - A)x_1 = Jx_1 - Ix_1 - Ax_1 = nx_1 - x_1 - kx_1 = (n - k - 1)x_1.$$

Thus x_1 is an eigenvector of $\overline{A}$ corresponding to the eigenvalue $n - k - 1$. Hence $n - k - 1$ is an eigenvalue of $\overline{G}$.

Next, consider x_i for $i \geq 2$. Since $\{x_1, x_2, \ldots, x_n\}$ is an orthonormal eigenbasis, we have

$$x_i \perp x_1 \qquad \text{for } i \geq 2.$$

But $x_1 = \frac{1}{\sqrt{n}}\mathbf{1}$, so $x_i \perp \mathbf{1}$. This means that the sum of the coordinates of x_i is zero, and therefore

$$Jx_i = 0.$$

Also, since x_i is an eigenvector of A corresponding to λ_i, we have

$$Ax_i = \lambda_i x_i.$$

Again using $\overline{A} = J - I - A$, we get

$$\overline{A}x_i = (J - I - A)x_i = Jx_i - Ix_i - Ax_i = 0 - x_i - \lambda_i x_i = (-1 - \lambda_i)x_i.$$

Thus x_i is an eigenvector of $\overline{A}$ with eigenvalue $-\lambda_i - 1$.

Therefore, the eigenvalues of $\overline{G}$ are

$$n - k - 1,\ -\lambda_2 - 1,\ -\lambda_3 - 1,\ \ldots,\ -\lambda_n - 1.$$

■

Example 4.3.5. The cycle C_4 is 2-regular and $\mathrm{Spec}(C_4) = \{2, 0, 0, -2\}$. Therefore $\mathrm{Spec}(\overline{C_4}) = \{4 - 2 - 1, -0 - 1, -0 - 1, -(-2) - 1\} = \{1, 1, -1, -1\}$. Since $\overline{C_4} \cong 2K_2$, this agrees with the fact that each copy of K_2 has eigenvalues 1 and -1.

4.3.2 Spectrum of Line Graphs

Let $V(G) = \{v_1, \ldots, v_n\}$, $E(G) = \{e_1, \ldots, e_m\}$. The *line graph* $G_\ell = L(G)$ has one vertex for each edge of G. Two distinct vertices of G_ℓ are adjacent precisely when their corresponding edges in G share an endpoint.

If G is k-regular, then every edge of G is incident with two vertices, and at each endpoint there are $k - 1$ other incident edges. Hence every vertex of G_ℓ has degree $(k - 1) + (k - 1) = 2k - 2$. Thus, for $k \geq 1$, the line graph G_ℓ is $(2k - 2)$-regular. This observation will also be reflected in its spectrum.

Lemma 4.3.6. *Let G be a simple graph, let A and B be the adjacency matrices of G and G_ℓ, respectively, and let M be the unoriented vertex-edge incidence matrix of G. Then $M^T M = B + 2I$. Moreover, $MM^T = D + A$, where D is the degree matrix of G. In particular, if G is k-regular, then $MM^T = A + kI$.*

Proof. Let G be a simple graph with vertex set $V(G) = \{v_1, \ldots, v_n\}$ and edge set $E(G) = \{e_1, \ldots, e_m\}$. Let M be its vertex-edge incidence matrix.

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Consider the matrix M^TM. Since every edge has exactly two endpoints, each column of M contains exactly two entries equal to 1. Hence every diagonal entry of M^TM is 2. For $i \neq j$, the entry $(M^TM)_{ij}$ counts the number of common endpoints of e_i and e_j. Since G is simple,

$$(M^TM)_{ij} = \begin{cases} 1, & \text{if } e_i \text{ and } e_j \text{ share an endpoint,} \\ 0, & \text{otherwise.} \end{cases}$$

These are precisely the off-diagonal entries of the adjacency matrix B of the line graph $L(G)$. Therefore,

$$M^TM = B + 2I.$$

Next, consider MM^T. Its diagonal entry $(MM^T)_{ii}$ counts the edges incident with v_i and hence equals $d(v_i)$. For $i \neq j$, the entry $(MM^T)_{ij}$ is 1 if v_i and v_j are adjacent, and 0 otherwise. Thus, the diagonal entries of MM^T coincide with those of the degree matrix D, while its off-diagonal entries coincide with those of the adjacency matrix A. Consequently,

$$MM^T = D + A.$$

If G is k-regular, then $D = kI$, and therefore

$$MM^T = A + kI.$$

■

Corollary 4.3.7. *For every simple graph G, each adjacency eigenvalue μ of $L(G)$ satisfies $\mu \geq -2$.*

Proof. Let B be the adjacency matrix of the line graph G_ℓ. By Lemma 4.3.6, we have $B + 2I = M^TM$. For any vector x,

$$x^TM^TMx = (Mx)^T(Mx) = \|Mx\|^2 \geq 0.$$

Thus, M^TM is positive semidefinite, and hence all its eigenvalues are nonnegative.

Now let μ be any eigenvalue of B. Since $B + 2I = M^TM$, it follows that $\mu + 2$ is an eigenvalue of M^TM. Therefore, $\mu + 2 \geq 0$, which gives $\mu \geq -2$, as required. ■

Theorem 4.3.8 (Characteristic Polynomial of a Line Graph)**.** *Let G be a k-regular graph of order n, where $k \geq 2$. Then $\phi(G_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(k-2)}\phi(G, \lambda + 2 - k)$.*

Proof. Put $m = |E(G)|$, and write $A = A(G)$ and $B = A(G_\ell)$. Let M be the $n \times m$ unoriented incidence matrix. By Lemma 4.3.6,

$$M^TM = B + 2I_m, \qquad MM^T = A + kI_n.$$

We use the determinant identity

$$\det(tI_m - M^TM) = t^{m-n}\det(tI_n - MM^T),$$

valid as a polynomial identity when $m \geq n$. To see this, apply Sylvester's determinant identity $\det(I_m - UV) = \det(I_n - VU)$ with $U = t^{-1}M^T$ and $V = M$, and then multiply by t^m. Since G is k-regular, the handshaking lemma gives $2m = nk$. In particular, $k \geq 2$ implies $m \geq n$.

Now put $t = \lambda + 2$. We obtain

$$\begin{aligned} \phi(G_\ell, \lambda) &= \det(\lambda I_m - B) \\ &= \det((\lambda + 2)I_m - M^TM) \\ &= (\lambda + 2)^{m-n}\det((\lambda + 2)I_n - MM^T) \\ &= (\lambda + 2)^{m-n}\det((\lambda + 2 - k)I_n - A) \\ &= (\lambda + 2)^{m-n}\phi(G, \lambda + 2 - k). \end{aligned}$$

Finally, $m - n = \frac{nk}{2} - n = \frac{n}{2}(k-2)$, which gives the required formula. ■

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Example 4.3.9 (Spectrum of the Line Graph of a Cycle)**.** For $n \geq 3$, the cycle C_n is 2-regular, and its line graph is again a cycle: $(C_n)_\ell \cong C_n$. The preceding theorem gives

$$\phi((C_n)_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(2-2)} \phi(C_n, \lambda + 2 - 2) = \phi(C_n, \lambda),$$

which agrees with $(C_n)_\ell \cong C_n$.

4.4 The Laplacian Spectrum of a Graph

Let $G = (V, E)$ be a simple undirected graph with $V(G) = \{v_1, \ldots, v_n\}$ and m edges. Recall that $A(G)$ denotes the adjacency matrix of G.

Definition 4.4.1 (Degree Matrix)**.** The **degree matrix** of G is the diagonal matrix $D(G) = \operatorname{diag}\big(d(v_1), d(v_2), \ldots, d(v$

Definition 4.4.2 (Laplacian Matrix)**.** The **Laplacian matrix** of G is $L(G) = D(G) - A(G)$. Thus

$$L_{ij} = \begin{cases} d(v_i), & i = j, \\ -1, & i \neq j \text{ and } v_i \sim v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Since both $D(G)$ and $A(G)$ are real symmetric matrices, $L(G)$ is also real and symmetric.

Definition 4.4.3 (Laplacian Spectrum)**.** The **Laplacian spectrum** of G, denoted by $\operatorname{Spec}_L(G)$, is the multiset of eigenvalues of $L(G)$. Since $L(G)$ is real and symmetric, all its eigenvalues are real. We order them as $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$.

The following quadratic-form identity is fundamental in the study of the Laplacian.

Theorem 4.4.4 (Positive Semidefinite Property)**.** *For every $x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$, $x^T L(G) x = \sum_{v_i v_j \in E(G)} (x_i - x_j)^2$. Consequently, $L(G)$ is positive semidefinite and every Laplacian eigenvalue is nonnegative.*

Proof. Using $L = D - A$, we obtain $x^T L x = x^T D x - x^T A x$. Now $x^T D x = \sum_{i=1}^{n} d(v_i) x_i^2$. Each edge $v_i v_j$ contributes x_i^2 once through the degree of v_i and x_j^2 once through the degree of v_j. Hence $x^T D x = \sum_{v_i v_j \in E(G)} (x_i^2 + x_j^2)$. Since A is symmetric, each edge contributes twice to $x^T A x$, so $x^T A x = 2 \sum_{v_i v_j \in E(G)} x_i x_j$. Therefore

$$\begin{aligned} x^T L x &= \sum_{v_i v_j \in E(G)} \left(x_i^2 + x_j^2 - 2 x_i x_j \right) \\ &= \sum_{v_i v_j \in E(G)} (x_i - x_j)^2. \end{aligned}$$

Every term in the last sum is nonnegative. Thus $x^T L x \geq 0$ for every $x \in \mathbb{R}^n$, so $L(G)$ is positive semidefinite. Hence all eigenvalues of $L(G)$ are nonnegative. ∎

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Remark 4.4.5. The identity $x^T Lx = \sum_{v_iv_j \in E(G)} (x_i - x_j)^2$ shows that the Laplacian measures how much the values assigned to adjacent vertices differ. The quadratic form vanishes exactly when x is constant along every edge.

Theorem 4.4.6 (Smallest Laplacian Eigenvalue)**.** *The smallest Laplacian eigenvalue of every graph is* 0.

Proof. For every vertex v_i, the sum of the entries in the ith row of $L(G)$ is $d(v_i) - d(v_i) = 0$. Therefore $L(G)\mathbf{1} = 0$, where $\mathbf{1} = (1, \ldots, 1)^T$. Thus 0 is a Laplacian eigenvalue. Since all Laplacian eigenvalues are nonnegative, 0 must be the smallest one. ■

Theorem 4.4.7 (Multiplicity of the Zero Eigenvalue)**.** *The multiplicity of* 0 *as a Laplacian eigenvalue equals the number of connected components of G. In particular,*

$$G \text{ is connected} \iff 0 \text{ is a simple Laplacian eigenvalue.}$$

Proof. Suppose first that $Lx = 0$. Then $x^T Lx = 0$. By the quadratic-form identity, $0 = \sum_{uv \in E(G)} (x_u - x_v)^2$. Every term in this sum is nonnegative, so each term must be zero. Hence $x_u = x_v$ for every edge uv.

If two vertices lie in the same connected component, they are joined by a path. Repeatedly applying the equality along that path shows that x has the same value at every vertex of the component. Thus every vector in $\ker L$ is constant on each connected component.

Conversely, suppose x is constant on every connected component. If v_i lies in a component on which the constant value is c, then all neighbors of v_i also have value c. Hence $(Lx)_i = d(v_i)c - \sum_{v_j \sim v_i} c = d(v_i)c - d(v_i)c = 0$. Therefore $Lx = 0$.

If G has r connected components, the characteristic vectors of these components form a basis for $\ker L$. Hence $\dim(\ker L) = r$. Since L is symmetric, the dimension of the nullspace is the multiplicity of the eigenvalue 0. Therefore the multiplicity of 0 is exactly the number of connected components. ■

Theorem 4.4.8 (Sum of Laplacian Eigenvalues)**.** *If G has m edges, then* $\sum_{i=1}^{n} \mu_i = 2m$.

Proof. The sum of the eigenvalues of a matrix equals its trace. Therefore $\sum_{i=1}^{n} \mu_i = \operatorname{tr}(L) = \sum_{i=1}^{n} d(v_i)$. By the Handshaking Lemma, $\sum_{i=1}^{n} d(v_i) = 2m$. Hence $\sum_{i=1}^{n} \mu_i = 2m$. ■

Remark 4.4.9. The average of the Laplacian eigenvalues is therefore $\frac{1}{n} \sum_{i=1}^{n} \mu_i = \frac{2m}{n}$, which is exactly the average degree of G.

Example 4.4.10 (Laplacian Spectrum of P_3)**.** For the path $P_3 = v_1v_2v_3$, the degree matrix and

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adjacency matrix are

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Hence

$$L = D - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is $\det(\lambda I - L) = \lambda(\lambda - 1)(\lambda - 3)$. Therefore $\mathrm{Spec}_L(P_3) = \{3, 1, 0\}$.

This agrees with the general results above: P_3 is connected, so 0 occurs once, and $3 + 1 + 0 = 4 = 2|E(P_3)|$.

4.5 Laplacian Energy

Before defining Laplacian energy, recall the usual notion of graph energy.

Definition 4.5.1 (Graph Energy)**.** If $\lambda_1, \ldots, \lambda_n$ are the adjacency eigenvalues of G, then the **energy** of G is $E(G) = \sum_{i=1}^{n} |\lambda_i|$.

Graph energy was introduced by Gutman in 1978. Laplacian energy was later introduced by Gutman and Zhou as an analogous quantity based on the Laplacian eigenvalues.

Since $\sum_{i=1}^{n} \mu_i = 2m$, the average Laplacian eigenvalue is $\frac{2m}{n}$, which is also the average degree of the graph.

Definition 4.5.2 (Laplacian Energy)**.** Let $\mu_1, \ldots, \mu_n$ be the Laplacian eigenvalues of a graph G with n vertices and m edges. The **Laplacian energy** of G is $LE(G) = \sum_{i=1}^{n} \left| \mu_i - \frac{2m}{n} \right|$.

Thus Laplacian energy measures the total deviation of the Laplacian eigenvalues from their mean value $2m/n$.

Theorem 4.5.3 (Laplacian Energy of Regular Graphs)**.** *If G is k-regular, then $LE(G) = E(G)$.*

Proof. For a k-regular graph, $D = kI$, and therefore $L = kI - A$. If $Ax = \lambda_i x$, then $Lx = (kI - A)x = (k - \lambda_i)x$. Hence the Laplacian eigenvalues are $\mu_i = k - \lambda_i$.

Since G is k-regular, $2m = nk$, so $\frac{2m}{n} = k$. Therefore

$$\begin{aligned} LE(G) &= \sum_{i=1}^{n} |\mu_i - k| \\ &= \sum_{i=1}^{n} |k - \lambda_i - k| \\ &= \sum_{i=1}^{n} |\lambda_i| \\ &= E(G). \end{aligned}$$

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■

Example 4.5.4 (Energy of K_3)**.** The adjacency spectrum of K_3 is $\text{Spec}(K_3) = \{2, -1, -1\}$. Hence $E(K_3) = |2| + |-1| + |-1| = 4$.

Since K_3 is 2-regular, its Laplacian eigenvalues are obtained from $\mu_i = 2 - \lambda_i$, giving $\text{Spec}_L(K_3) = \{3, 3, 0\}$. Also, $\frac{2m}{n} = \frac{2 \cdot 3}{3} = 2$. Therefore $LE(K_3) = |3-2| + |3-2| + |0-2| = 4$. Thus $LE(K_3) = E(K_3)$, as predicted by the preceding theorem.

Example 4.5.5 (Laplacian Energy of P_3)**.** The path P_3 is not regular, so its Laplacian energy need not equal its ordinary graph energy.

From the preceding example, $\text{Spec}_L(P_3) = \{3, 1, 0\}$. Since P_3 has $n = 3$ vertices and $m = 2$ edges, $\frac{2m}{n} = \frac{4}{3}$. Hence

$$\begin{aligned} LE(P_3) &= \left|3 - \frac{4}{3}\right| + \left|1 - \frac{4}{3}\right| + \left|0 - \frac{4}{3}\right| \\ &= \frac{5}{3} + \frac{1}{3} + \frac{4}{3} \\ &= \frac{10}{3}. \end{aligned}$$

This example illustrates that the equality $LE(G) = E(G)$ is a special property of regular graphs.

Theorem 4.5.6 (Basic Properties of Laplacian Energy)**.** *For every graph G:*

(i) $LE(G) \geq 0$;

(ii) $LE(G) = 0$ *if and only if G has no edges.*

Proof. Since Laplacian energy is a sum of absolute values, $LE(G) = \sum_{i=1}^{n} \left|\mu_i - \frac{2m}{n}\right| \geq 0$. This proves (i).

Suppose first that G has no edges. Then $m = 0$ and $L = 0$. Thus every Laplacian eigenvalue is 0, and consequently $LE(G) = 0$.

Conversely, suppose $LE(G) = 0$. A sum of nonnegative terms can equal zero only when every term is zero. Hence $\mu_i = \frac{2m}{n}$ for every i. But 0 is always a Laplacian eigenvalue. Therefore $\frac{2m}{n} = 0$, which implies $m = 0$. Thus G has no edges. ■

4.5.1 Laplacian Spectra of Some Standard Graphs

Theorem 4.5.7 (Laplacian Spectrum of K_n)**.** *For $n \geq 2$, $\text{Spec}_L(K_n) = \{0^{(1)}, n^{(n-1)}\}$.*

Proof. For the complete graph K_n, every vertex has degree $n-1$. Hence $D = (n-1)I$. Also, $A = J - I$, so $L = D - A = (n-1)I - (J - I) = nI - J$.

The all-ones vector $\mathbf{1}$ satisfies $J\mathbf{1} = n\mathbf{1}$. Therefore $L\mathbf{1} = (nI - J)\mathbf{1} = 0$, so 0 is an eigenvalue.

If $x \perp \mathbf{1}$, then the sum of the coordinates of x is zero, so $Jx = 0$. Hence $Lx = (nI - J)x = nx$. The subspace orthogonal to $\mathbf{1}$ has dimension $n-1$. Therefore n is an eigenvalue with multiplicity $n-1$, and $\text{Spec}_L(K_n) = \{0^{(1)}, n^{(n-1)}\}$. ■

Theorem 4.5.8 (Laplacian Spectrum of $K_{r,s}$)**.** *For $r, s \geq 1$, $\text{Spec}_L(K_{r,s}) = \{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$.*

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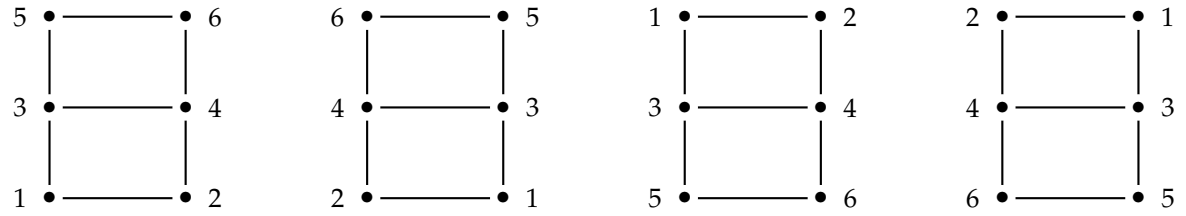


Figure 4.4: Automorphism group of $G_{2,3}$.

Proof. Let the two partite sets be $V_1 = \{u_1, \ldots, u_r\}$, $V_2 = \{v_1, \ldots, v_s\}$. Every vertex in V_1 has degree s, while every vertex in V_2 has degree r.

Since $K_{r,s}$ is connected, 0 is a simple Laplacian eigenvalue, with eigenvector $\mathbf{1}$.

Consider a vector x supported on V_1 whose coordinates on V_1 sum to zero. Thus $x = (x_1, \ldots, x_r, 0, \ldots, 0)^T$, $\sum_{i=}$ 0. For a vertex $u_i \in V_1$, all neighbors lie in V_2, where the coordinates of x are zero. Hence $(Lx)_{u_i} = sx_i$. For a vertex $v_j \in V_2$, $(Lx)_{v_j} = -\sum_{i=1}^{r} x_i = 0$. Thus $Lx = sx$. The space of vectors on V_1 whose coordinates sum to zero has dimension $r-1$. Therefore s is an eigenvalue with multiplicity $r-1$.

Similarly, vectors supported on V_2 whose coordinates sum to zero form an $(s-1)$-dimensional eigenspace corresponding to the eigenvalue r. Hence r has multiplicity $s-1$.

It remains to find one eigenvalue. Consider the vector

$$z = (\underbrace{s, \ldots, s}_{r \text{ entries}}, \underbrace{-r, \ldots, -r}_{s \text{ entries}})^T.$$

For $u_i \in V_1$, $(Lz)_{u_i} = s(s) - \sum_{v_j \in V_2} (-r) = s^2 + rs = (r+s)s$. For $v_j \in V_2$, $(Lz)_{v_j} = r(-r) - \sum_{u_i \in V_1} s = -r^2 - rs = (r+s)(-r)$. Therefore $Lz = (r+s)z$. Thus $r+s$ is an eigenvalue.

The multiplicities now sum to $1 + (r-1) + (s-1) + 1 = r+s$, which is the order of $K_{r,s}$. Hence all Laplacian eigenvalues have been accounted for: $\text{Spec}_L(K_{r,s}) = \{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$. ■

4.6 Automorphism Groups of Graphs

Definition 4.6.1. A graph automorphism of G is a permutation φ on the set of vertices $V(G)$ that satisfies the property that $\{u_i, u_j\} \in E(G)$ if and only if $\{\varphi(u_i), \varphi(u_j)\} \in E(G)$.

Example 4.6.2. For example, the grid graph $G_{2,3}$ has four automorphisms: $(1,2,3,4,5,6)$, $(2,1,4,3,6,5)$, $(5,6,3,4,1,2)$, $(6,5,4,3,2,1)$. These correspond to the graph itself, the graph flipped from left to right, the graph flipped up down, and the graph flipped left to right and up down, respectively.

The star graph S_4 has six automorphisms: $(1,2,3,4)$, $(1,3,2,4)$, $(2,1,3,4)$, $(2,3,1,4)$, $(3,1,2,4)$, $(3,2,1,4)$.

Theorem 4.6.3. *The set* $\text{Aut}(G)$ *of all graph automorphisms of a graph* G *forms a group under function composition.*

Definition 4.6.4 (Identity Graph)**.** A graph G whose automorphism group if $Aut(G) \cong S_1$ is called the identity graph.

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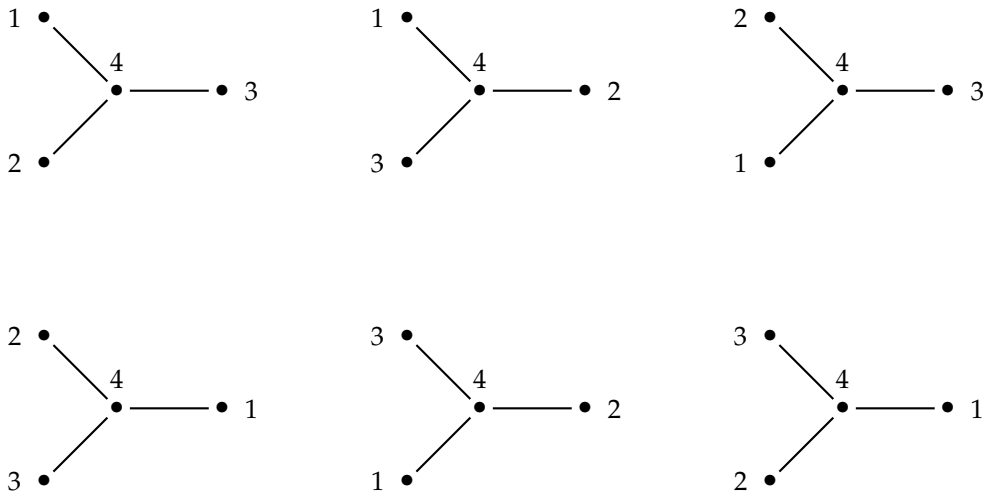


Figure 4.5: Automorphisms of star graph S_4.

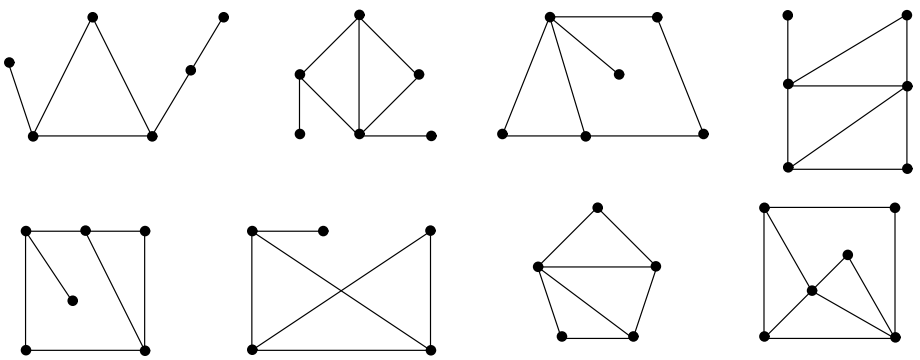


Figure 4.6: The Eight Identity Graphs of order Six.

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The elements of $Aut(G)$ induce a group of permutations on the edge set E of G in the obvious way:

If $e = uv \in E$ and $\alpha \in Aut(G)$, then $\alpha_1(e) = \alpha_0(u)\alpha_0(v)$ where α_0 and α_1 denote the action of α on V and E respectively. This permutation group on E with degree $m = |E|$ and order $|Aut(G)|$ is called the induced edge group of G and is denoted by $Aut_I(G)$.

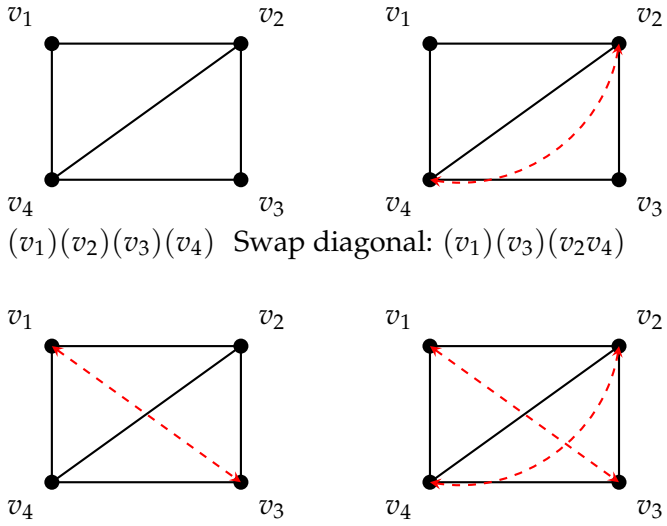


Figure 4.7: Automorphisms of $K_4 - e$

Example 4.6.5. Consider the graph $K_4 - e$ with vertices, Figure 4.7, labeled v_1, v_2, v_3, v_4 and edges e_1, e_2, e_3, e_4, e_5. The vertex group $Aut(K_4 - e)$ consists of the four permutations:

$$(v_1)(v_2)(v_3)(v_4), \quad (v_1)(v_3)(v_2v_4), \quad (v_2)(v_4)(v_1v_3), \quad (v_1v_3)(v_2v_4)$$

or

$$\begin{pmatrix} v_1 & v_2 & v_3 & v_4 \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix}, \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_1 & v_3 & v_4 & v_2 \end{pmatrix}, \begin{pmatrix} v_2 & v_4 & v_1 & v_3 \\ v_2 & v_4 & v_3 & v_1 \end{pmatrix}, \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_3 & v_1 & v_4 & v_2 \end{pmatrix}.$$

The identity permutation of the vertex-group induces the identity permutation on the edges, while $(v_1)(v_3)(v_2v_4)$ induces a permutation on the edges which fixes e_5, interchanges e_1 with e_4, and e_2 with e_3. In this way, the edge-group consists of the following permutations, induced respectively by the above members of the vertex-group.

$$(e_1)(e_2)(e_3)(e_4)(e_5), \ (e_1e_4)(e_2e_3)(e_5), \ (e_1e_2)(e_3e_4)(e_5), \quad (e_1e_3)(e_2e_4)(e_5)$$

or

$$\begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_1 & e_2 & e_3 & e_4 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_4 & e_2 & e_3 & e_5 \\ e_4 & e_1 & e_3 & e_2 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_2 & e_1 & e_4 & e_3 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_3 & e_1 & e_4 & e_2 & e_5 \end{pmatrix}.$$

Of course, the edge-group and the vertex-group of $K_4 - e$ are isomorphic. But they are not identical permutation groups since $Aut_I(K_4 - e)$ has degree 5 and $Aut(K_4 - e)$ has degree 4. Note that edge e_5 is fixed by every member of the edge-group. Even the permutation group obtained from $Aut_I(K_4 - e)$ by restricting its object set to e_1, e_2, e_3, e_4 is not identical with $Aut(K_4 - e)$, since these two isomorphic permutation groups of the same degree have different cycle structure.

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Theorem 4.6.6. *The vertex automorphism group $Aut(G)$ and the induced edge automorphism group $Aut_I(G)$ of a graph G are isomorphic if and only if G contains at most one isolated vertex and no connected component isomorphic to K_2.*

Proof. Let $\phi : \mathrm{Aut}(G) \to \mathrm{Aut}_I(G)$ be the natural homomorphism defined by $\phi(\alpha) = \alpha_I$, where α_I is the edge automorphism induced by α, that is,

$$\alpha_I(\{u, v\}) = \{\alpha(u), \alpha(v)\}.$$

By the definition of $\mathrm{Aut}_I(G)$, the map ϕ is surjective. Hence $\mathrm{Aut}(G) \cong \mathrm{Aut}_I(G)$ if and only if ϕ is injective, or equivalently, if and only if $\ker \phi = \{\mathrm{id}\}$.

Suppose first that G contains two distinct isolated vertices u and v. Let α be the permutation of $V(G)$ that interchanges u and v and fixes every other vertex. Since u and v are isolated, α is a nonidentity automorphism of G and fixes every edge of G. Thus $\alpha \in \ker \phi$, so ϕ is not injective.

Similarly, suppose that G has a connected component isomorphic to K_2, with vertices u and v and edge $\{u, v\}$. The automorphism that interchanges u and v and fixes every other vertex fixes the edge $\{u, v\}$ as an unordered pair and fixes all other edges. Hence it is a nonidentity element of $\ker \phi$. Therefore, if ϕ is injective, then G has at most one isolated vertex and no connected component isomorphic to K_2.

Conversely, suppose that G has at most one isolated vertex and no connected component isomorphic to K_2. Let $\alpha \in \ker \phi$. Then every edge is fixed as an unordered pair, so for each $\{x, y\} \in E(G)$,

$$\{\alpha(x), \alpha(y)\} = \{x, y\}.$$

We show that $\alpha = \mathrm{id}$. Suppose, to the contrary, that $\alpha(u) = v \neq u$ for some vertex u. Since automorphisms preserve degree, $\deg(u) = \deg(v)$. If $\deg(u) = 0$, then u and v are two distinct isolated vertices, contrary to the hypothesis. Hence $\deg(u) \geq 1$.

Choose a neighbor w of u. Since the edge $\{u, w\}$ is fixed,

$$\{\alpha(u), \alpha(w)\} = \{u, w\}.$$

Using $\alpha(u) = v$, we obtain

$$\{v, \alpha(w)\} = \{u, w\}.$$

Therefore $v \in \{u, w\}$. Since $v \neq u$, it follows that $v = w$. Hence $\{u, v\} \in E(G)$. Substituting $w = v$ in the preceding equality gives

$$\{v, \alpha(v)\} = \{u, v\},$$

and therefore $\alpha(v) = u$.

We now show that u has no neighbor other than v. Suppose that $z \neq v$ is another neighbor of u. Since $\{u, z\}$ is fixed,

$$\{\alpha(u), \alpha(z)\} = \{u, z\}.$$

Thus

$$\{v, \alpha(z)\} = \{u, z\}.$$

The equality of these two sets implies $v \in \{u, z\}$, which is impossible because $v \neq u$ and $v \neq z$. Hence v is the only neighbor of u, so $\deg(u) = 1$. Since $\deg(v) = \deg(u)$, we also have $\deg(v) = 1$. Thus $\{u, v\}$ is a connected component isomorphic to K_2, contrary to the hypothesis.

Therefore no vertex is moved by α, so $\alpha = \mathrm{id}$. Hence $\ker \phi = \{\mathrm{id}\}$, and ϕ is injective. Since ϕ is also surjective, it is an isomorphism. Therefore,

$$\mathrm{Aut}(G) \cong \mathrm{Aut}_I(G).$$

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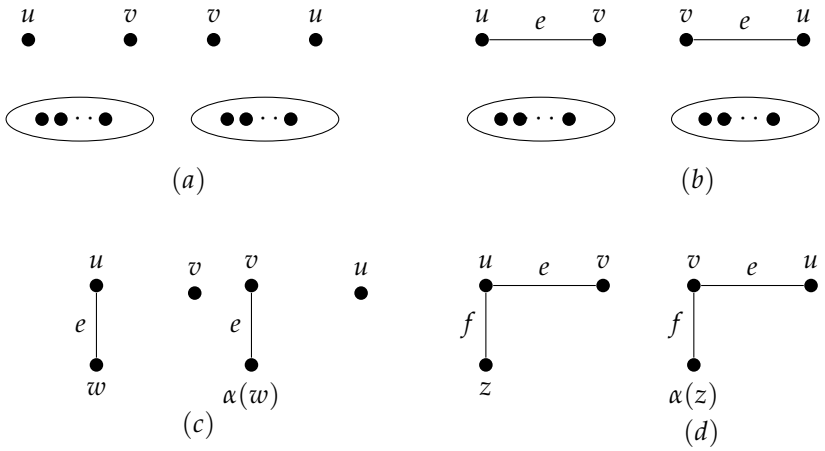


Figure 4.8: Illustration for Theorem 4.6.6. In (a) two isolated vertices may be interchanged; in (b) the two vertices of a K_2-component may be interchanged; in (c) the fixed edge $\{u, w\}$ together with $\alpha(u) = v$ forces $w = v$; and in (d) the existence of another neighbour $z \neq v$ leads to a contradiction.

Corollary 4.6.7. *If G is a connected graph with $n \geq 3$, then $Aut(G) \cong Aut_I(G)$.*

Theorem 4.6.8. *Given any graph G, $\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G})$.*

Proof. Let V be the vertex set of G. Recall that for the complement graph $\overline{G}$, an edge $\{u, v\}$ exists in $E(\overline{G})$ if and only if $\{u, v\} \notin E(G)$.

Let σ be a permutation of V. We show that $\sigma \in \mathrm{Aut}(G)$ if and only if $\sigma \in \mathrm{Aut}(\overline{G})$. By the definition of an automorphism, $\sigma \in \mathrm{Aut}(G)$ if and only if for all distinct vertices $u, v \in V$:

$$\{u, v\} \in E(G) \iff \{\sigma(u), \sigma(v)\} \in E(G).$$

Taking the logical negation of both sides of the equivalence, we obtain:

$$\{u, v\} \notin E(G) \iff \{\sigma(u), \sigma(v)\} \notin E(G).$$

By the definition of the complement graph, $\{x, y\} \notin E(G)$ is equivalent to $\{x, y\} \in E(\overline{G})$. Substituting this into the expression above yields:

$$\{u, v\} \in E(\overline{G}) \iff \{\sigma(u), \sigma(v)\} \in E(\overline{G}).$$

This is precisely the condition for σ to be an automorphism of $\overline{G}$. Thus, $\sigma \in \mathrm{Aut}(G) \iff \sigma \in \mathrm{Aut}(\overline{G})$, proving that $\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G})$. ■

Definition 4.6.9. Let S_n be the symmetric group of degree n, and let $\mathbb{Z}_2 = \{0, 1\}$ be the cyclic group of order 2 (under addition modulo 2).

The **semidirect product** $S_n \rtimes_\varphi \mathbb{Z}_2$ is the set of ordered pairs $S_n \times \mathbb{Z}_2$ equipped with a group operation defined by a homomorphism:

$$\varphi : \mathbb{Z}_2 \to \mathrm{Aut}(S_n)$$

Let $\varphi_0 = \mathrm{id}_{S_n}$ and $\varphi_1 = \theta$, where θ is an automorphism of S_n such that $\theta^2 = \mathrm{id}$ (an involution).

The group operation for two elements (σ, a) and (τ, b) in $S_n \rtimes_\varphi \mathbb{Z}_2$ is defined as:

$$(\sigma, a) \cdot (\tau, b) = (\sigma \circ \varphi_a(\tau), a + b)$$

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where $\circ$ denotes permutation multiplication and $+$ denotes addition modulo 2.

Note:

- If φ is the trivial homomorphism (maps everything to the identity automorphism), the group becomes the **Direct Product** $S_n \times \mathbb{Z}_2$.
- To obtain a non-trivial semidirect product, θ is typically chosen to be an inner automorphism (conjugation by an element of order 2) or, for $n = 6$, an outer automorphism.

Theorem 4.6.10. *The following hold.*

1. *For $n \geq 1$, $\mathrm{Aut}(K_n) \cong S_n$.*
2. *For $n \geq 3$, $\mathrm{Aut}(K_n - e) \cong S_{n-2} \times S_2$.*
3. *For $n \geq 4$, the automorphism group of K_n with two adjacent edges removed is isomorphic to $S_{n-3} \times S_2$.*
4. *For $n \geq 4$, the automorphism group of K_n with two non-adjacent edges removed is isomorphic to $S_{n-4} \times D_4$, where S_0 is understood to be the trivial group.*
5. *For $n \geq 3$, $\mathrm{Aut}(C_n) \cong D_n$.*
6. *For $m, n \geq 1$,*
$$\mathrm{Aut}(K_{m,n}) \cong \begin{cases} S_m \times S_n, & m \neq n, \\ (S_n \times S_n) \rtimes \mathbb{Z}_2, & m = n. \end{cases}$$
7. *For $n \geq 2$, $\mathrm{Aut}(P_n) \cong S_2$.*

Proof. We prove the assertions separately.

1. Since every pair of distinct vertices of K_n is adjacent, every permutation of its vertex set preserves adjacency. Conversely, every automorphism of K_n is a permutation of its n vertices. Hence
$$\mathrm{Aut}(K_n) \cong S_n.$$

2. Let $G = K_n - e$, where $e = \{u, v\}$. In G, the vertices u and v have degree $n - 2$, while each of the remaining $n - 2$ vertices has degree $n - 1$. Since automorphisms preserve degrees, every automorphism of G preserves the set $\{u, v\}$ and its complement $W = V(G) \setminus \{u, v\}$. The vertices of W may be permuted arbitrarily, giving a subgroup isomorphic to S_{n-2}, while u and v may either be fixed or interchanged, giving a subgroup isomorphic to S_2. These two actions are independent. Therefore
$$\mathrm{Aut}(K_n - e) \cong S_{n-2} \times S_2.$$

3. Let G be obtained from K_n by deleting the adjacent edges $\{u, v\}$ and $\{v, w\}$. Then
$$\deg(v) = n - 3, \qquad \deg(u) = \deg(w) = n - 2,$$
while every other vertex has degree $n - 1$. Hence every automorphism fixes v, preserves the set $\{u, w\}$, and preserves the remaining $n - 3$ vertices as a set. The vertices u and w may be interchanged, giving S_2, while the remaining $n - 3$ vertices may be permuted arbitrarily, giving S_{n-3}. These actions are independent. Thus
$$\mathrm{Aut}(G) \cong S_{n-3} \times S_2.$$

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4. Let G be obtained from K_n by deleting two non-adjacent edges $\{u, v\}$ and $\{x, y\}$. The four vertices u, v, x, y have degree $n-2$, while every other vertex has degree $n-1$. Therefore every automorphism preserves the sets

$$X = \{u, v, x, y\}, \qquad Z = V(G) \setminus X.$$

The vertices of Z may be permuted arbitrarily, giving S_{n-4}. The subgraph induced by X is K_4 with two independent edges removed, and hence is a cycle C_4. Its automorphism group is D_4, the dihedral group of order 8. Since the actions on X and Z are independent,

$$\mathrm{Aut}(G) \cong S_{n-4} \times D_4.$$

5. Let the vertices of C_n be $v_0, v_1, \ldots, v_{n-1}$ in cyclic order. An automorphism is completely determined by the image of v_0 and by the choice of which neighbor of its image is the image of v_1. There are n choices for the image of v_0 and, for each such choice, two choices for the image of v_1. Thus $\mathrm{Aut}(C_n)$ has $2n$ elements, namely the n rotations and n reflections of the cycle. Therefore

$$\mathrm{Aut}(C_n) \cong D_n,$$

where D_n denotes the dihedral group of order $2n$.

6. Let $K_{m,n}$ have partite sets V_1 and V_2, with $|V_1| = m$ and $|V_2| = n$.

Suppose first that $m \neq n$. Every vertex of V_1 has degree n, whereas every vertex of V_2 has degree m. Since these degrees are different, every automorphism preserves V_1 and V_2 setwise. The vertices within each part may be permuted arbitrarily, and any such pair of permutations preserves adjacency. Hence

$$\mathrm{Aut}(K_{m,n}) \cong S_m \times S_n.$$

Now suppose that $m = n$. Let $G = K_{n,n}$ with partite sets V_1 and V_2. Since a permutation of the vertex set is an automorphism of a graph if and only if it is an automorphism of its complement,

$$\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G}).$$

The complement $\overline{G}$ is the disjoint union of two copies of K_n, one induced by V_1 and the other by V_2. We may independently permute the vertices within these two components, giving the subgroup $S_n \times S_n$. In addition, the two isomorphic components may be interchanged. This interchange has order 2 and swaps the two factors of $S_n \times S_n$. Therefore

$$\mathrm{Aut}(K_{n,n}) \cong (S_n \times S_n) \rtimes \mathbb{Z}_2,$$

which is also the wreath product $S_n \wr S_2$.

7. Let the vertices of P_n be $v_1, v_2, \ldots, v_n$ in their natural order. The vertices v_1 and v_n are the only vertices of degree 1, so every automorphism preserves the set $\{v_1, v_n\}$.

If an automorphism fixes v_1, then it must fix its unique neighbor v_2. Continuing along the path, it follows successively that every vertex is fixed. Hence the automorphism is the identity.

If an automorphism sends v_1 to v_n, then adjacency forces

$$v_2 \mapsto v_{n-1}, \qquad v_3 \mapsto v_{n-2}, \qquad \ldots, \qquad v_i \mapsto v_{n-i+1}.$$

Thus the only nonidentity automorphism is the reflection σ defined by

$$\sigma(v_i) = v_{n-i+1}.$$

Therefore

$$\mathrm{Aut}(P_n) = \{\mathrm{id}, \sigma\}, \qquad \sigma^2 = \mathrm{id},$$

and hence

$$\mathrm{Aut}(P_n) \cong S_2.$$

■

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4.7 Edge Isomorphisms and Automorphisms

Definition. Two graphs $G(V, E)$ and $H(U, F)$ are said to be **edge-isomorphic** if there exists a bijection $\eta : E \to F$ that preserves edge-adjacency and edge-non-adjacency. Specifically, two edges $e, f \in E$ share a common vertex in G if and only if their images $\eta(e), \eta(f) \in F$ share a common vertex in H. Such a map η is called an *edge-isomorphism*.

An edge-isomorphism from a graph to itself is called an **edge-automorphism**. The set of all edge-automorphisms of a graph G forms a group under composition, known as the *edge-automorphism group* of G, denoted by $\mathrm{Aut}_E(G)$.

It is important to observe that for a given graph G, the vertex automorphism group $\mathrm{Aut}(G)$ and the edge-automorphism group $\mathrm{Aut}_E(G)$ need not be isomorphic. This discrepancy arises because there may exist edge-automorphisms that are not induced by any vertex automorphism (i.e., they are not in the induced edge group $\mathrm{Aut}_I(G)$).

Figure 1 illustrates three specific cases—$K_{1,3} + e$, $K_4 - e$, and K_4—where each graph admits an edge-automorphism η that cannot be induced by any permutation of the vertices.

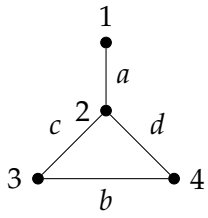


Figure 4.9: *
$K_{1,3} + e$
$\eta = (a\ b)(c\ d)$

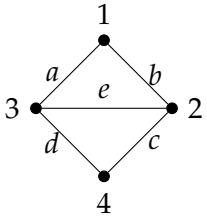


Figure 4.10: *
$K_4 - e$
$\eta = (a\ b\ c\ d)(e)$

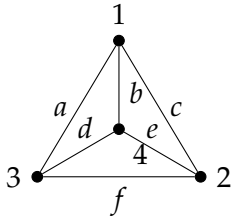


Figure 4.11: *
K_4
$\eta = (b\ f)(d)(e)(a)(c)$

Figure 4.12: Edge-automorphisms not induced by vertex automorphisms.

Theorem 4.7.1. *Let $\eta : E(G) \to E(H)$ be an edge-isomorphism from a connected graph G onto a connected graph H. If G is not isomorphic to any of the graphs*

$$K_3,\ K_{1,3},\ K_{1,3} + e,\ K_4 - e,\ K_4,$$

then η is induced by a vertex isomorphism from G to H.

Proof. For $v \in V(G)$, let

$$E_G(v) = \{e \in E(G) : e \text{ is incident with } v\}.$$

Since η is an edge-isomorphism, two edges of G are adjacent if and only if their images under η are adjacent in H.

We shall use the following simple fact: if at least four edges of a simple graph are pairwise adjacent, then they have a common end-vertex. For three pairwise adjacent edges there are only two possibilities: either they have a common end-vertex or they are the three edges of a triangle.

Let $v \in V(G)$ with $\deg(v) \geq 3$. The edges in $E_G(v)$ are pairwise adjacent, so the edges in $\eta(E_G(v))$ are also pairwise adjacent. If $\deg(v) \geq 4$, they must therefore have a common end-vertex in H.

Now suppose that $\deg(v) = 3$. Again the three edges in $\eta(E_G(v))$ are pairwise adjacent. If they do not have a common end-vertex, they must form a triangle in H. We show that this can occur only when G is one of the excluded graphs.

Write $N_G(v) = \{v_1, v_2, v_3\}$. Suppose that some v_i is incident with an edge $f \neq vv_i$ whose other end is not one of v_1, v_2, v_3. Then f is adjacent to exactly one of the three edges vv_1, vv_2, vv_3. Hence $\eta(f)$

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would have to be adjacent to exactly one edge of the triangle $\eta(E_G(v))$. This is impossible, because an edge outside a triangle is adjacent to either none or exactly two of its edges. Therefore every edge incident with one of v_1, v_2, v_3, apart from vv_i, must join two of the vertices v_1, v_2, v_3.

Since G is connected, no other vertices can occur. Thus G consists of the vertex v, its three neighbors v_1, v_2, v_3, and some of the possible edges among v_1, v_2, v_3. According as there are $0, 1, 2$, or 3 such edges, G is isomorphic to $K_{1,3}$, $K_{1,3} + e$, $K_4 - e$, or K_4, respectively. All of these are excluded. Hence, for every vertex v with $\deg(v) \geq 3$, the edges in $\eta(E_G(v))$ have a common end-vertex.

If $\deg(v) = 2$, the two edges in $E_G(v)$ are adjacent, and therefore their images have a unique common end-vertex. Thus, for every vertex v with $\deg(v) \geq 2$, define $\varphi(v)$ to be the common end-vertex of the edges in $\eta(E_G(v))$.

We next show that if $uv \in E(G)$ and $\deg(u), \deg(v) \geq 2$, then $\varphi(u) \neq \varphi(v)$. Suppose, to the contrary, that $\varphi(u) = \varphi(v)$. Choose an edge $e \neq uv$ incident with u and an edge $f \neq uv$ incident with v. Since $\eta(e)$ and $\eta(f)$ are both incident with $\varphi(u)$, they are adjacent in H. Hence e and f are adjacent in G. It follows that $e = uw$ and $f = vw$ for some vertex w, so u, v, w form a triangle.

If either u or v had another incident edge, the same argument would force that edge to be adjacent to both uw and vw, which is impossible in a simple graph. Hence $\deg(u) = \deg(v) = 2$. If w had an edge other than wu and wv, then the images of all edges incident with w would have to meet at the same vertex $\varphi(u)$, making the image of that additional edge adjacent to $\eta(uv)$. Its preimage would then have to be adjacent to uv, which is impossible. Thus $\deg(w) = 2$. Since G is connected, $G \cong K_3$, contrary to the hypothesis. Therefore $\varphi(u) \neq \varphi(v)$.

Now let $uv \in E(G)$ with $\deg(u), \deg(v) \geq 2$. The edge $\eta(uv)$ belongs to both $\eta(E_G(u))$ and $\eta(E_G(v))$. Hence it is incident with both $\varphi(u)$ and $\varphi(v)$. Since these vertices are distinct,

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

It remains to define φ on vertices of degree 1. Let u be such a vertex and let v be its unique neighbor. If $G \cong K_2$, then $H \cong K_2$ and the result is immediate. Otherwise $\deg(v) \geq 2$. Since $\eta(uv)$ is incident with $\varphi(v)$, define $\varphi(u)$ to be the other end-vertex of $\eta(uv)$. Then again

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

Thus, for every edge $uv \in E(G)$,

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

Hence φ preserves adjacency and induces η.

Finally, φ is bijective. Indeed, if two distinct vertices had the same image, two distinct incident edges in H would have adjacent preimages in a way incompatible with the construction above. Also, every vertex of H is incident with some edge because H is connected and nontrivial, and every edge of H is the image under η of an edge of G. Hence every vertex of H lies in the image of φ. Therefore φ is a vertex isomorphism from G onto H, and η is induced by φ. ■

Theorem 4.7.2. *For a non-empty graph G, $\Gamma_I(G) \cong \Gamma_E(G)$ if and only if the following conditions hold:*

(a) K_3 and $K_{1,3}$ do not both occur as components of G;

(b) none of the graphs $K_{1,3} + e$, $K_4 - e$, and K_4 is a component of G.

Proof. (See Figure 4.13.) We first prove necessity.

Suppose that both K_3 and $K_{1,3}$ occur as components of G. In both graphs every pair of edges is adjacent, so their edge sets are edge-isomorphic. Hence an edge-automorphism of G can interchange the edge sets of these two components while fixing all other edges. Such an edge-automorphism cannot be induced by a vertex automorphism, because a vertex automorphism maps a connected component only onto an isomorphic connected component, whereas $K_3 \ncong K_{1,3}$. Thus condition (a) is necessary.

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Now suppose that $K_{1,3}+e$ is a component of G. Let v be its vertex of degree 3, let u_1, u_2, u_3 be its neighbors, and suppose that u_1u_2 is the additional edge. The edge permutation which interchanges vu_3 and u_1u_2 and fixes vu_1 and vu_2 preserves edge adjacency, but it cannot be induced by a vertex automorphism. Hence $K_{1,3}+e$ cannot be a component.

If $K_4 - e$ is a component, let x, y be its vertices of degree 3 and let u, v be its vertices of degree 2. The four edges xu, yu, yv, xv form a 4-cycle in the edge-adjacency graph, while the edge xy is adjacent to all four. A rotation of this 4-cycle through one step, fixing xy, gives an edge-automorphism which is not induced by any vertex automorphism. Hence $K_4 - e$ cannot be a component.

If K_4 is a component, label its vertices $1, 2, 3, 4$. The edge permutation

$$12 \leftrightarrow 34,\ 13 \leftrightarrow 24,\ 14 \leftrightarrow 23$$

preserves edge adjacency but is not induced by any vertex automorphism of K_4. Therefore K_4 cannot be a component. This proves necessity of condition (b).

Conversely, assume that conditions (a) and (b) hold, and let $\alpha \in \Gamma_E(G)$ be an arbitrary edge-automorphism. We show that α is induced by a vertex automorphism of G.

Let C be a connected component of G containing at least one edge. Since α preserves adjacency between edges, the set $\alpha(E(C))$ is the edge set of some connected component C' of G. Thus

$$\alpha|_{E(C)} : E(C) \to E(C')$$

is an edge-isomorphism.

If $C \cong K_3$ or $C \cong K_{1,3}$, then condition (a) ensures that C' is of the same type as C. Indeed, K_3 and $K_{1,3}$ are the only two nonisomorphic connected graphs among the present exceptions whose edge-adjacency graphs are both K_3. Every permutation of the three edges of K_3 is induced by a vertex automorphism of K_3, and every permutation of the three edges of $K_{1,3}$ is induced by a permutation of its three leaves. Hence $\alpha|_{E(C)}$ is induced by a vertex isomorphism from C to C'.

If C is not isomorphic to K_3 or $K_{1,3}$, then condition (b) ensures that it is also not isomorphic to $K_{1,3}+e$, $K_4 - e$, or K_4. Therefore the preceding theorem applies, and $\alpha|_{E(C)}$ is induced by a vertex isomorphism

$$\varphi_C : C \to C'.$$

Thus, on every component of G containing an edge, the restriction of α is induced by a vertex isomorphism. Since distinct components are disjoint, these componentwise isomorphisms combine to give a vertex automorphism on the union of all nontrivial components of G. The isolated vertices of G are not incident with any edge, so they may be fixed, or permuted arbitrarily among themselves. Extending the componentwise maps in this way gives a vertex automorphism φ of G whose induced edge-automorphism is exactly α.

Hence every edge-automorphism of G is induced by a vertex automorphism. Therefore

$$\Gamma_I(G) \cong \Gamma_E(G).$$

■

Corollary 4.7.3. *If G is a connected graph of order $n \geq 3$, then $Aut(G)$, $Aut_I(G)$ and $Aut_E(G)$ are all isomorphic if and only if G is none of the graphs $K_{1,3}+e, K_4 - e, K_4$.*

Proof. Since G is connected and has order $n \geq 3$, it has no isolated vertices and no component isomorphic to K_2. Hence, by the preceding theorem,

$$\mathrm{Aut}(G) \cong \mathrm{Aut}_I(G).$$

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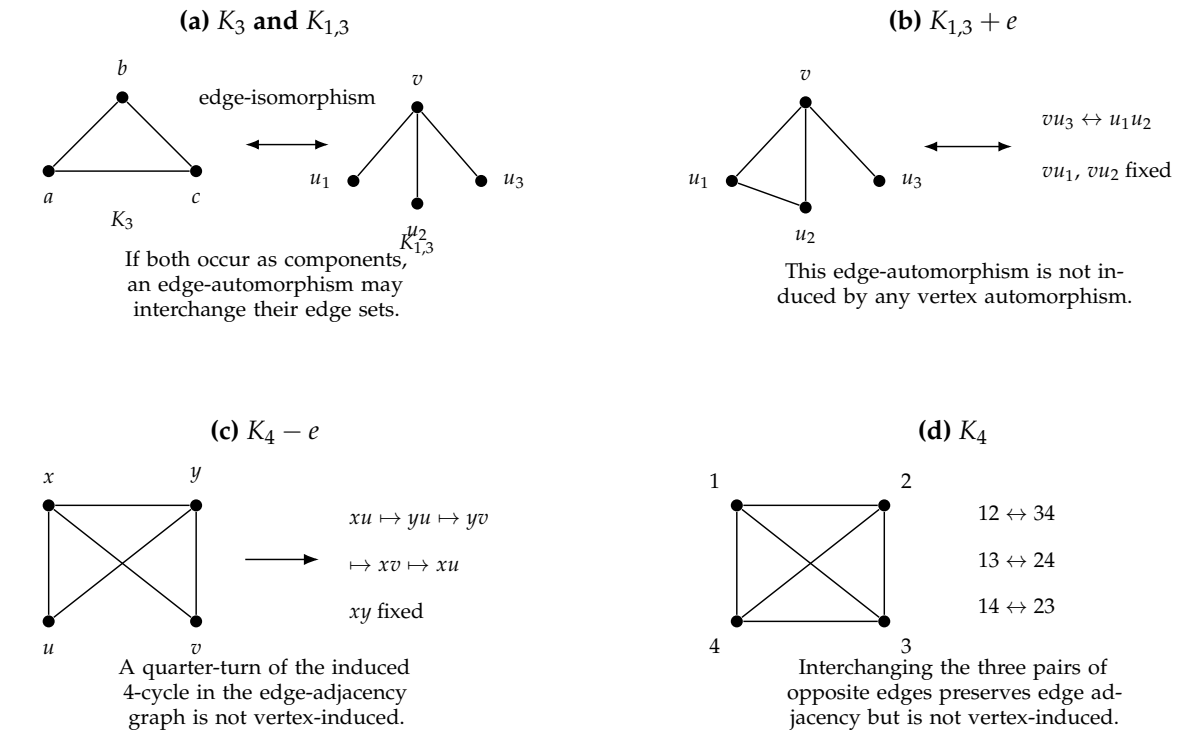


Figure 4.13: The proof of the criterion for $\Gamma_I(G) \cong \Gamma_E(G)$. Panels (a)–(d) illustrate the exceptional edge-automorphisms appearing in the necessity part. Panel (e) shows the componentwise construction used in the sufficiency part.

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Also, since G is connected, K_3 and $K_{1,3}$ cannot both occur as components of G. Therefore, by the preceding theorem,

$$\mathrm{Aut}_I(G) \cong \mathrm{Aut}_E(G)$$

if and only if G is none of the graphs $K_{1,3}+e$, K_4-e, and K_4. Consequently,

$$\mathrm{Aut}(G) \cong \mathrm{Aut}_I(G) \cong \mathrm{Aut}_E(G)$$

if and only if G is none of the graphs $K_{1,3}+e$, K_4-e, and K_4. ■

4.8 Cayley Graphs of Groups

Let G be a group and let $S \subseteq G$ be a generating set of G. The **Cayley graph** $\Gamma = \mathrm{Cay}(G, S)$ is a graph constructed as follows:

1. The set of vertices $V(\Gamma)$ corresponds to the elements of the group G. That is, $V(\Gamma) = G$.
2. The set of edges $E(\Gamma)$ is defined such that there is a directed edge from g to h if and only if $h = gs$ for some $s \in S$.

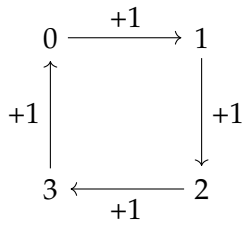
Notation: An edge is typically denoted as (g, gs) and acts as a directed edge colored or labeled by the generator s.

- Usually, the generating set S is chosen such that the identity element $1 \notin S$.
- If S is symmetric (i.e., $s \in S \implies s^{-1} \in S$), the graph is viewed as undirected, because an edge (g, gs) implies the existence of an edge $(gs, gss^{-1}) = (gs, g)$.

Example 4.8.1 (The Cyclic Group $\mathbb{Z}_4$)**.** Let $G = \mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. Let the generating set be $S = \{1\}$.

- **Vertices:** $0, 1, 2, 3$.
- **Edges:** Since the operation is $+$, edges are $(g, g+1)$.

$$0 \to 1, \quad 1 \to 2, \quad 2 \to 3, \quad 3 \to 0$$



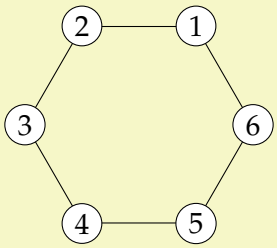
This forms a directed cycle C_4.

Example 4.8.2 (Cyclic Group $\mathbb{Z}_6$). Let $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ (under addition modulo 6). Let the generating set be $S = \{1\}$.

- Vertices: $\{0, 1, 2, 3, 4, 5\}$
- Edges: $(i, i+1 \pmod 6)$.

This forms a simple directed cycle C_6.

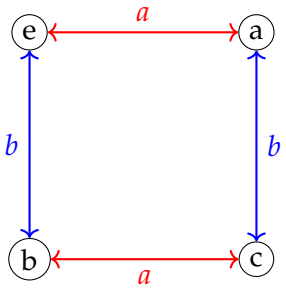
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Example 4.8.3 (The Klein 4-Group (V_4)). Let $G = V_4 = \{e, a, b, c\}$ where $a^2 = b^2 = c^2 = e$ and $ab = c$. Let the generating set be $S = \{a, b\}$.

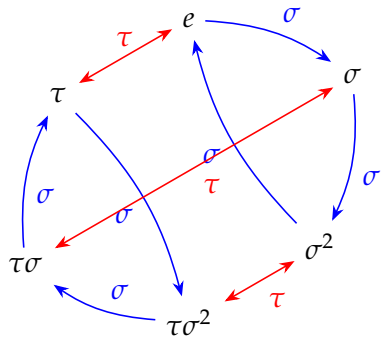
- Edges generated by a (Red): $(e, a), (a, e), (b, c), (c, b)$.
- Edges generated by b (Blue): $(e, b), (b, e), (a, c), (c, a)$.

Since $a = a^{-1}$ and $b = b^{-1}$, for every arrow $x \to y$, there is an arrow $y \to x$. We often draw these as undirected edges.



Example 4.8.4 (Symmetric Group S_3). Let $G = S_3$, the permutations of $\{1, 2, 3\}$. Let $S = \{\tau, \sigma\}$ where $\tau = (12)$ and $\sigma = (123)$.

- σ has order 3 (forms hexagons).
- τ has order 2 (forms bi-directional swaps).



Note: The red edges represent τ. Since $\tau^2 = e$, these are technically bidirectional arrows $x \leftrightarrow y$.

4.9 Undirected Cayley Graphs

While the definition implies a digraph (directed graph), we often treat them as undirected graphs under specific conditions.

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Definition 4.9.1. $\text{Cay}(G, S)$ is considered an undirected graph if S is inverse-closed. That is, for every $s \in S$, we have $s^{-1} \in S$.

In this case, if there is an edge $g \to gs$, there is also an edge $gs \to (gs)s^{-1} = g$. These pairs of directed edges are treated as a single undirected edge.

Theorem 4.9.2. *If S generates the group G, then the Cayley graph $Cay(G, S)$ is connected.*

Proof. Let $u, v \in G$ be any two vertices in the graph. We wish to show there is a path from u to v.

Consider the element $g = u^{-1}v$. Since S generates G, the element g can be written as a product of generators and their inverses (if we consider the undirected case or if S is closed under inverses).

$$g = s_1 s_2 \dots s_k \quad \text{where } s_i \in S \cup S^{-1}$$

Therefore, $v = u(s_1 s_2 \dots s_k)$. We can construct a walk starting at u:

$$\begin{aligned} w_0 &= u \\ w_1 &= us_1 \\ w_2 &= us_1 s_2 \\ &\vdots \\ w_k &= us_1 s_2 \dots s_k = v \end{aligned}$$

Since there is a sequence of edges connecting u to v, the graph is connected. ■

Theorem 4.9.3. *The Cayley graph $\Gamma = Cay(G, S)$ is $|S|$-regular (in terms of out-degree).*

Proof. Let $g \in G$ be any vertex. The edges originating from g are determined by the definition:

$$E_g = \{(g, gs) \mid s \in S\}$$

Since group cancellation holds ($gs_1 = gs_2 \implies s_1 = s_2$), the number of distinct edges starting at g is exactly equal to the number of distinct elements in S. Thus, the out-degree of every vertex is $|S|$. ■

This is the most fundamental property of Cayley graphs. It implies that the graph "looks the same" from every vertex.

Definition 4.9.4. A graph Γ is said to be *vertex-transitive* if, for every pair of vertices $u, v \in V(\Gamma)$, there exists an automorphism $\phi \in \text{Aut}(\Gamma)$ such that $\phi(u) = v$.

Theorem 4.9.5. *Every Cayley graph $\text{Cay}(G, S)$ is vertex-transitive.*

Proof. Let $\Gamma = \text{Cay}(G, S)$. Thus the vertices of Γ are the elements of the group G, and two vertices $x, y \in G$ are adjacent whenever $y = xs$ for some $s \in S$.

Take arbitrary vertices $u, v \in G$. We shall construct an automorphism of Γ which maps u to v. Put $h = vu^{-1}$ and define $\lambda_h : G \to G$ by $\lambda_h(x) = hx$. Thus λ_h is simply left multiplication by the group element h.

The map λ_h is a bijection, since its inverse is $\lambda_{h^{-1}}$. We now show that it preserves adjacency. Suppose that x and y are adjacent in Γ. By the definition of a Cayley graph, $y = xs$ for some $s \in S$. Therefore

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$\lambda_h(y) = hy = h(xs) = (hx)s = \lambda_h(x)s$. Hence $\lambda_h(x)$ and $\lambda_h(y)$ are also adjacent. Thus λ_h is an automorphism of Γ.

Finally, $\lambda_h(u) = hu = (vu^{-1})u = v$. Thus, for every pair of vertices $u, v \in G$, there exists an automorphism of Γ mapping u to v. Hence $\mathrm{Cay}(G, S)$ is vertex-transitive. ■

Definition 4.9.6. Let G be a group and let $\Lambda \subseteq G$ be a generating set. The *Cayley digraph* $\Gamma = D(G, \Lambda)$ has vertex set $V(\Gamma) = G$, and there is an arc from g to k if and only if $k = gh$ for some $h \in \Lambda$. Equivalently, (g, k) is an arc of Γ if and only if $g^{-1}k \in \Lambda$.

If (g, k) is an arc of Γ, then the element $g^{-1}k \in \Lambda$ is called the *color* of the arc (g, k). Thus, the arc from g to gh, where $h \in \Lambda$, has color h.

An automorphism $\phi \in \mathrm{Aut}(\Gamma)$ is said to be *color-preserving* if it preserves the color of every arc. More precisely, whenever (g, k) is an arc of color h, the image arc $(\phi(g), \phi(k))$ also has color h. Thus, $\phi(k) = \phi(g)h$ then $h = \phi^{-1}(g)\phi(k)$, that is,

$$g^{-1}k = \phi(g)^{-1}\phi(k)$$

for every arc (g, k) of Γ.

Theorem 4.9.7 (Automorphism Theorem for Colored Cayley Digraphs)**.** *Let $D(\Gamma, \Lambda)$ be a Cayley digraph, where Λ generates Γ. Then the group of all color-preserving automorphisms of $D(\Gamma, \Lambda)$ is isomorphic to Γ.*

Proof. Let $C(D)$ denote the group of all color-preserving automorphisms of $D(\Gamma, \Lambda)$. We shall show that the elements of $C(D)$ are precisely the left multiplications by elements of Γ.

For each $g \in \Gamma$, define $\alpha_g : \Gamma \to \Gamma$ by $\alpha_g(x) = gx$. Since left multiplication by g is a bijection, α_g is a permutation of the vertex set of D.

We first show that α_g preserves the color of every arc. Suppose that (u, v) is an arc of color $h \in \Lambda$. Then $u^{-1}v = h$, or equivalently, $v = uh$. Under α_g, the arc (u, v) is mapped to (gu, gv). The color of the image arc is

$$(gu)^{-1}(gv) = u^{-1}g^{-1}gv = u^{-1}v = h.$$

Thus the color of every arc is preserved. Hence $\alpha_g \in C(D)$ for every $g \in \Gamma$.

If

$$L(\Gamma) = \{\alpha_g : g \in \Gamma\},$$

then we have $L(\Gamma) \subseteq C(D)$.

We now prove the reverse inclusion. Let $\phi \in C(D)$ be arbitrary, let e denote the identity element of Γ, and suppose that $\phi(e) = g$. We claim that $\phi(x) = gx$ for every $x \in \Gamma$.

Since Λ generates Γ, every vertex can be reached from e by following arcs whose colors are elements of Λ. We show that once the image of one vertex is known, the image of the next vertex along such an arc is completely determined.

Suppose that $\phi(u) = gu$ and that (u, v) is an arc of color $h \in \Lambda$. Since $v = uh$ and ϕ preserves colors, the image arc $(\phi(u), \phi(v))$ also has color h. Therefore $\phi(u)^{-1}\phi(v) = h$. Substituting $\phi(u) = gu$, we obtain $(gu)^{-1}\phi(v) = h$, and hence $\phi(v) = guh = gv$.

Starting with $\phi(e) = g = ge$ and applying this argument successively along a path from e to an arbitrary vertex x, we obtain $\phi(x) = gx$. Therefore $\phi = \alpha_g$, and hence $C(D) \subseteq L(\Gamma)$.

Consequently,

$$C(D) = L(\Gamma).$$

It remains to identify this group with Γ. Define $\Psi : \Gamma \to C(D)$ by $\Psi(g) = \alpha_g$. The map Ψ is bijective. Moreover, for any $g, k, x \in \Gamma$,

$$(\alpha_g \circ \alpha_k)(x) = \alpha_g(kx) = gkx = \alpha_{gk}(x).$$

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Thus $\alpha_g \circ \alpha_k = \alpha_{gk}$, and therefore $\Psi(gk) = \Psi(g)\Psi(k)$. Hence Ψ is a group isomorphism. Consequently,

$$C(D) \cong \Gamma.$$

■

Remark 4.9.8. The essential point of the preceding theorem is that a color-preserving automorphism is completely determined by the image of the identity element. Indeed, if $\phi(e) = g$, then necessarily $\phi(x) = gx$ for every $x \in \Gamma$. Thus the color-preserving automorphisms are exactly the left translations of the group.

Theorem 4.9.9 (Frucht, 1939)**.** *Let Γ be a finite group. Then there exists a finite undirected graph G whose automorphism group is isomorphic to Γ; that is,*

$$\mathrm{Aut}(G) \cong \Gamma.$$

Proof. Let $\Lambda = \{h_1, \ldots, h_n\}$ be a generating set for the finite group Γ, and let $D = D(\Gamma, \Lambda)$ be the corresponding Cayley digraph. Its colour-preserving automorphism group $C(D)$ is isomorphic to Γ. Thus it is enough to construct a finite undirected graph G such that $\mathrm{Aut}(G) \cong C(D)$.

For each directed arc $g_i \to g_j$ of colour h_k, replace the arc by a path

$$g_i - u - v - g_j.$$

To record the direction and the colour of the arc, attach to u a pendant path of length $2k$ and to v a pendant path of length $2k+1$. Carrying out this replacement for every arc of D produces a finite undirected graph G.

Every colour-preserving automorphism of D extends naturally to an automorphism of G, since arcs of the same colour are replaced by identical subgraphs. Hence

$$C(D) \leq \mathrm{Aut}(G).$$

Conversely, every automorphism of G preserves the set of original vertices of D, since these vertices are structurally distinguishable from the newly introduced vertices. Moreover, in each replacement subgraph the two pendant paths have different lengths, namely $2k$ and $2k+1$. Therefore an automorphism cannot interchange the two ends of the replaced arc, so the direction is preserved. The pair of lengths also determines the index k, and hence the colour h_k. Thus every automorphism of G restricts to a colour-preserving automorphism of D.

It follows that

$$\mathrm{Aut}(G) \cong C(D).$$

Since $C(D) \cong \Gamma$, we conclude that

$$\mathrm{Aut}(G) \cong \Gamma.$$

Hence every finite group is isomorphic to the automorphism group of a finite undirected graph. ■

Detailed Explanation of the Above Proof. Let $\Lambda = \{h_1, h_2, \ldots, h_n\}$ be a generating set for the group Γ, and let $D = D(\Gamma, \Lambda)$ be the corresponding Cayley digraph. We have already shown that the group $C(D)$ of colour-preserving automorphisms of D is isomorphic to Γ. Thus it is enough to construct a finite undirected graph G whose full automorphism group is precisely $C(D)$.

The idea is to replace every directed, coloured arc of the Cayley digraph by an undirected subgraph whose shape records both the direction and the colour of that arc.

Consider an arc (g_i, g_j) of D having colour h_k. Thus $g_i^{-1} g_j = h_k$. Replace this arc by a path $g_i - u - v - g_j$ of length three. To distinguish the two internal vertices and to record the colour h_k,

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attach to u a pendant path P_{2k} of length $2k$, and attach to v a pendant path P_{2k+1} of length $2k+1$. Performing this replacement for every arc of D produces a finite undirected graph G. The replacement corresponding to an arc $g_i \xrightarrow{h_k} g_j$ is shown in Figure 4.9.

We first show that every colour-preserving automorphism of D induces an automorphism of G. Let $\alpha \in C(D)$. If α maps the arc (g_i, g_j) of colour h_k to the arc (g_x, g_y), then the latter arc also has colour h_k, since α preserves colours. Consequently, the subgraph replacing (g_i, g_j) and the subgraph replacing (g_x, g_y) have exactly the same structure: in both cases the first internal vertex has an attached path of length $2k$ and the second has an attached path of length $2k+1$.

Hence α extends naturally to the new vertices of G by mapping the internal vertices and the corresponding pendant paths of one replacement subgraph onto those of the other. This extension preserves all adjacencies and is therefore an automorphism of G. Thus $C(D)$ is isomorphic to a subgroup of $\mathrm{Aut}(G)$.

Conversely, let $\beta \in \mathrm{Aut}(G)$. We show that its action on the original vertices induces a colour-preserving automorphism of D.

The original vertices of the Cayley digraph can be distinguished from the newly introduced vertices by their structural position in G. The internal vertices u and v of each replacement subgraph have degree three, while the vertices lying on the attached pendant paths have degree one or two. The original group vertices, on the other hand, are precisely the vertices at which the replacement subgraphs corresponding to the incident arcs meet. Hence every automorphism of G must preserve the set of original vertices.

Now consider the subgraph replacing an arc $g_i \xrightarrow{h_k} g_j$. The vertex adjacent to g_i has an attached path of length $2k$, whereas the vertex adjacent to g_j has an attached path of length $2k+1$. Since these two path lengths are different, an automorphism cannot interchange the two ends of the replacement subgraph. Thus the asymmetry of the gadget preserves the direction of the original arc.

The same construction also preserves its colour. Indeed, the lengths $2k$ and $2k+1$ uniquely determine the index k. Therefore a replacement subgraph corresponding to the colour h_k can be mapped only to another replacement subgraph corresponding to the same colour h_k.

It follows that the restriction of β to the original vertices maps arcs of D to arcs of the same direction and the same colour. Hence this restriction is a colour-preserving automorphism of D, and so it belongs to $C(D)$.

Therefore every automorphism of G arises uniquely from a colour-preserving automorphism of D. Consequently,

$$\mathrm{Aut}(G) \cong C(D).$$

Since $C(D) \cong \Gamma$, we conclude that

$$\mathrm{Aut}(G) \cong \Gamma.$$

■

Corollary 4.9.10. *Let Γ be a finite group of order n, and let Λ be a generating set of size $s = |\Lambda|$. The Frucht graph $G(\Gamma)$ (Fig 4.13) has exactly $n(s+1)(2s+1)$ vertices.*

Proof. To determine the order of the Frucht graph $G(\Gamma)$, we consider the vertices contributed by the original group elements and those introduced by the topological construction. The underlying Cayley digraph $D(\Gamma, \Lambda)$ contains n vertices, and each vertex possesses exactly s outgoing arcs corresponding to the k generators indexed $k = 1$ to s. In the transition to $G(\Gamma)$, each arc associated with the k-th generator is replaced by a subgraph that introduces $4k+1$ new internal vertices. Consequently, for every element in the group, the set of outgoing arcs contributes a total of $\sum_{k=1}^{s}(4k+1)$ additional vertices. Evaluating this sum yields $4\frac{s(s+1)}{2} + s = 2s^2 + 3s$ new vertices per group element. Across the entire group of n

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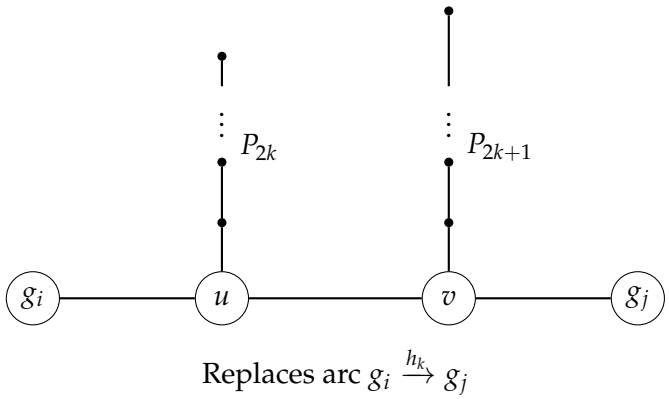


Figure 4.14: Frucht Graph: The subgraph replacing the directed edge from g_i to g_j with color h_k. The asymmetry of the paths P_{2k} and P_{2k+1} enforces the direction and color preservation.

elements, the total number of added vertices is $n(2s^2 + 3s)$. Adding the n original vertices to this count, we find that the total order of the graph is $n(2s^2 + 3s + 1)$. Factoring the quadratic expression, we conclude that $|V(G(\Gamma))| = n(s + 1)(2s + 1)$. ■

Example 4.9.11. Frucht Graph Order for $n = 3$, and $s = 2$.

Solution. First, calculating directly using the formula:

$$|V(G)| = 3(2 + 1)(2(2) + 1) = 3(3)(5) = 45.$$

We now construct the count step-by-step based on the topological definition of the graph.

Every vertex in Γ has $s = 2$ outgoing arcs, one for each generator. The construction replaces these arcs with "gadgets" of different sizes depending on the index k. The number of internal vertices added for the k-th generator is $4k + 1$.

- For generator h_1 ($k = 1$): The gadget adds $4(1) + 1 = 5$ vertices.
- For generator h_2 ($k = 2$): The gadget adds $4(2) + 1 = 9$ vertices.

Since every group element (e, x, y) has exactly one outgoing arc for h_1 and one outgoing arc for h_2, the total number of vertices added **per group element** is:

$$\text{Added per vertex} = 5 + 9 = 14.$$

Therefore,

$$\begin{aligned}\text{Total Vertices} &= n + n\sum_{k=1}^{2}(4k + 1) \\ &= 3 + 3[(4(1) + 1) + (4(2) + 1)] \\ &= 3 + 3[5 + 9] \\ &= 3 + 42 \\ &= 45.\end{aligned}$$

■

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